

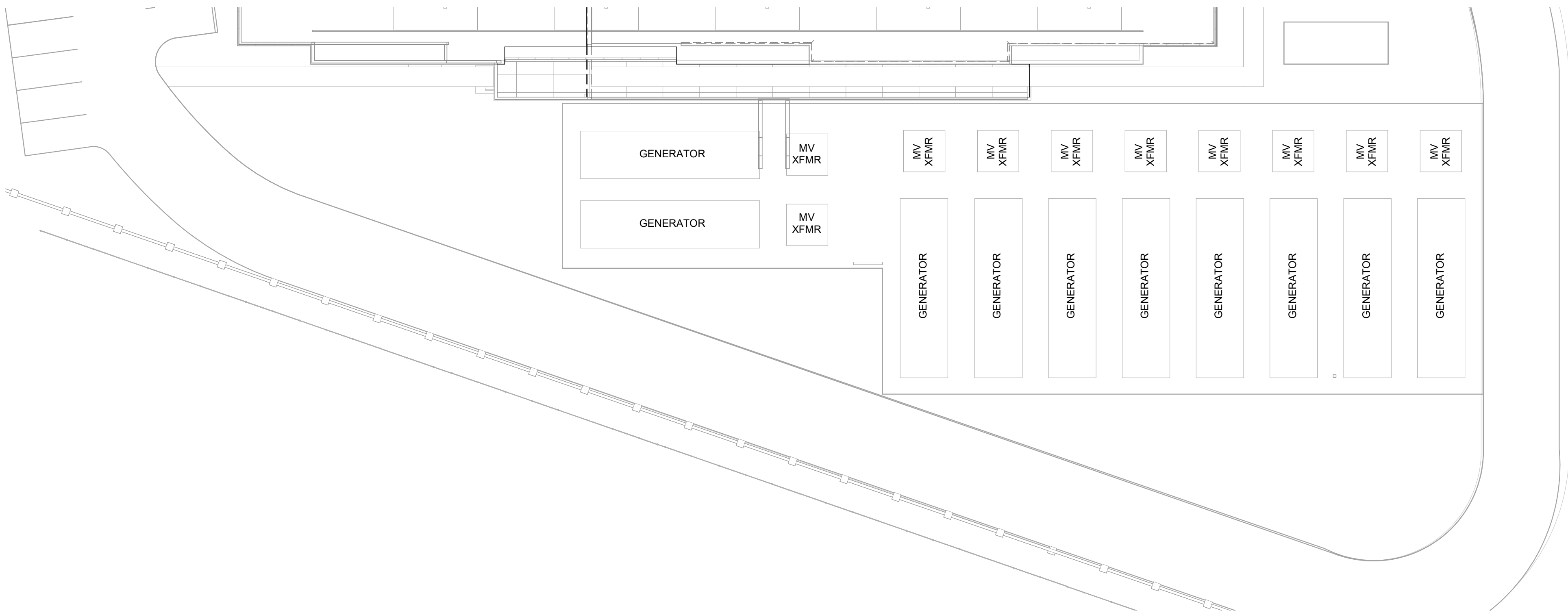
GENERAL STRUCTURAL NOTES									
PART I - DESIGN CRITERIA		PART IV - BUILDING EARTHWORK (CONTINUED)		PART VI - STRUCTURAL STEEL (CONTINUED)	PART VIII - STRUCTURAL TESTING AND INSPECTIONS	PART VIII - STRUCTURAL TESTING AND INSPECTIONS (CONTINUED)			
A. GENERAL BUILDING CODE		C. SOIL MATERIALS		7. WELDING PROCEDURE SPECIFICATION: ALL WELDING SHALL BE PERFORMED IN ACCORDANCE WITH A WELDING PROCEDURE SPECIFICATION (WPS) AS REQUIRED IN AWS D1.1 AND APPROVED BY THE TESTING AGENCY AND THE ENGINEER. THE WPS VARIABLES SHALL BE WITHIN THE PARAMETERS ESTABLISHED BY THE FILLER-METAL MANUFACTURER. THE CONTRACTOR SHALL ENGAGE THE SERVICES OF AN INDEPENDENT TESTING AGENCY TO PROVIDE THE QUALIFICATION TESTING REQUIRED BY AWS D 1.1, CHAPTER 4, PART 8 TO QUALIFY ANY NON-REGULATED WPS NEEDED FOR THE PROJECT. THE CONTRACTOR'S TESTING AGENCY SHALL PREPARE WELDING PROCEDURE QUALIFICATION RECORDS (WPQR) DOCUMENTING THE SUCCESSFUL QUALIFICATION OF EACH WELDING PROCEDURE SPECIFICATION.		A. GENERAL		7. NONDESTRUCTIVE EXAMINATION, IF REQUIRED, CONDUCTED IN ACCORDANCE WITH THE SPECIFIC REQUIREMENTS FOR THE ITEM BEING EXAMINED INCLUDING ULTRASONIC (UT) OR MAGNETIC PARTICLE (MT). NONDESTRUCTIVE INSPECTION PROCEDURES SHALL CONFORM TO AWS D1.1.	
1. THE CONSTRUCTION DOCUMENTS ARE BASED ON THE REQUIREMENTS OF THE 2021 VIRGINIA CONSTRUCTION CODE AND SUBSEQUENT REFERENCE STANDARDS.		1. GENERAL SOIL MATERIALS: SATISFACTORY SOIL MATERIALS ARE DEFINED AS THOSE COMPLYING WITH ASTM D2487 SOIL CLASSIFICATION GROUPS GW, GP, GM, SM, SW, AND SP.		1.1. CHAPTER 4, PART 8 TO QUALIFY ANY NON-REGULATED WPS NEEDED FOR THE PROJECT. THE CONTRACTOR'S TESTING AGENCY SHALL PREPARE WELDING PROCEDURE QUALIFICATION RECORDS (WPQR) DOCUMENTING THE SUCCESSFUL QUALIFICATION OF EACH WELDING PROCEDURE SPECIFICATION.		a. RE-INSPECTION OF WORK SHALL BE AT CONTRACTOR'S EXPENSE. COSTS OF RE-INSPECTION SHALL BE PAID FOR BY THE OWNER. THESE COSTS SHALL BE REIMBURSED TO THE OWNER BY THE CONTRACTOR.		8. FILLET WELDS: MT TEST A MINIMUM OF 10% OF THE LENGTH OF EACH FILLET WELD EXCEEDING 5/16". THE FOLLOWING TESTING REGIMEN IS NOT REQUIRED BY SPECIAL INSPECTION. THOUGH RECOMMENDED, IT MAY BE DELETED WITHOUT VIOLATING CODE REQUIREMENTS IF SO DESIRED BY THE OWNER. PERIODIC MT TESTING OF REPRESENTATIVE FILLET WELDS 5/16" AND LESS BUT NEED NOT EXCEED 10% OF ALL SUCH WELDS, EXCEPT AS REQUIRED FOR HIGH REJECTION RATES AS INDICATED IN THE FOLLOWING PARAGRAPH. INCREASE MT TESTING RATE FOR WELDERS HAVING A HIGH REJECTION RATE AS REQUIRED TO ENSURE ACCEPTABLE WELDS.	
B. DEAD LOADS		D. SUB-BASE		E. BASEPLATE GROUT		2. SPECIAL INSPECTIONS: THE TESTING AGENCY SHALL SERVE AS A SPECIAL INSPECTOR TO PROVIDE SPECIAL INSPECTION SERVICES, AS LEGALLY REQUIRED BY THE REFERENCED BUILDING CODE FOR THE ITEMS LISTED IN SECTION "SPECIAL INSPECTIONS."		9. COMPLETE JOINT PENETRATION (CJP) WELDS: ALL CJP WELDS EXCEEDING 5/16" THICKNESS SHALL BE 100% UT TESTED. THE TESTING AGENCY SHALL REVIEW THE CJP JOINTS TO DETERMINE WHERE GEOMETRY OR ACCESSIBILITY PRECLUDES THE USE OF STANDARD SCANNING PATTERNS. AT THESE LOCATIONS THE TESTING AGENCY SHALL DEVELOP AND SUBMIT FOR APPROVAL A WRITTEN TESTING PROCEDURE TO EXAMINE THESE AREAS. PERIODIC MT TESTING OF REPRESENTATIVE CJP WELDS 5/16" AND LESS NOT TO EXCEED 10% OF ALL SUCH WELDS, EXCEPT AS REQUIRED FOR HIGH REJECTION RATES AS INDICATED IN THE FOLLOWING PARAGRAPH.	
1. SEE LOAD PLANS ON S-004 AND S-005		1. INSTALL, TAMP, AND LEVEL NEW SAND OR AGGREGATE SUB-BASE TO (±2%) TO 85% RELATIVE DENSITY FOR COHESIONLESS MATERIAL PRIOR TO INSTALLING VAPOR BARRIER.		1. NON-METALLIC NON-SHRINK GROUT: PREMIXED, FACTORY PACKAGED, NON-METALLIC, NON-SHRINK, NON-CORROSIVE, NON-STAINING, PRODUCT CONTAINING PORTLAND CEMENT, SILICA SANDS, SHRINKAGE COMPENSATING AGENTS, AND FLUIDITY IMPROVING COMPOUNDS. GROUT SHALL CONFORM TO ASTM C1107 AND HAVE A MINIMUM 28 DAY ASTM C109 COMPRESSIVE STRENGTH OF 8,000 PSI. ACCEPTABLE GROUT MATERIALS FOR GROUTING BASEPLATES ARE: a. SIKKA CORPORATION; MASTERFLOW 555 OR 4316 b. SIKKA CORPORATION; SIKAGROUT 212 OR 528 c. EUCILID CHEMICAL COMPANY; NS GROUT OR NS GROUT d. MAPEI CORPORATION; PLANIGROUT 755 OR 728		1. CONTRACTOR OBLIGATIONS: 1. CONTRACTOR RESPONSIBILITY: THE CONTRACTOR SHALL PROVIDE THE OWNER'S TESTING AGENCY WITH THE FOLLOWING: a. THE CONTRACTOR SHALL PROVIDE TO THE TESTING AGENCY CERTIFICATES AND REPRESENTATIVE SAMPLES OF MATERIALS PROPOSED FOR USE IN THE WORK IN QUANTITIES SUFFICIENT FOR ACCURATE TESTING AS SPECIFIED b. DOCUMENTATION OF WELDERS CERTIFICATION TO PERFORM THE INDICATED WORK AND THE APPLICABLE WPS DOCUMENTATION AS INDICATED ABOVE. c. PROPER FACILITIES, INCLUDING SCAFFOLDING, TEMPORARY WORK PLATFORMS, HOISTING FACILITIES, CASUAL LABOR, ETC, FOR INSPECTION OF WORK IN THE MILLS, SHOP, AND FIELD.		10. MARK FOR REMEDIATION ANY AREA NOT MEETING CONTRACT DOCUMENT REQUIREMENTS. CORRECTION OF REJECTED WELDS SHALL BE MADE IN ACCORDANCE WITH AWS D1.1.	
C. LIVE LOADS		E. BELOW GRADE VAPOR BARRIER		F. CORROSION RESISTANCE		2. OWNER'S TESTING AGENCY RESPONSIBILITY: THE TESTING AGENCY SHALL SAMPLE AND TEST MATERIALS AS THEY ARE BEING INSTALLED IN ACCORDANCE WITH THE SPECIFIED ACCEPTANCE CRITERIA. THE TESTING AGENCY WILL REPORT AND INTERPRET THE TEST RESULTS. THE TESTING AGENCY SHALL MONITOR AND REPORT ON THE INSTALLATION OF CONSTRUCTION WORK AND SHALL PERFORM TESTS ON THE COMPLETED CONSTRUCTION AS REQUIRED TO INDICATE CONTRACTOR'S COMPLIANCE WITH THE VARIOUS PROJECT REQUIREMENTS. a. THE TESTING AGENCY SHALL MEET THE BASIC REQUIREMENTS OF ASTM E520 AND SHALL SUBMIT TO THE OWNER (ARCHITECT), AND ENGINEER EVIDENCE OF CURRENT ACCREDITATION FROM THE AMERICAN ASSOCIATION FOR LABORATORY ACCREDITATION, THE ANSITO ACCREDITATION PROGRAM OR THE "NIST" NATIONAL VOLUNTARY LABORATORY ACCREDITATION PROGRAM. b. THE TESTING AGENCY SHALL BE AN APPROVED AGENCY BY THE BUILDING OFFICIAL OF THE CITY WHEREIN THE PROJECT IS LOCATED TO PERFORM SPECIAL INSPECTIONS AND OTHER TESTS AND INSPECTIONS AS OUTLINED IN THE APPLICABLE BUILDING CODE. c. THE INSPECTION BY THE TESTING AGENCY OF THE WELDER'S WORK SHALL BE IN SEQUENCE, TIMELY, AND PERFORMED IN SUCH A MANNER SO THAT CORRECTIONS CAN BE MADE WITHOUT DELAYING THE PROGRESS OF THE WORK. INSPECTIONS OF THE WELDER'S WORK SHALL BE PERFORMED BY CERTIFIED WELDING INSPECTORS IN ACCORDANCE WITH AWS D1.1 SECTION 6.1.4 FOR VISUAL WELD INSPECTION AND AWS D1.1, SECTION 6.1.4.6 FOR NONDESTRUCTIVE EXAMINATION. QUALIFIED TECHNICIANS WITH A MINIMUM OF TWO YEARS' EXPERIENCE IN STRUCTURAL STEEL TESTING AND INSPECTION. d. INSPECTIONS OF STRUCTURAL STEEL AND BOLTED CONNECTIONS SHALL BE PERFORMED BY QUALIFIED TECHNICIANS WITH A MINIMUM OF TWO YEAR'S EXPERIENCE IN STRUCTURAL STEEL TESTING AND INSPECTION. e. THE TESTING AGENCY SHALL PROVIDE REPORTS OF ALL TESTING AND INSPECTIONS. ALL REPORTS SHALL INDICATE TYPES AND LOCATIONS OF ALL DEFECTS FOUND DURING INSPECTION. THE MEASURING PERSONNEL AND PERFORMED TO CORRECT SUCH DEFECTS, STATEMENTS OF FINAL APPROVAL OF ALL WELDING OF FIELD CONNECTIONS, AND OTHER FABRICATION AND ERECTION DATA PERTINENT TO THE SAFE AND PROPER WELDING OF FIELD CONNECTIONS.		11. RE-EXAMINE ALL REWORKED AREAS AND INTERPRET, RECORD, AND REPORT THE RESULTS OF EXAMINATIONS OF REWORKED WELDS.	
D. WIND LOADS		1. MATERIALS		3. ZINC COATING: THE ZINC COATING FOR STEEL SHAPES AND PLATES SHALL CONFORM TO ASTM A123. WEIGHT OF ZINC COATING PER SQUARE FT OF SURFACE FOR 1/8" AND 3/16" THICK STEELS SHALL AVERAGE NOT LESS THAN 2.0 OZ WITH NO INDIVIDUAL THICKNESS LESS THAN 1.8 OZ. THE COATING WEIGHT SHALL AVERAGE NOT LESS THAN 2.3 OZ WITH NO INDIVIDUAL THICKNESS LESS THAN 2.0 OZ FOR 1/4" THICK AND HEAVIER STEEL.		F. INSPECTION OF POST-INSTALLED ANCHOR, BOLTS, AND REINFORCING BARS INSTALLED IN HARDENED CONCRETE: THE TESTING AGENCY OR MANUFACTURE SHALL MAKE THE FOLLOWING INSPECTIONS OF POST-INSTALLED ANCHOR INSTALLATIONS IN ACCORDANCE WITH THE PUBLISHED, CURRENTLY VALID, EVALUATION SERVICE REPORT (ESR) FOR EACH ANCHOR PRODUCT. INSPECTIONS SHALL BE PERFORMED AT THE FREQUENCY REQUIRED BY THE ESR.		12. VERIFY THAT QUALITY OF WELDS MEET THE REQUIREMENTS OF AWS D1.1.	
1. WIND PRESSURES ARE BASED ON THE PROVISIONS OF THE AMERICAN SOCIETY OF CIVIL ENGINEERS, MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22 AND THE FOLLOWING CRITERIA: a. BASIC DESIGN WIND SPEED (V): 119 MPH (3 SECOND GUST) b. WIND EXPOSURE CATEGORY: B c. INTERNAL PRESSURE COEFFICIENT (GCP1): +0.18/-0.18		a. PLASTIC SHEET VAPOR BARRIER AND RELATED MATERIALS SHALL ALL BE FROM THE SAME MANUFACTURER. b. PLASTIC SHEET VAPOR BARRIER: 15 MIL THICK PLASTIC SHEET CONFORMING TO ASTM E745 CLASS A. ASTM F1249 WATER VAPOR PERMEANCE SHALL BE LESS THAN 0.01 US PERMS AFTER MANDATORY CONDITIONING. ACCEPTABLE SYSTEMS ARE: 1) STEGO INDUSTRIES, LLC; STEGO WRAP VAPOR BARRIER (15 MIL) 2) EPRO; ECOSHIELD-E (15 MIL) 3) RAVEN INDUSTRIES; VAPORBLOCK V615 4) W.R. MEADOWS; PERMIMATON MIL 5) FORTIFIBER BUILDING SYSTEMS GROUP; MOISTOP ULTRA 15 c. SEAM TAPE FOR PLASTIC VAPOR RETARDERS: HIGH-DENSITY POLYETHYLENE TAPE WITH PRESSURE SENSITIVE ADHESIVE HAVING A MINIMUM WIDTH OF 0.75" HAVING A MAXIMUM WATER VAPOR PERMEANCE RATE OF 0.3 PERMS. ACCEPTABLE SYSTEMS ARE: 1) STEGO INDUSTRIES, LLC; STEGO TAPE 2) EPRO; ECOTAPE 3) RAVEN INDUSTRIES; VAPORBOND TAPE 4) W.R. MEADOWS; PERMIMATON TAPE 5) FORTIFIBER BUILDING SYSTEMS GROUP; MOISTOP TAPE		4. ALL FIELD WELDS AND PENETRATIONS IN ZINC COATING SHALL BE PAINTED IN THE FIELD WITH ZRC GALVALITE REPAIR PAINT OR ENGINEER APPROVED EQUAL.		13. THE COSTS OF REWORKING ALL DEFECTIVE WELDS AND THE COSTS OF RETESTING BY THE TESTING AGENCY PROVIDING SERVICES FOR THE OWNER SHALL BE BORNE BY THE CONTRACTOR.			
E. SNOW LOADS		2. INSTALLATION		5. PAINTED FINISH: ALL STEEL SHAPES, PLATES, ANGLES, BOLTS, ANCHOR RODS, NUTS, WASHERS AND OTHER EXPOSED STEEL SHALL BE SHOP PAINTED AND FIELD TOUCHED-UP WHERE REQUIRED USING THE FOLLOWING SURFACE PREPARATION AND PAINTS: a. SURFACE PREPARATION: COMMERCIAL BLAST (SSPC-SP6) b. SHOP PRIMER: GLID-GUARD ALKALOY METAL PRIMER 4570 AS MANUFACTURED BY GLIDDEN, 2.0 MILS MINIMUM DRY FILM OR ENGINEER APPROVED EQUAL. c. FINISH COAT: GLID-GUARD ALKALOY INDUSTRIAL ENAMEL NO. 4550 SERIES AS MANUFACTURED BY GLIDDEN (COLOR AS SELECTED BY ARCHITECT) 2.0 MILS MINIMUM DRY THICKNESS OR ENGINEER APPROVED EQUAL.		2. GENERAL ANCHORS INSPECTION REQUIREMENTS: VERIFY THAT THE ANCHOR IS INSTALLED IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTALLATION INSTRUCTIONS, ESR REPORT SPECIAL INSPECTION REQUIREMENTS, AS WELL AS THE FOLLOWING DESIGN REQUIREMENTS: a. ANCHOR MANUFACTURER AND PRODUCT, INCLUDING MATERIAL, IS IN ACCORDANCE WITH DESIGN DRAWINGS OR APPROVED SUBSTITUTION. b. ANCHOR DIAMETER, LENGTH, AND INSTALLED EMBEDMENT DEPTH. c. DRILL BIT TYPE AND DIAMETER. d. ANCHOR LOCATION, EDGE DISTANCE, AND SPACING. e. HOLE DIAMETER, DEPTH, CLEANING PROCEDURE, AND CLEANLINESS. f. ANCHOR MAXIMUM TIGHTENING TORQUE. g. CONCRETE TYPE, CONCRETE STRENGTH, AND CONCRETE THICKNESS ARE IN ACCORDANCE WITH DESIGN DRAWINGS.		12. VERIFY THAT QUALITY OF WELDS MEET THE REQUIREMENTS OF AWS D1.1.	
F. SEISMIC LOADS		3. REPAIRS		6. SUBSTRATE CLEANING OF HOT-DIPPED GALVANIZED SURFACES TO RECEIVE PAINT		3. ADHESIVE ANCHORS INSPECTION REQUIREMENTS: IN ADDITION TO THE REQUIREMENTS FOR GENERAL ANCHORS INSPECTIONS, VERIFY ADHESIVE IDENTIFICATION AND EXPIRATION DATE.		13. THE COSTS OF REWORKING ALL DEFECTIVE WELDS AND THE COSTS OF RETESTING BY THE TESTING AGENCY PROVIDING SERVICES FOR THE OWNER SHALL BE BORNE BY THE CONTRACTOR.	
1. THE STRUCTURAL ALTERATIONS HAVE BEEN DESIGNED TO WITHSTAND THE SEISMIC FORCES SPECIFIED IN SECTION 1613.0 OF THE INTERNATIONAL BUILDING CODE, USING THE FOLLOWING DESIGN DATA: DESIGN SPECTRAL RESPONSE: SS = 0.0,xxx DESIGN SHORT PERIOD RESPONSE: SDS = 0.0,xxx DESIGN ONE SECOND PERIOD RESPONSE: SD1 = 0.0,xxx SITE CLASS: C SEISMIC IMPORTANCE FACTOR: I = 1.25 SEISMIC DESIGN CATEGORY: B SEISMIC RESPONSE COEFFICIENT: CSD=0.01 SEISMIC BASE SHEAR: V= X.XX KIPS (NORTH SOUTH) V= X.XX KIPS (EAST WEST) SEISMIC-FORCE-RESISTING SYSTEM: INTERMEDIATE PRECAST SHEAR WALLS STEEL SYSTEMS NOT SPECIFICALLY DETAILED FOR SEISMIC RESISTANCE: OMEGA=3.0 DEFLECTION AMPLIFICATION FACTOR: CDF=0.0 ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE		a. INSTALL NEW VAPOR RETARDER IN ACCORDANCE WITH ASTM E1643 AND MANUFACTURER'S INSTRUCTIONS. b. DO NOT PUNCTURE VAPOR BARRIER WITH REINFORCEMENT. c. OVERLAP NEW VAPOR BARRIER WITH EXISTING VAPOR BARRIER AT LEAST 4" AND TAPE ALL SEAM.		7. GALVANIZED SURFACES: ALL GALVANIZED SURFACES SHALL BE CLEANED BY ACID WASHING WITH 10% PHOSPHORIC ACID SOLUTION. REMOVE THE ACID FROM THE GALVANIZED SURFACE WITH WATER AND ALLOW THE SURFACE TO DRY. PROPERLY ETCHED SURFACES WITH PHOSPHORIC ACID APPEAR BLACK. AREAS CONTAMINATED WITH CHROMIUM OR OTHER CHEMICALS WILL NOT APPEAR BLACK AND SHALL BE RE-ETCHED UNTIL THE SURFACE APPEARS BLACK. WASHING WITH TSP (TRI SODIUM PHOSPHATE) SOLUTION MAY BE USED IN LIEU OF PHOSPHORIC ACID WASHING. PRIME THE ETCHED SURFACE WITH A FLAT EPOXY PRIMER COMPATIBLE WITH APPLIED CEMENTITIOUS FIREPROOFING AFTER THE SURFACE HAS DRIED.		1. INSPECTION FREQUENCY: PROVIDE PERIODIC INSPECTION OF REINFORCING STEEL PLACEMENT. INSPECT 100% OF REINFORCEMENT BEFORE EACH CONCRETE PLACEMENT TO VERIFY THE INFORMATION NOTED BELOW. a. REINFORCEMENT HAS CORRECT SIZE, SPACING, AND PROPER COVER. b. SUFFICIENT SPACING BETWEEN REINFORCEMENT FOR CONCRETE PLACEMENT. c. REINFORCEMENT IS PROPERLY SUPPORTED AND BRACED TO FORMWORK TO PREVENT MOVEMENT DURING CONCRETE PLACEMENT. d. DOWEL REINFORCEMENT IS OF PROPER SIZE, AT PROPER SPACING, AND HAS PROPER LAP LENGTH AND EMBEDMENT LENGTH. e. PROPER HOOKS ARE PROVIDED AT BAR ENDS AS DETAILED. f. WELDED WIRE REINFORCEMENT IS COMPOSED OF FLAT SHEETS, HAS PROPER WIRE GAGE AND SPACING, IS PROPERLY SUPPORTED, AND IS PROPERLY LAPPED WITH A LENGTH OF ONE SQUARE PLUS TWO IN.. g. PROPER CONSTRUCTION/CONTROL/EXPANSION JOINT SPACING AND REINFORCEMENT. h. REINFORCEMENT AROUND EMBEDDED ITEMS IS PLACED PER THE DRAWINGS. i. PROPER INSTALLATION OF FLAT SLAB SHEAR HEAD REINFORCEMENT. j. MECHANICAL TENSION SPLICES: THE TESTING AGENCY SHALL PROVIDE 100% VISUAL INSPECTION OF MECHANICAL TENSION SPLICES ON THE PROJECT AND CONSULT WITH THE MANUFACTURER REGARDING RECOMMENDATIONS FOR INSTALLATION. INSPECTION SHALL VERIFY COMPLIANCE WITH THE REQUIREMENTS IN THE GENERAL NOTES AND CONFORMANCE WITH THE MANUFACTURER'S RECOMMENDATIONS FOR INSTALLATION AFTER CONSULTING WITH THE MANUFACTURER, WHO IS TO BE PRESENT FOR THE FIRST INSTALLATION OF THE SPLICE ON THE PROJECT. k. WELDED REINFORCEMENT HAS BEEN DONE PER AWS REQUIREMENTS.		1. THE CONTRACTOR SHALL PREPARE A DETAILED LIST AND SCHEDULE OF ALL SUBMITTAL ITEMS TO BE SENT TO THE STRUCTURAL ENGINEER PRIOR TO THE START OF CONSTRUCTION. THIS LIST SHALL BE UPDATED AND REVISED AND KEPT CURRENT AS THE JOB PROGRESSES. THE SUBMITTAL LIST SHALL BE ORGANIZED AS SHOWN BELOW: a. SHOP DRAWINGS b. DESIGN CALCULATIONS c. PRODUCT TECHNICAL DATA d. PRODUCT MSDS e. CERTIFICATES, REPORTS, AND OTHER LITERATURE	
PART II - NON-DESTRUCTIVE EVALUATION		A. CONCRETE REPAIR MATERIALS		8. INSPECTION OF REINFORCEMENT STEEL PLACEMENT: THE TESTING AGENCY SHALL MAKE THE FOLLOWING INSPECTIONS OF REINFORCEMENT STEEL PLACEMENT:		PART IX - SUBMITTALS			
1. ALL CONCRETE SHALL CONFORM TO THE REQUIREMENTS AS SPECIFIED IN SPECIFICATION SECTION "CONCRETE REPAIR MATERIALS."		1. HORIZONTAL CONSTRUCTION JOINTS IN CONCRETE POURS		1. THERE SHALL BE NO HORIZONTAL CONSTRUCTION JOINTS IN ANY CONCRETE POURS UNLESS SHOWN ON THE DRAWINGS. THE ENGINEER SHALL APPROVE ALL DEVIATIONS OR ADDITIONAL JOINTS IN WRITING.		A. SUBMITTAL LIST AND SCHEDULE			
A. ITEMS EMBEDDED IN CONCRETE STRUCTURES		1. REINFORCING STEEL		1. ALL REINFORCING STEEL SHALL BE ASTM A615 GRADE 60 UNLESS NOTED OTHERWISE ON THE DRAWINGS OR IN THESE NOTES.		1. THE CONTRACTOR SHALL PREPARE A DETAILED LIST AND SCHEDULE OF ALL SUBMITTAL ITEMS TO BE SENT TO THE STRUCTURAL ENGINEER PRIOR TO THE START OF CONSTRUCTION. THIS LIST SHALL BE UPDATED AND REVISED AND KEPT CURRENT AS THE JOB PROGRESSES. THE SUBMITTAL LIST SHALL BE ORGANIZED AS SHOWN BELOW: a. SHOP DRAWINGS b. DESIGN CALCULATIONS c. PRODUCT TECHNICAL DATA d. PRODUCT MSDS e. CERTIFICATES, REPORTS, AND OTHER LITERATURE			
1. ITEMS EMBEDDED IN CONCRETE STRUCTURES SHALL NOT BE DAMAGED DURING REPAIR WORK OR INSTALLATION OF NEW MEMBERS REQUIRING POST-INSTALLED ANCHORS. EMBEDDED ITEMS MAY INCLUDE MILD REINFORCEMENT, PRESTRESSING REINFORCEMENT, DOWELS, EMBEDDED CONNECTIONS, ELECTRICAL CONDUITS, PLUMBING, ETC.		2. REINFORCING STEEL COVERAGE		1. REINFORCING STEEL COVERAGE SHOULD CONFORM TO THE REQUIREMENTS SPECIFIED ON THE DRAWINGS. COVER SPECIFIED SHALL BE CONSIDERED MINIMUM, HOWEVER EXISTING REINFORCEMENT MAY HAVE A SMALLER COVER THAN SPECIFIED IN DRAWING DETAILS. CONCRETE PATCHES CAN BE BUILT UP TO PROVIDE THE REQUIRED COVER AS LONG AS THE PATCH APPEARANCE IS ACCEPTABLE TO OWNER AND IT DOES NOT REPRESENT A TRIPPING HAZARD TO PEDESTRIANS OR A BUMP TO VEHICLES. COVER IN STRUCTURAL MEMBERS NOT SPECIFIED IN THE DETAILS SHALL CONFORM TO THE REQUIREMENTS OF ACI 318 UNLESS SPECIFIED OTHERWISE ON THE DRAWINGS.		B. SUBMITTALS TO BE PROVIDED TO STRUCTURAL ENGINEER			
2. ITEMS EMBEDDED IN CONCRETE SHALL BE LOCATED BY NON-DESTRUCTIVE EVALUATION PRIOR TO PERFORMING ANY WORK. CONTRACTOR SHALL MARK ON THE STRUCTURE THE LOCATION OF EMBEDDED ITEMS AND PROVIDE A REPORT TO THE ENGINEER.		E. SPLICES AND HOOKS IN REINFORCING STEEL		1. REFER TO DRAWING DETAILS FOR SPLICE AND HOOK REQUIREMENTS OF REINFORCING STEEL BEING SUPPLEMENTED.		1. PRODUCT SUBMITTALS: IN ADDITION TO THE SUBMITTALS REQUIRED BY THE PROJECT SPECIFICATIONS, THE FOLLOWING SUBMITTALS SHALL BE PROVIDED: a. SELECT FILL MATERIALS. b. EPOXY-MODIFIED CEMENTITIOUS COATING FOR EMBEDDED REINFORCEMENT. c. CONCRETE REPAIR MATERIALS. d. FIREPROOFING SYSTEM FOR STRUCTURAL STEEL.			
3. CONTRACTOR SHALL NOT START FABRICATION OF NEW MEMBERS UNTIL ITEMS EMBEDDED IN CONCRETE HAVE BEEN LOCATED. CONTACT ENGINEER IF EXISTING EMBEDDED ITEMS INTERFERE WITH LOCATION OF POST-INSTALLED ANCHORS SPECIFIED IN DRAWINGS.		PART VI - STRUCTURAL STEEL		A. MATERIAL		2. DEFERRED SUBMITTALS: a. THE FOLLOWING ITEMS ARE CONSIDERED DEFERRED SUBMITTALS BY THE REGISTERED DESIGN PROFESSIONAL IN RESPONSIBLE CHARGE: 1) FORMWORK, SHORING, RESHORING, AND BACKSHORING (\$8S, REC) 2) DISTRIBUTED SYSTEM (PIPING AND CONDUIT) ANCHORAGE AND BRACING 3) UNISTRUT FRAMING 4) STRUCTURAL CEILING GRIDS 5) COLD FORMED STEEL FRAMING 6) STRUCTURAL STEEL CONNECTIONS 7) SOIL IMPROVEMENTS 8) SLIDE AND THERMAL BEARING PADS 9) PRE-MANUFACTURED COMPONENTS, INCLUDING ATTACHMENT TO PRIMARY STRUCTURE 10) CONTRACTOR SHALL OBTAIN DELEGATED DESIGN SUBMITTAL FOR INDUSTRIAL RACKING AND ANCHORAGE, TO BE INSTALLED BY CONTRACTOR. ANY DEVIATIONS FROM THE DESIGN SHALL NOT BE PERMITTED WITHOUT APPROVAL FROM THE INDUSTRIAL RACKING DESIGN ENGINEER AND OWNER 11) RAISED ACCESS FLOOR SYSTEM 12) EQUIPMENT ANCHORAGE TO SUPPORTS, SUPPORT CURBS, HANGERS. 13) PRE-FABRICATED METAL STAIRS AND PLATFORMS, ARCHITECTURAL METALS, ACCESS LADDER AND FALL ARREST SYSTEM			
4. ENGINEER MAY REQUIRE A PARTICULAR NON-DESTRUCTIVE EVALUATION METHOD FOR THE LOCATION OF EMBEDDED ITEMS.		1. CONNECTION DETAILS ARE INDICATED ON THE DRAWINGS.		B. CONNECTIONS		NOTES: (\$8S) ITEMS MARKED THUS SHALL HAVE THE SHOP DRAWINGS AND DELEGATED DESIGN SUBMITTALS (INCLUDING CALCULATIONS) SEALED PER THE PROJECT SPECIFICATIONS BY AN ENGINEER REGISTERED IN THE STATE WHERE THE PROJECT IS LOCATED. (REC) ITEMS MARKED THUS SHALL BE SUBMITTED TO ENGINEER FOR RECORD ONLY AND WILL NOT HAVE THE ENGINEER'S SHOP DRAWING STAMP AFFIXED. b. DOCUMENTS FOR DEFERRED SUBMITTAL ITEMS SHALL BE SUBMITTED TO THE REGISTERED DESIGN PROFESSIONAL AND SHALL BE FORWARDED TO THE BUILDING OFFICIAL. c. DEFERRED SUBMITTAL ITEMS SHALL NOT BE INSTALLED UNTIL THE DEFERRED SUBMITTAL DOCUMENTS HAVE BEEN APPROVED BY THE BUILDING OFFICIAL.			
PART III - FOUNDATION		C. STRUCTURAL BOLTS AND THREADED FASTENERS		1. ALL BOLTS, 3/4" DIAMETER AND GREATER, FOR STRUCTURAL CONNECTIONS, SHALL CONFORM TO ASTM F3125 GRADE A325 TYPE 1 PLAIN, UNLESS INDICATED OTHERWISE ON THE DRAWINGS. THE NUTS, BOLTS, AND WASHERS SHALL BE SHIPPED TOGETHER AS AN ASSEMBLY, UNLESS INDICATED OTHERWISE ON THE DRAWINGS.		3. SUBMITTAL REQUIREMENTS: a. ALL SHOP DRAWINGS MUST BE REVIEWED AND ELECTRONICALLY STAMPED BY THE CONTRACTOR PRIOR TO SUBMITTAL b. CONTRACTOR SHALL PROVIDE THE SUBMITTAL IN ELECTRONIC PORTABLE DOCUMENT FORMAT (PDF) PER THE SPECIFICATIONS. c. THE OMISSION FROM THE SHOP DRAWINGS OF ANY MATERIALS REQUIRED BY THE CONTRACT DOCUMENTS TO BE FURNISHED SHALL NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY OF FURNISHING AND INSTALLING SUCH MATERIALS, REGARDLESS OF WHETHER THE SHOP DRAWINGS HAVE BEEN REVIEWED AND APPROVED.			
A. GEOTECHNICAL REPORT		1. TRENCHING AND EXCAVATION		2. ALL BOLTS SMALLER THAN 5/8" DIAMETER BUT NOT SMALLER THAN 0.25" DIAMETER, FOR STRUCTURAL CONNECTIONS, SHALL CONFORM TO ASTM A307 GR. A, UNLESS INDICATED OTHERWISE ON THE DRAWINGS.		1. THE USE OF ELECTRONIC FILES OR REPRODUCTIONS OF THESE CONTRACT DOCUMENTS BY ANY CONTRACTOR, SUBCONTRACTOR, ERECTOR, FABRICATOR, OR MATERIAL SUPPLIER IN LIEU OF PREPARATION OF SHOP DRAWINGS SIGNIFIES THEIR ACCEPTANCE OF ALL INFORMATION SHOWN HEREON AS CORRECT, AND OBLIGATES THEMSELVES TO ANY JOB EXPENSE, REAL OR IMPLIED, ARISING DUE TO ANY ERRORS THAT MAY OCCUR HEREON.			
1. FOUNDATION DESIGN IS BASED ON THE GEOTECHNICAL INVESTIGATION REPORT AS FOLLOWS 480 SPRINGPARK PLACE - GENERATOR ADDITION PREPARED BY EGS MID-ATLANTIC, LLC DATED JANUARY 31, 2025. THE STRUCTURAL FOUNDATION DESIGN IS PENDING AN ADDENDUM TO THE GEOTECHNICAL REPORT CONTAINING REQUIRED UPDATED FOUNDATION RECOMMENDATIONS.		1. NOTIFY UTILITY COMPANIES AND OWNER OF INTENT TO EXCAVATE AND REQUEST SUBGRADE UTILITY IDENTIFICATION MARKING.		3. BOLTS, NUTS, AND WASHERS SPECIFIED THIS PROJECT SHALL BE TYPE 1 PLAIN. GALVANIZED FASTENER NUTS SHALL BE OVER-TAPPED BY THE MINIMUM AMOUNT REQUIRED FOR ASSEMBLY AND SHALL BE LUBRICATED WITH A LUBRICANT CONTAINING A VISIBLE DYE.		2. SUBMITTAL REQUIREMENTS: a. ALL SHOP DRAWINGS MUST BE REVIEWED AND ELECTRONICALLY STAMPED BY THE CONTRACTOR PRIOR TO SUBMITTAL b. CONTRACTOR SHALL PROVIDE THE SUBMITTAL IN ELECTRONIC PORTABLE DOCUMENT FORMAT (PDF) PER THE SPECIFICATIONS. c. THE OMISSION FROM THE SHOP DRAWINGS OF ANY MATERIALS REQUIRED BY THE CONTRACT DOCUMENTS TO BE FURNISHED SHALL NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY OF FURNISHING AND INSTALLING SUCH MATERIALS, REGARDLESS OF WHETHER THE SHOP DRAWINGS HAVE BEEN REVIEWED AND APPROVED.			
2. THE GEOTECHNICAL REPORT IS AVAILABLE TO THE GENERAL CONTRACTOR UPON REQUEST TO THE OWNER. THE INFORMATION INCLUDED THEREIN MAY BE USED BY THE GENERAL CONTRACTOR FOR HIS GENERAL INFORMATION ONLY. THE ARCHITECT AND ENGINEER WILL NOT BE RESPONSIBLE FOR THE ACCURACY OR APPLICABILITY OF SUCH DATA THEREIN.		2. EXCAVATE EXISTING SUBGRADE EARTH TO DEPTHS INDICATED ON THE DRAWINGS.		4. TWIST-OFF-TYPE TENSION-CONTROL BOLT ASSEMBLIES: WHERE INDICATED ON THE DRAWINGS ASTM F3125 GRADE A325 TENSION CONTROLLED TYPE 1 PLAIN BOLTS SHALL BE PROVIDED. TENSION CONTROLLED BOLTS SHALL ADDITIONALLY CONFORM TO ASTM F3125 GRADE F182.		1. THE USE OF ELECTRONIC FILES OR REPRODUCTIONS OF THESE CONTRACT DOCUMENTS BY ANY CONTRACTOR, SUBCONTRACTOR, ERECTOR, FABRICATOR, OR MATERIAL SUPPLIER IN LIEU OF PREPARATION OF SHOP DRAWINGS SIGNIFIES THEIR ACCEPTANCE OF ALL INFORMATION SHOWN HEREON AS CORRECT, AND OBLIGATES THEMSELVES TO ANY JOB EXPENSE, REAL OR IMPLIED, ARISING DUE TO ANY ERRORS THAT MAY OCCUR HEREON.			
3. THE FOUNDATION FOR THE STRUCTURE HAS BEEN DESIGNED FOR THE FOLLOWING NET ALLOWABLE SOIL BEARING PRESSURES AT A BEARING DEPTHS AS NOTED ON PLAN.		3. ENGINEER WILL INSPECT AND APPROVE FOUNDATION SOIL PRIOR TO PLACEMENT OF FILL.		D. WELDING		2. SUBMITTAL REQUIREMENTS: a. ALL SHOP DRAWINGS MUST BE REVIEWED AND ELECTRONICALLY STAMPED BY THE CONTRACTOR PRIOR TO SUBMITTAL b. CONTRACTOR SHALL PROVIDE THE SUBMITTAL IN ELECTRONIC PORTABLE DOCUMENT FORMAT (PDF) PER THE SPECIFICATIONS. c. THE OMISSION FROM THE SHOP DRAWINGS OF ANY MATERIALS REQUIRED BY THE CONTRACT DOCUMENTS TO BE FURNISHED SHALL NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY OF FURNISHING AND INSTALLING SUCH MATERIALS, REGARDLESS OF WHETHER THE SHOP DRAWINGS HAVE BEEN REVIEWED AND APPROVED.			
4. SHALLOW FOUNDATIONS 3,000 PSF (CONTINUOUS AND SPREAD FOOTINGS)		4. SHORE TRENCHING AND EXCAVATIONS IN STABLE SOILS DEEPER THAN 5 FT. ADDITIONALLY SHORE ALL TRENCHING AND EXCAVATIONS IN UNSTABLE SOILS. SLOPED OR STEPPED-BACKED EXCAVATION MEETING OSHA 29 CFR 1926 SUBPART "P" MAY BE USED INSTEAD OF SHORING. SHORING, SLOPED EXCAVATION, AND STEPPED-BACKED EXCAVATION SHALL BE DESIGNED BY THE CONTRACTOR. SHORING, SLOPED EXCAVATION, AND STEPPED-BACKED EXCAVATION BEYOND THE PRESCRIPTIVE LIMITS OF OSHA 29 CFR 1926 SUBPART "P" SHALL BE DESIGNED BY THE CONTRACTOR'S ENGINEER.		1. UNLESS NOTED OTHERWISE, ELECTRODES FOR WELDING SHALL CONFORM TO E70XX (SMWV).		3. DEFERRED SUBMITTAL ITEMS SHALL NOT BE INSTALLED UNTIL THE DEFERRED SUBMITTAL DOCUMENTS HAVE BEEN APPROVED BY THE BUILDING OFFICIAL.			
PART IV - BUILDING EARTHWORK		B. SOIL PREPARATION		2. PROVIDE ELECTRODES THAT COMPLY WITH AWS D1.1, "STRUCTURAL WELDING CODE - STEEL" ELECTRODES FOR SHOP AND FIELD WELDS SHALL CONFORM TO AWS A5.1 OR AWS A5.5, CLASS EXXX, LOW HYDROGEN..		4. CHECK JOINT PREPARATION.			
1. SCARIFY THE TOP 6 IN. OF THE SUBGRADE BELOW BOTTOM OF SLAB.		1. RETURN OVER-EXCAVATION BEYOND THE SPECIFIED LIMITS TO GRADE AT NO COST TO THE OWNER.		3. ELECTRODES SHALL BE COMPATIBLE WITH PARENT METAL JOINED.		5. REVIEW WELDING SEQUENCE.			
2. REMOVE ALL UNSUITABLE MATERIALS AND SOFT SPOTS AND REPLACE WITH SELECT FILL.		2. CORRECT GRADING THAT RESULTS IN STANDING WATER AT NO COST TO THE OWNER.		4. GALVANIZING MAINTENANCE PAINT: GALVANIZING MAINTENANCE PAINT SHALL BE "ZRC GALVALITE GALVANIZING REPAIR COMPOUND" AS MANUFACTURED BY ZRC		6. ALL WELDS SHALL BE 100% VISUALLY INSPECTED. UNLESS SPECIFICALLY NOTED OTHERWISE, ALL WELDING SHALL BE CONSIDERED STATICALLY LOADED NONTUBULAR CONNECTIONS.			
3. RETURN OVER-EXCAVATION BEYOND THE SPECIFIED LIMITS TO GRADE AT NO COST TO THE OWNER.		3. CONSTRUCT TO LINES, GRADES AND DIMENSIONS SHOWN ON THE PLANS.		CODE: ALL WELDING SHALL CONFORM TO ALL REQUIREMENTS IN THE "STRUCTURAL WELDING CODE - STEEL", ANSI/AWS D1.1, AS PUBLISHED BY THE AMERICAN WELDING SOCIETY (AWS).		7. WELDER CERTIFICATION: a. ALL WELDERS SHALL BE CERTIFIED ACCORDING TO AWS PROCEDURES FOR THE WELDING PROCESS AND WELDING POSITION USED. EACH WELDER SHALL BE ASSIGNED AN IDENTIFYING SYMBOL OR MARK AND ALL WELDED CONNECTIONS CONTAINING COMPLETE OR PARTIAL JOINT PENETRATION WELDS, MULTI-PASS FILLET WELDS AND FILLET WELDS GREATER THAN 5/16" SHALL BE IDENTIFIED BY THE SYMBOL OR MARK OF THE WELDER OR WELDERS RESPONSIBLE FOR THE CONNECTION.			
4. CORRECT GRADING THAT RESULTS IN STANDING WATER AT NO COST TO THE OWNER.		4. CONSTRUCT TO LINES, GRADES AND DIMENSIONS SHOWN ON THE PLANS.		6. WELDER CERTIFICATION: a. ALL WELDERS SHALL					

[illegible]

GROUND LEVEL MEP WEIGHT	
DESCRIPTION	OPERATIONAL WEIGHT
MV XFMR	22,170 LBS
GENERATOR	209,500 LBS

NOTE:

1. GENERATOR WEIGHT INCLUDES THE 3,000 KW GENERATOR AND THE BASE FUEL TANK AND FUEL LOCATED BENEATH IT.



10 GROUND LEVEL- SITE LOAD DIAGRAM
3/64" = 1'-0"

LEVEL 01 UPPER LEVEL PLAN			
DESCRIPTION	SUPER IMPOSED DEAD LOAD	LIVE LOAD	HATCH PATTERN
DATA HALL	65 PSF	350 PSF	

NOTES:

1.  INDICATES OPENING IN FLOOR.
2. ALL LOADS ARE UNFACTORED SERVICE LOADS.

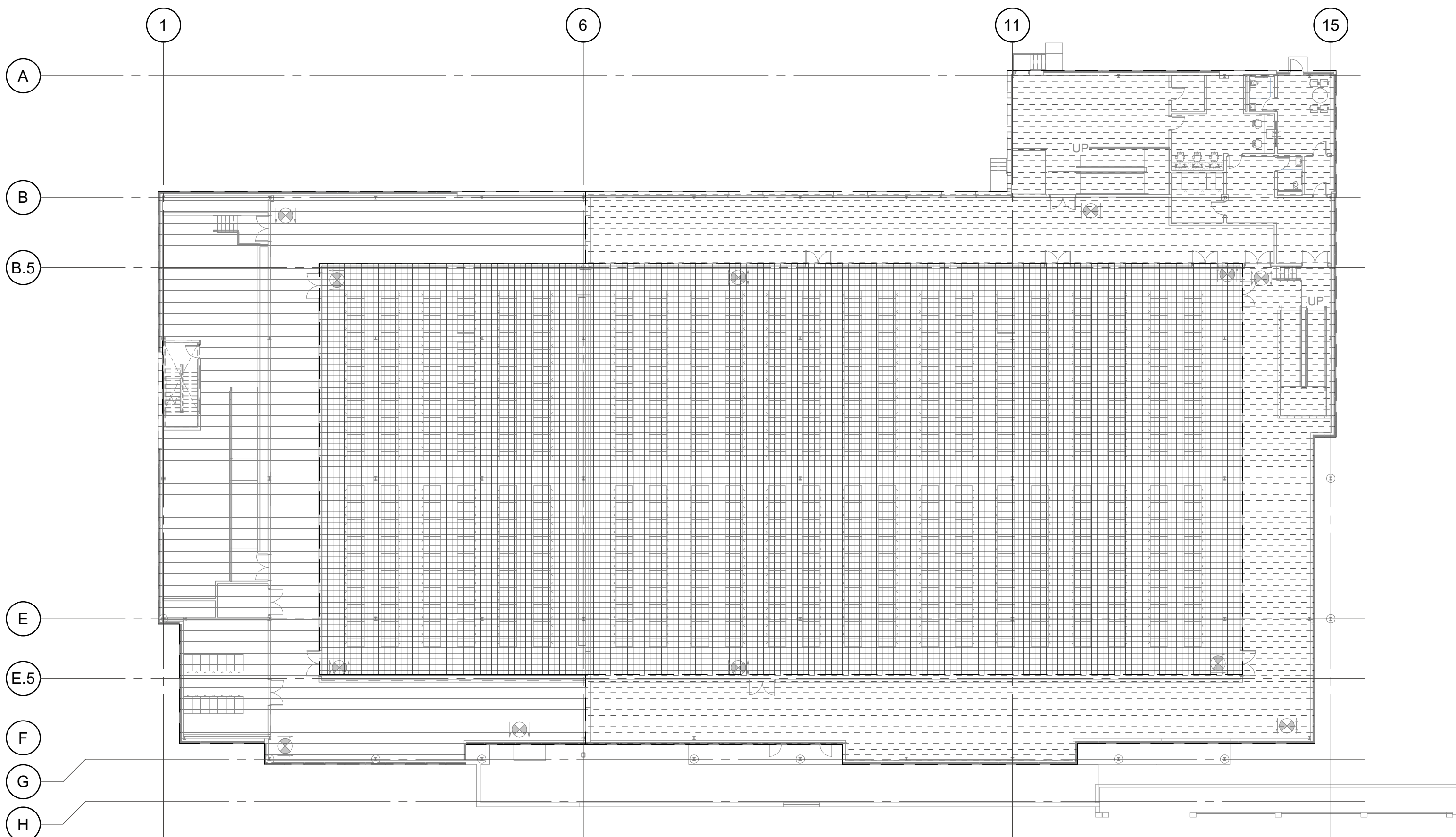


15 LEVEL 01 - LOAD DIAGRAM
3/64" = 1'-0"

LEVEL 02 UPPER LEVEL PLAN			
DESCRIPTION	SUPER IMPOSED DEAD LOAD	LIVE LOAD	HATCH PATTERN
DATA HALL	65 PSF	350 PSF	
GALLERY-OVER THE GRADE	-	250 PSF	
GALLERY-ELEVATED	65 PSF	200 PSF	

NOTES:

1.  INDICATES OPENING IN FLOOR.
2. DATA HALL DEAD LOAD INCLUDES MISCELLANEOUS RAISED FLOOR, HUNG GRID SUPPORT SYSTEM, COLLATERAL LOADS, OWNER-FURNISHED EQUIPMENT, AND STRUCTURAL STRENGTHENING SYSTEMS.
3. DATA HALL LIVE LOAD INCLUDES THE WEIGHT OF RACKS.
4. ALL LOADS ARE UNFACTORED SERVICE LOADS.



20 LEVEL 02 - LOAD DIAGRAM
3/64" = 1'-0"

VSW

V Squared Workshop
7900 Westpark Dr, Suite A205
McLean, VA 22102
301.590.8900
www.vsw.co

MEP

AHA

3700 Mansell Road, Suite 200
Alpharetta, GA 30022
www.aha-engineers.com

STRUCTURAL

walter p moore

1700 K Street NW, Suite 1050,
Washington, DC, 20006
202.481.8861
www.walterpmoore.com

CIVIL

VIA

8180 Greensboro Dr.
Suite 200
Tysons, VA 22102



H5 DATA CENTERS

H5 Herndon
480 Springpark Pl, Herndon, VA 20170
H5 Data Centers

S-004 - LOADING DIAGRAMS / DESIGN CRITERIA

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Professional Certification:
I hereby certify that these
documents were prepared by
me, and that I am a duly
licensed architect under the
laws of the State of Maryland.

Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026

VSW JOB NO. 250202 SCALE 3/64" = 1'-0"
LOADING DIAGRAMS / DESIGN CRITERIA

DRAWING

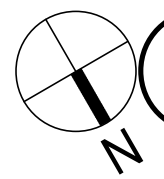
S-004

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LEVEL 03 ROOF PLAN		
DESCRIPTION	DEAD LOAD	HATCH PATTERN
SUB-FRAMING	30 PSF	

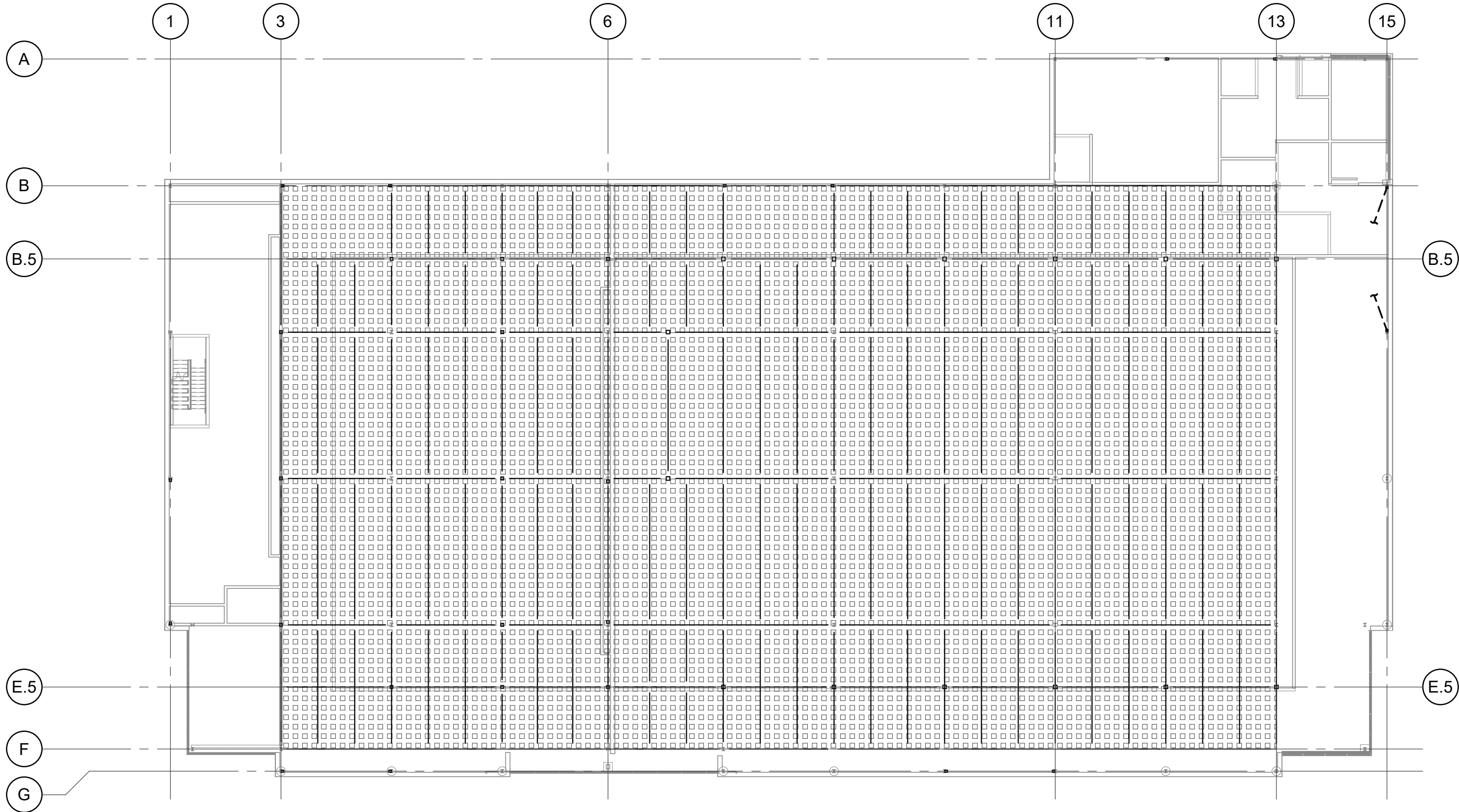
NOTES:

- ALL LOADS ARE UNFACTORED SERVICE LOADS.
- SUB-FRAMING LOAD APPLIES ON THE COLUMNS BELOW IT.



10 LEVEL 03 - SUB-FRAMING LOAD DIAGRAM

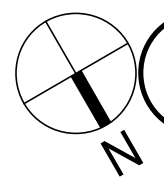
3/64" = 1'-0"



LEVEL 03 ROOF PLAN				
DESCRIPTION	DEAD LOAD	LIVE LOAD	FLAT ROOF SNOW LOAD	HATCH PATTERN
ROOF BELOW DUNNAGE	20 PSF	30 PSF	-	
ROOF - NO DUNNAGE ABOVE	20 PSF	30 PSF	56.7 PSF	

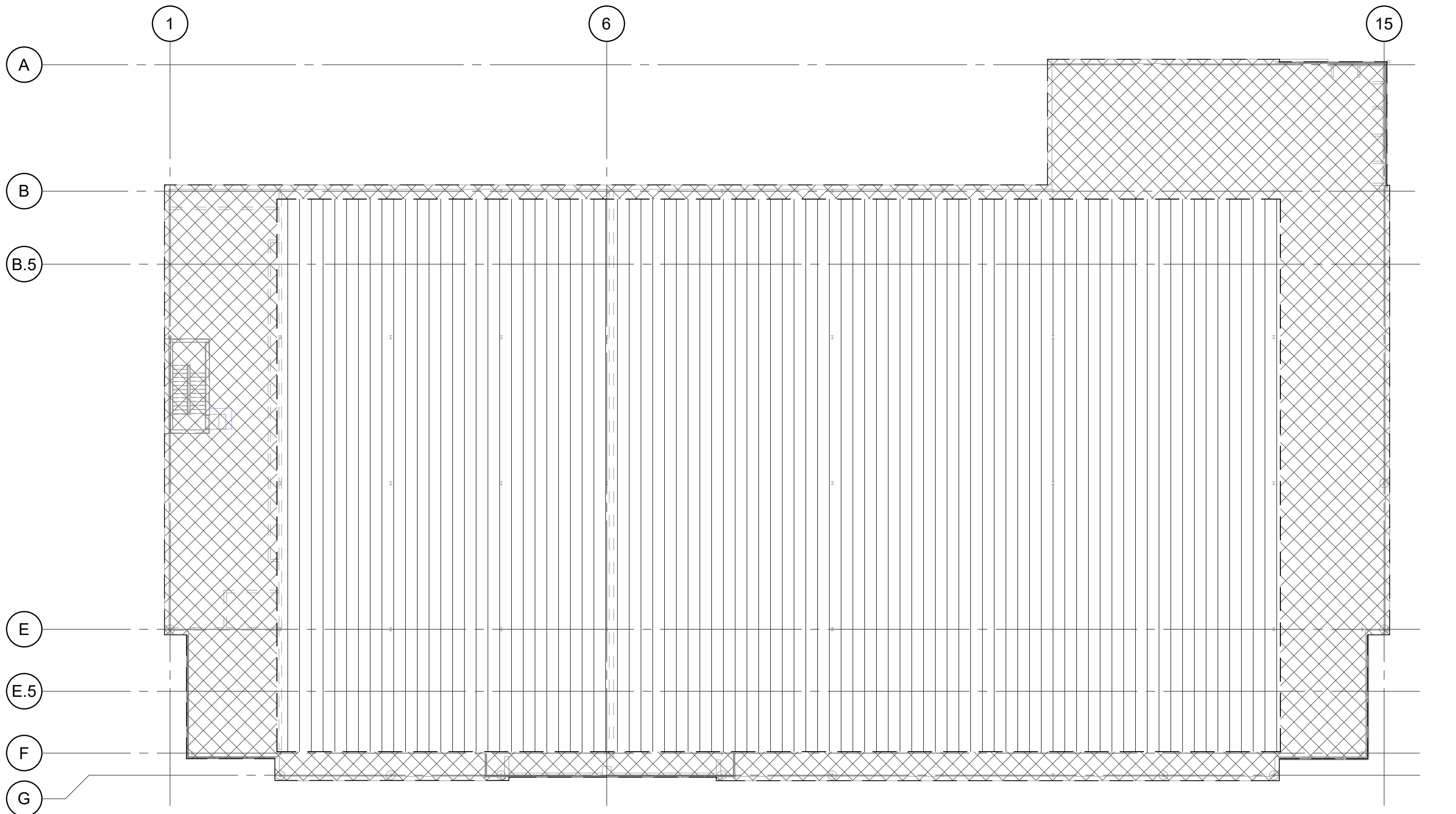
NOTES:

- ALL LOADS ARE UNFACTORED SERVICE LOADS.



15 LEVEL 03 - ROOF LOAD DIAGRAM

3/64" = 1'-0"

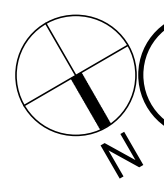


LEVEL 04 MEP WEIGHT ON DUNNAGE	
DESCRIPTION	OPERATIONAL WEIGHT
CHILLER	42,600 LBS
TES	65,200 LBS

NOTES:

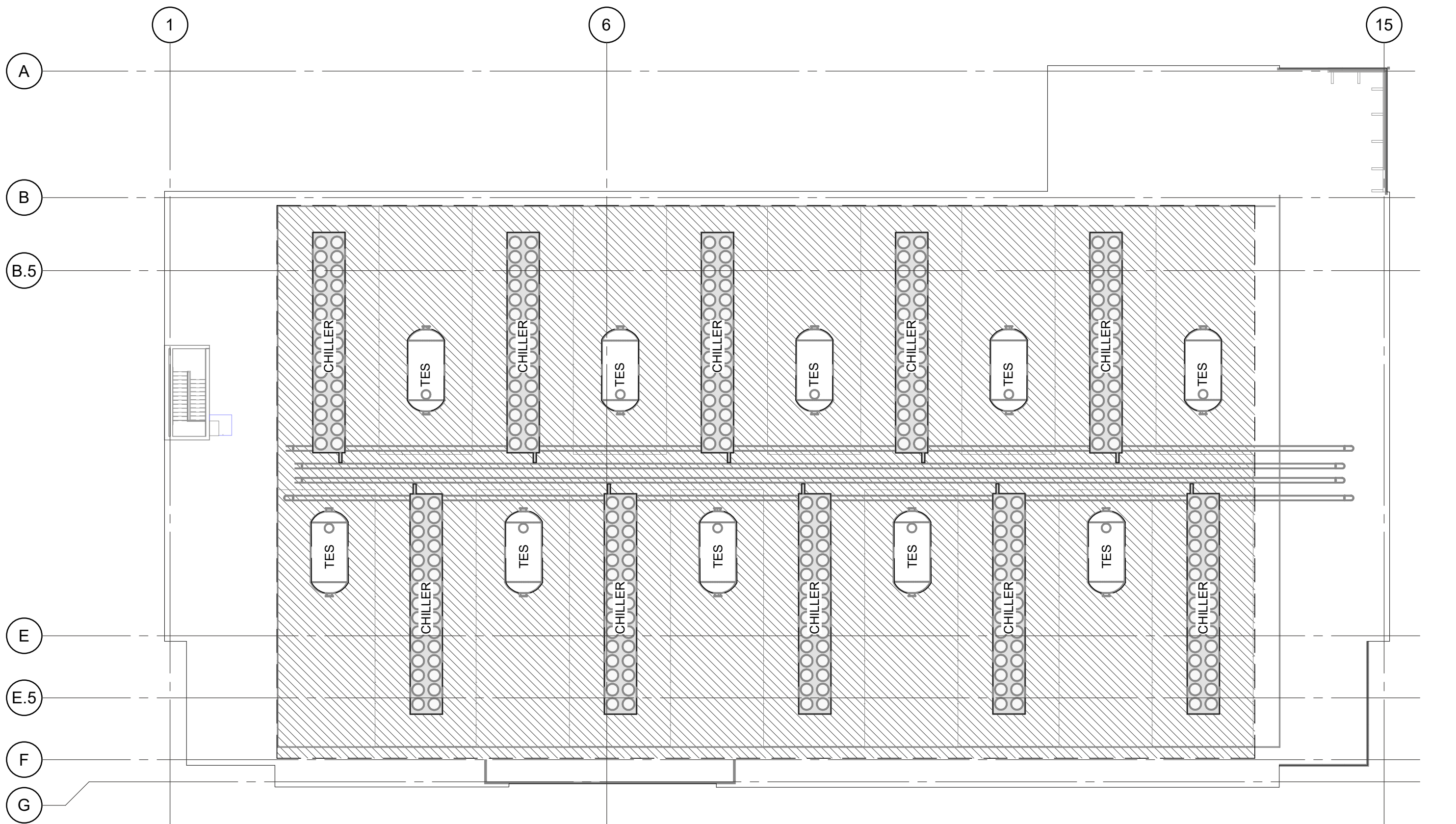
- ICE LOAD IS APPLIED AS A UNIFORM LINE LOAD (PLF) ALONG THE LENGTH OF EXPOSED STEEL DUNNAGE MEMBERS.
- ALL LOADS ARE UNFACTORED SERVICE LOADS.

LEVEL 04 DUNNAGE PLAN				
DESCRIPTION	SUPER IMPOSED DEAD LOAD	LIVE LOAD	FLAT ROOF SNOW LOAD	HATCH PATTERN
DUNNAGE	10 PSF	40 PSF	56.7 PSF	



20 LEVEL 04 - DUNNAGE LOAD DIAGRAM

3/64" = 1'-0"



VSW

V Squared Workshop
7900 Westpark Dr, Suite A205
McLean, VA 22102
301.590.8900
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MEP

AHA

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CIVIL

VIKI

8180 Greensboro Dr.
Suite 200
Tysons, VA 22102

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H5 Herndon
480 Springpark Pl, Herndon, VA 20170

H5 Data Centers

S-005 - LOADING DIAGRAMS / DESIGN CRITERIA

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Professional Certification:
I hereby certify that these
documents were prepared by
me, and that I am a duly
licensed architect under the
laws of the State of Maryland,

Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026

VSW JOB NO. 250202 SCALE 3/64" = 1'-0"
LOADING DIAGRAMS / DESIGN CRITERIA

DRAWING
S-005

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EXISTING COLUMN SCHEDULE	
MARK	COLUMN SIZE
C-1	W8x24
C-2	W8x24
C-3	W8x31
C-4	W8x24
C-5	W8x28
C-6	W8x35
C-7	W8x28
C-8	W8x40
C-9	w8x5
C-10	W8x31
C-11	W8x48

RI-X OR RE-X
STRUCTURAL STEEL
COLUMN REINFORCING
DETAIL, SEE SCHEDULES ON
SHEET S-800


$$3/32" = 1'-0"$$

EXISTING COLUMN SCHEDULE	
MARK	COLUMN SIZE
C-1	W8x24
C-2	W8x24
C-3	W8x31
C-4	W8x224
C-5	W8x28
C-6	W8x35
C-7	W8x28
C-8	W 8 X 40
C-9	W8x58
C-10	W8x31
C-11	W8x48

RI-X OR RE-X
STRUCTURAL STEEL
COLUMN REINFORCING
DETAIL, SEE SCHEDULES ON
SHEET S-800



S-101



UPPER LEVEL PLAN NOTES:

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING DIMENSIONS, ELEVATIONS, AND CONDITIONS SHOWN ON THE DRAWINGS. EXISTING INFORMATION IS BASED ON ORIGINAL STRUCTURAL DRAWINGS BY "CAD CONCEPTS INC." DATED OCTOBER 1, 1997. NOTIFY THE ARCHITECT AND STRUCTURAL ENGINEER OF ANY DISCREPANCIES.
2. TOP OF SLAB DUNNAGE ELEVATION = 1' - 0" UNLESS NOTED OTHERWISE.
3. CONTRACTOR TO NOTE THAT HEAVY WELDING OF SEVERAL MISCELLANEOUS PLATES ARE REQUIRED AT ALL COLUMN TO BEAM INTERFACES TO LOAD THROUGH 2ND AND ROOF LEVEL (NOT CURRENTLY SHOWN), TO BE DESIGNED AND DETAILED IN FUTURE SUBMISSIONS.
4. CONTRACTOR TO NOTE: MAJORITY OF THE EXISTING SLAB ON GROUND WILL NEED TO BE DEMOLISHED AND REPLACED TO MATCH EXISTING IN ORDER TO INSTALL NEW FOOTINGS AND FOOTING ENLARGEMENTS. EXISTING SLAB ON GROUND IS A STRUCTURAL MEMBER THAT BRACES THE EXISTING PERIMETER RETAINING WALLS. CONTRACTOR TO PROVIDE TEMPORARY LATERAL BRACING OF EXISTING PERIMETER RETAINING WALLS AS REQUIRED TO ENSURE STRUCTURAL STABILITY. DEMO OF SLAB ON GROUND CAN ONLY BE PERFORMED IN LIMITED SECTIONS AND STRIPS AND NOT ALL AT ONCE. SEQUENCING REQUIREMENTS AND LIMITATIONS OF SLAB ON GROUND TO BE PROVIDED IN FUTURE SUBMISSIONS.
5. BF-X ON PLAN, DENOTES (N)INE-FOOT-HEIGHT STEEL BRACED FRAME.


$$3/32'' = 1'-0''$$

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Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026

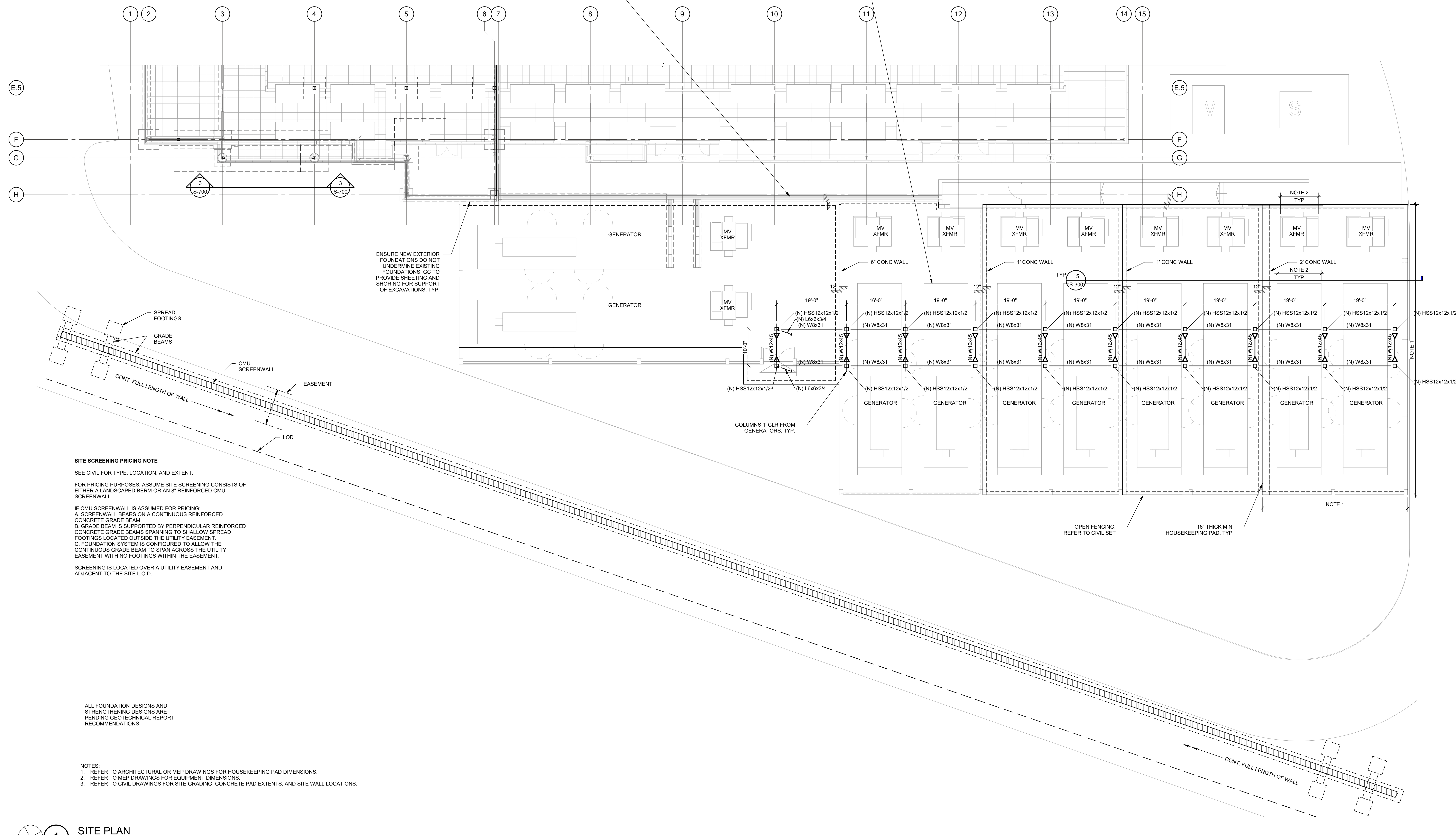
VSW JOB NO. SCALE

250202 3/32" = 1'-0"

SITE PLAN

DRAWING

S-102

EXISTING BUILDING FOUNDATION
PROTECTION - PRICING NOTECIVIL GRADING IS PRELIMINARY. NEW
CONSTRUCTION IS LOCATED ADJACENT TO
EXISTING BUILDING FOUNDATIONS.FOR PRICING PURPOSES, CONTRACTOR SHALL
ACCOUNT FOR MEANS AND METHODS
REQUIRED TO PREVENT UNDERMINING OR
LOSS OF SUPPORT OF EXISTING FOUNDATIONS,
INCLUDING TEMPORARY AND/OR PERMANENT
SUPPORT SYSTEMS AS NECESSARY.FINAL GRADING AND FOUNDATION DETAILS
WILL BE CONFIRMED IN FINAL DESIGN.OVERHEAD ELECTRICAL SUPPORT PILLARS -
PRICING NOTECONDUIT ROUTING, ELEVATION, AND
SUPPORT REQUIREMENTS ARE TBD. REFER
TO ELECTRICAL DRAWINGS.FOR PRICING PURPOSES, ASSUME OVERHEAD
CONDUITS ARE SUPPORTED BY (2) HSS 8x8
STEEL TUBE COLUMNS, APPROXIMATELY 15
FT TALL, WITH W8 STEEL BEAMS
APPROXIMATELY 6 FT LONG. THE POSTS ARE
SUPPORTED ON 24" DIAMETER REINFORCED
CONCRETE DRILLED PIERS OR ON THE
GENERATOR PAD.ASSUME (4) SUPPORT POSTS PER
GENERATOR LINEUP FOR PRICING ONLY.

SITE SCREENING PRICING NOTE

SEE CIVIL FOR TYPE, LOCATION, AND EXTENT.

FOR PRICING PURPOSES, ASSUME SITE SCREENING CONSISTS OF
EITHER A LANDSCAPED BERM OR AN 8" REINFORCED CMU
SCREENWALL.IF CMU SCREENWALL IS ASSUMED FOR PRICING:
A. SCREENWALL BEARS ON A CONTINUOUS REINFORCED
CONCRETE GRADE BEAM.
B. GRADE BEAM IS SUPPORTED BY PERPENDICULAR REINFORCED
CONCRETE GRADE BEAMS SPANNING TO SHALLOW SPREAD
FOOTINGS LOCATED OUTSIDE THE UTILITY EASEMENT.
C. FOUNDATION SYSTEM IS CONFIGURED TO ALLOW THE
CONTINUOUS GRADE BEAM TO SPAN ACROSS THE UTILITY
EASEMENT WITH NO FOOTINGS WITHIN THE EASEMENT.SCREENING IS LOCATED OVER A UTILITY EASEMENT AND
ADJACENT TO THE SITE L.O.D.ALL FOUNDATION DESIGNS AND
STRENGTHENING DESIGNS ARE
PENDING GEOTECHNICAL REPORT
RECOMMENDATIONSNOTES:
1. REFER TO ARCHITECTURAL OR MEP DRAWINGS FOR HOUSEKEEPING PAD DIMENSIONS.
2. REFER TO MEP DRAWINGS FOR EQUIPMENT DIMENSIONS.
3. REFER TO CIVIL DRAWINGS FOR SITE GRADING, CONCRETE PAD EXTENTS, AND SITE WALL LOCATIONS.

Revisions

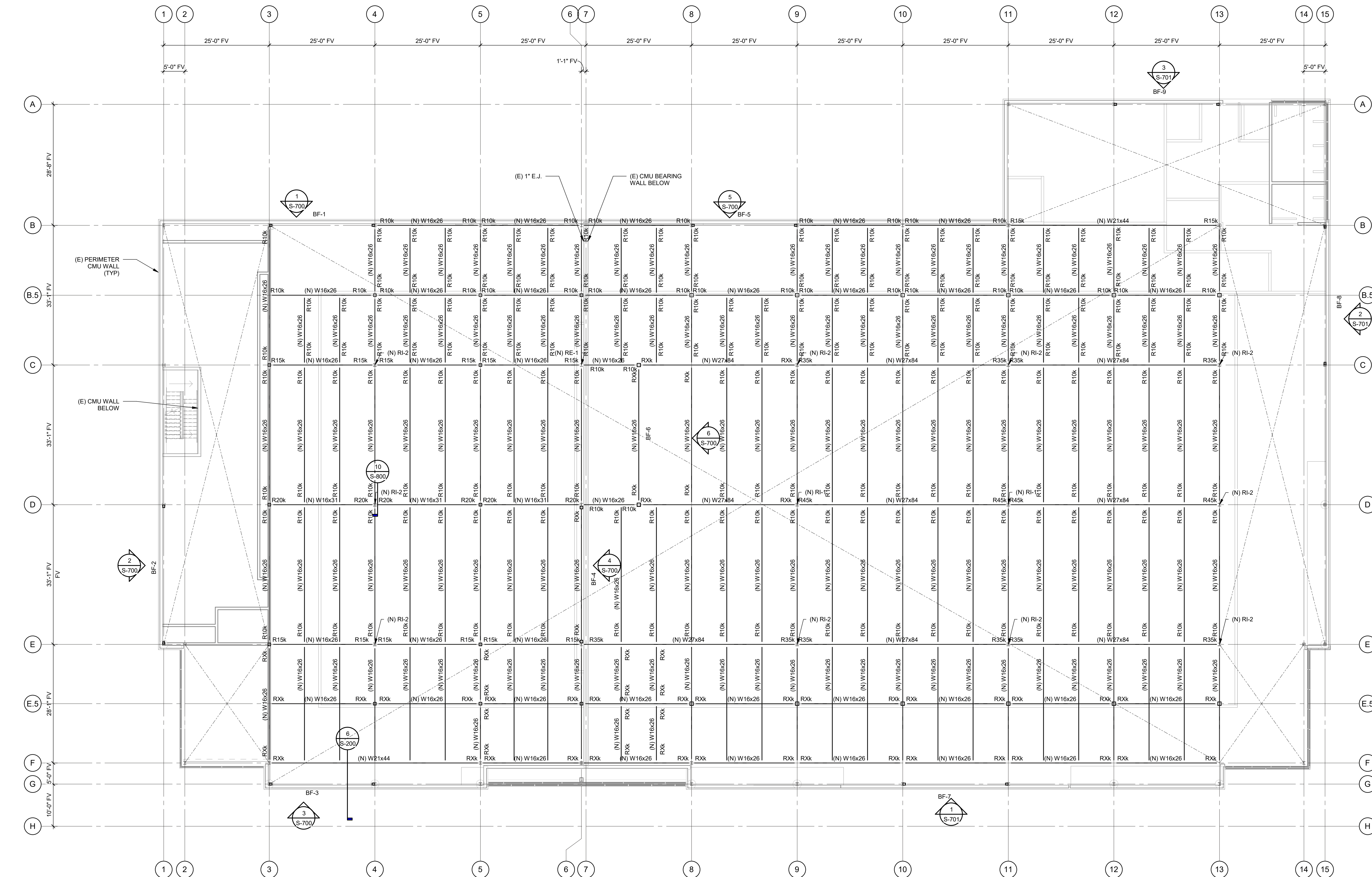
02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026VSW JOB NO. 250202 SCALE
SUB-FRAMING PLAN As indicated

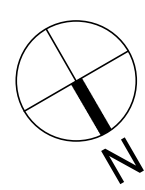
DRAWING

S-103

PLAN SYMBOLS:

R1-X OR RE-X
STRUCTURAL STEEL
COLUMN REINFORCING
DETAIL, SEE SCHEDULES ON
SHEET S-800

- NOTES:
- STRUCTURAL STEEL IS INTENDED FOR INTERIOR, NON-OCCUPIABLE COLLATERAL SUPPORT FRAMING ONLY.
 - MAINTAIN 15'-6\"/>



1 SUB-FRAMING PLAN

3/32\"/>

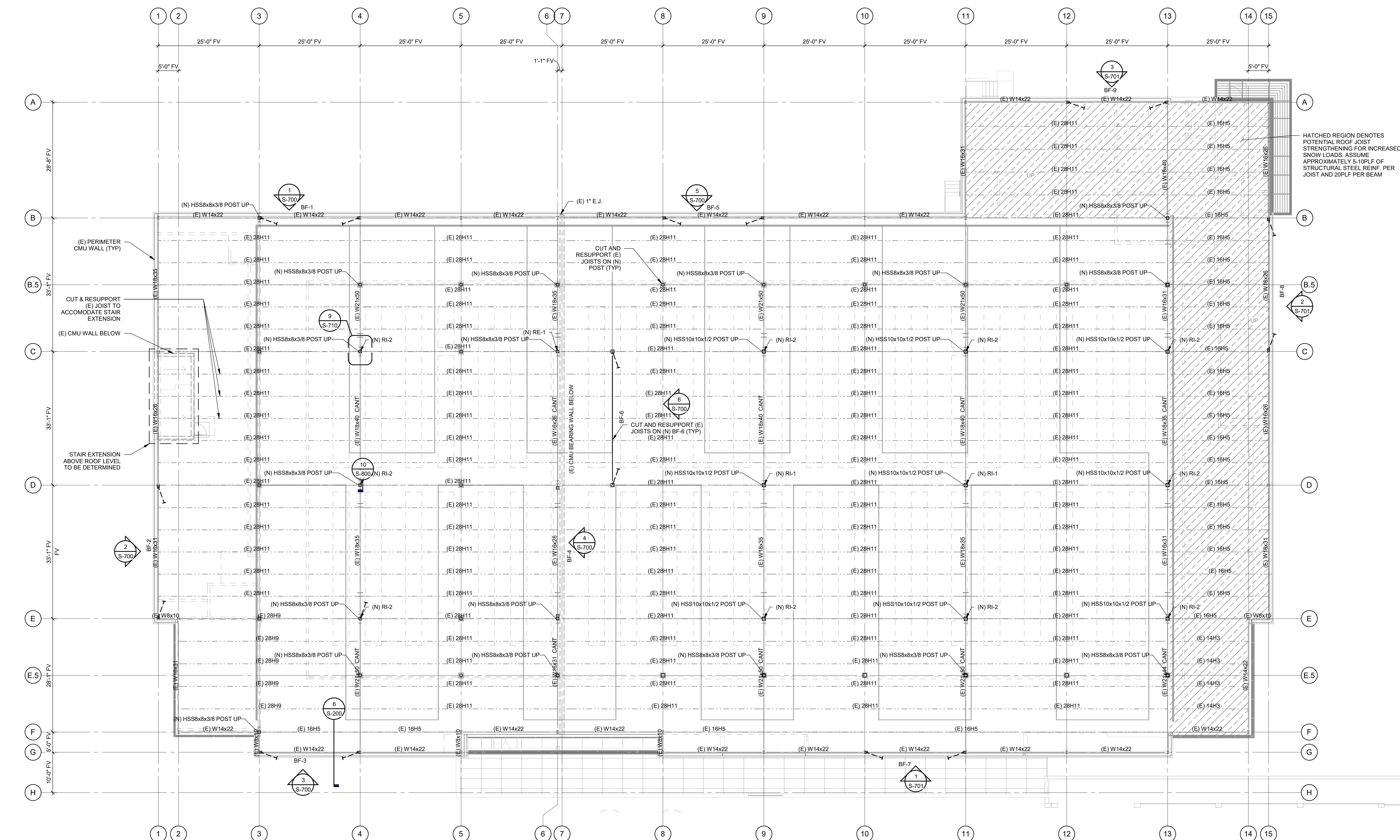
Revisions

Revisions	DATE	DESCRIPTION
02/03/26	DESIGN DEVELOPMENT / PRICING	
1 04/06/26	DD SUBMISSION	

ISSUED:
DD SUBMISSION
04/06/2026VSW JOB NO. SCALE
250202 3/32" = 1'-0"

ROOF FRAMING PLAN

DRAWING

S-104

[illegible]

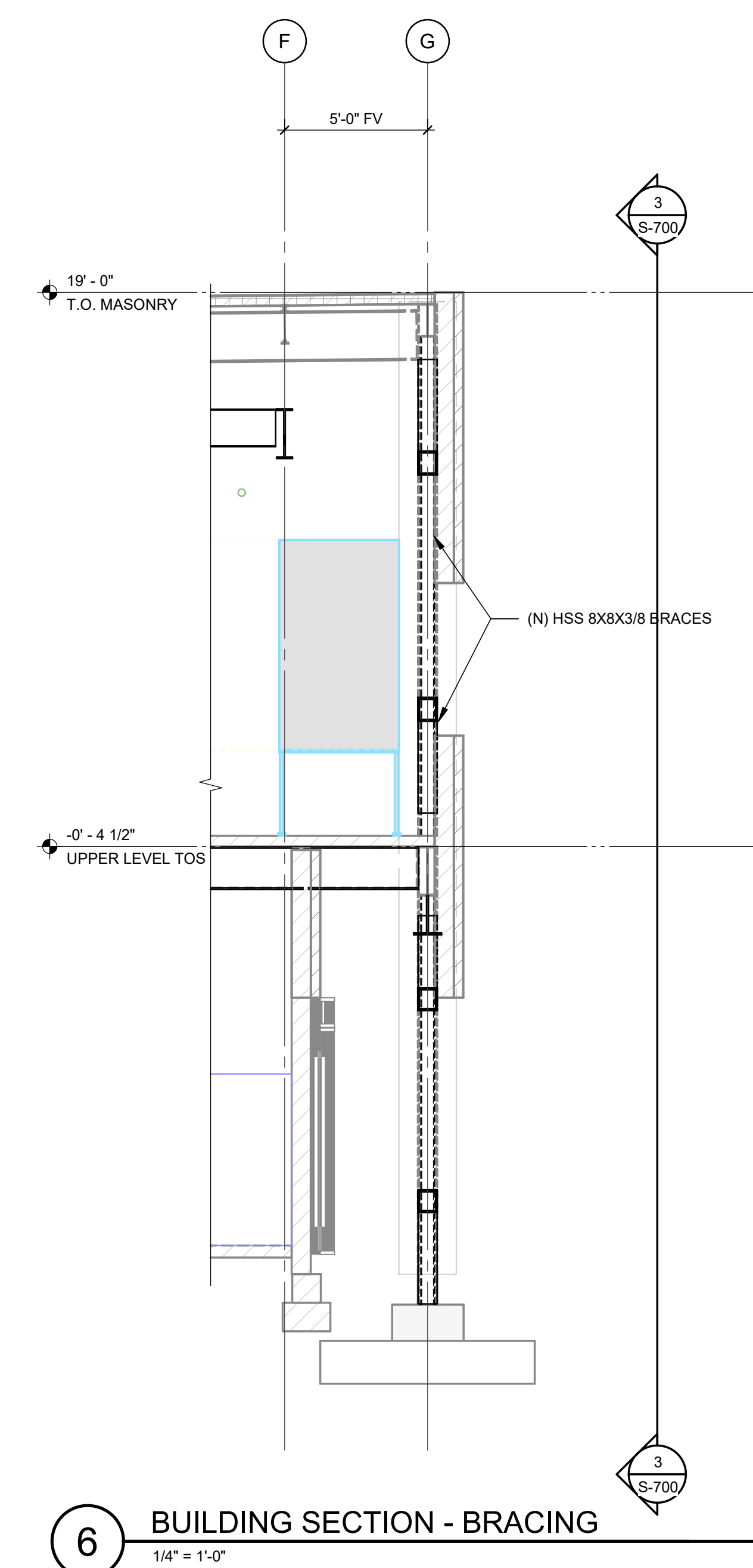
ISSUED:
DD SUBMISSION
04/06/2026

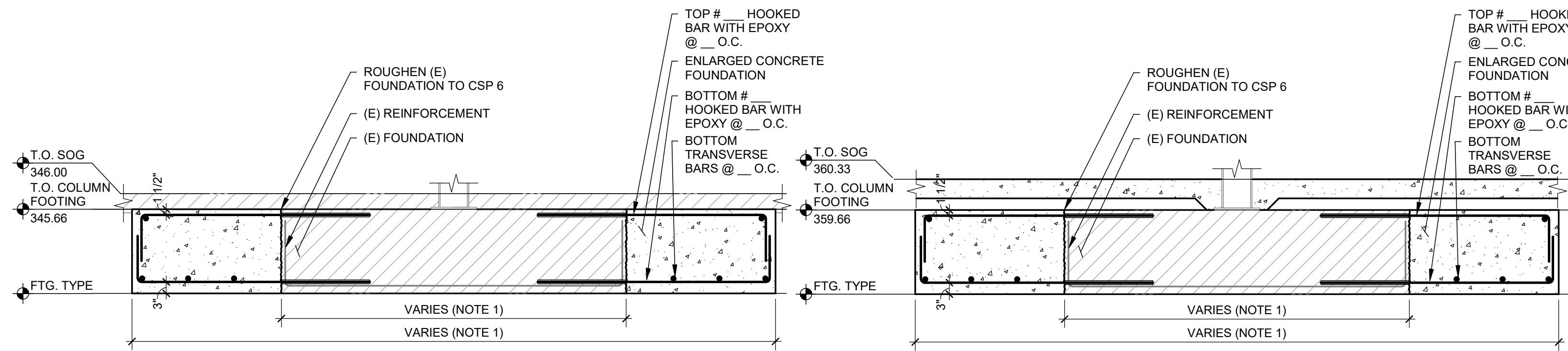
VSW JOB NO. 250202	SCALE 1/4" = 1'-0"
ELEVATIONS AND SECTIONS	

DRAWING

S-200

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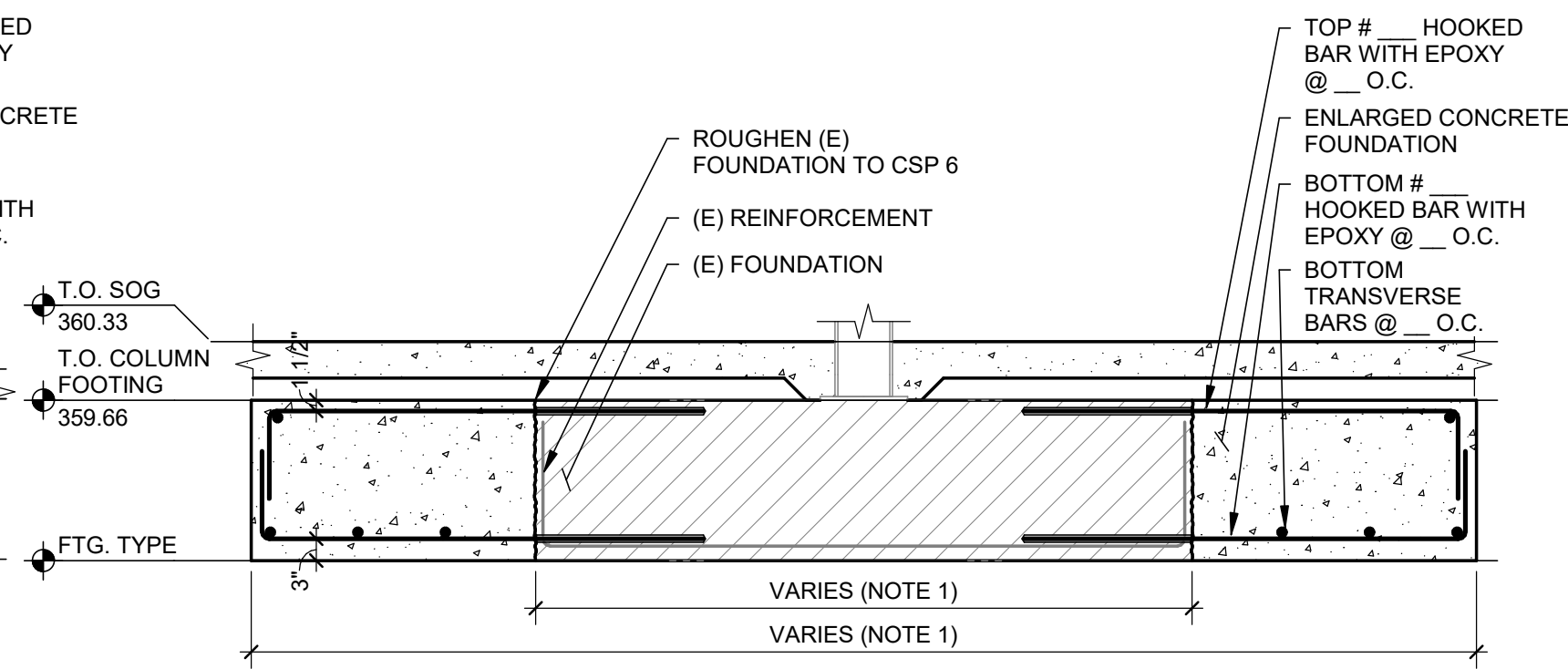




- NOTES:
1. REFER TO FOUNDATION PLAN (S-101) FOR EXISTING AND NEW FOUNDATION DIMENSIONS.
 2. REFER TO "CONCRETE REPAIR MATERIAL" SECTION FOR MORE INFORMATION.
 3. MAINTAIN 3" COVER OVER NEW BARS WHERE THE CONCRETE SURFACE IS IN CONTACT WITH SOIL.
 4. COMPACT SOIL AROUND EXISTING FOUNDATION AS SPECIFIED IN CONSTRUCTION DOCUMENTS BEFORE ADDING NEW CONCRETE SECTION.
 5. CONTRACTOR SHALL LOCATE EXISTING REINFORCEMENT PRIOR TO DRILLING INTO THE FOUNDATION TO AVOID DAMAGING EXISTING REINFORCEMENT.

4 TYPICAL DETAIL AT G1-G7

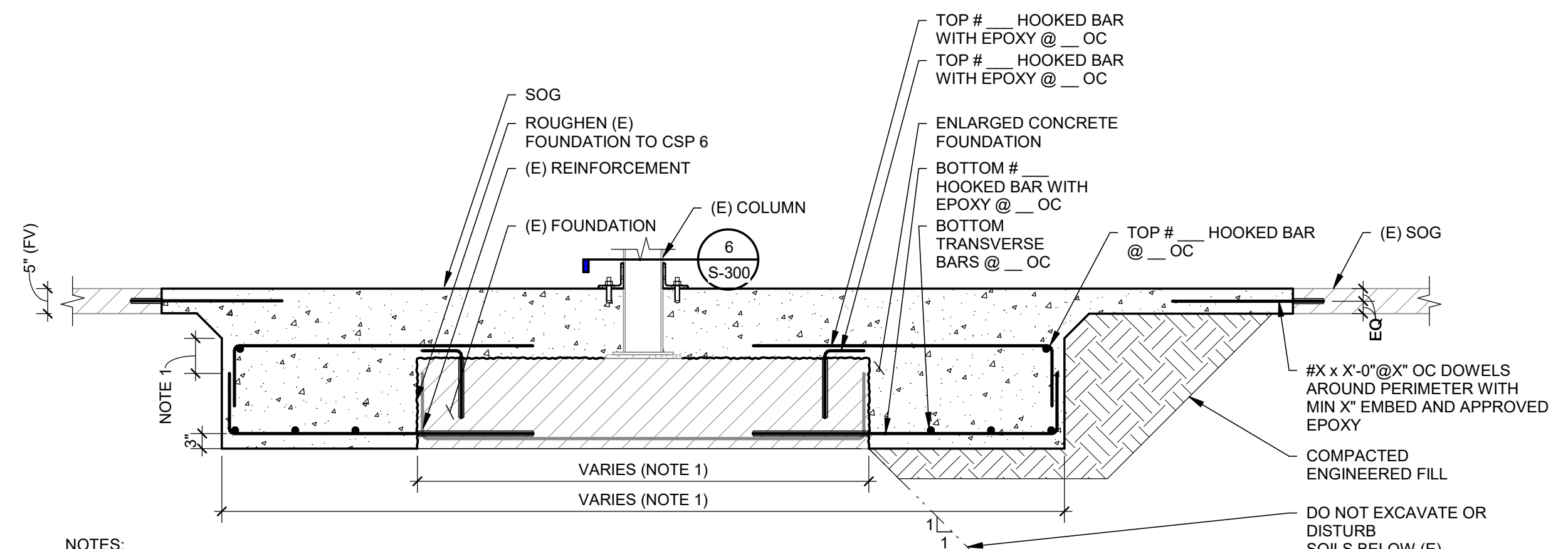
1/2" = 1'-0"



- NOTES:
1. REFER TO FOUNDATION PLAN (S-101) FOR EXISTING AND NEW FOUNDATION DIMENSIONS.
 2. REFER TO "CONCRETE REPAIR MATERIAL" SECTION FOR MORE INFORMATION.
 3. MAINTAIN 3" COVER OVER NEW BARS WHERE THE CONCRETE SURFACE IS IN CONTACT WITH SOIL.
 4. COMPACT SOIL AROUND EXISTING FOUNDATION AS SPECIFIED IN CONSTRUCTION DOCUMENTS BEFORE ADDING NEW CONCRETE SECTION.
 5. CONTRACTOR SHALL LOCATE EXISTING REINFORCEMENT PRIOR TO DRILLING INTO THE FOUNDATION TO AVOID DAMAGING EXISTING REINFORCEMENT.

3 TYPICAL DETAIL AT G7-G15

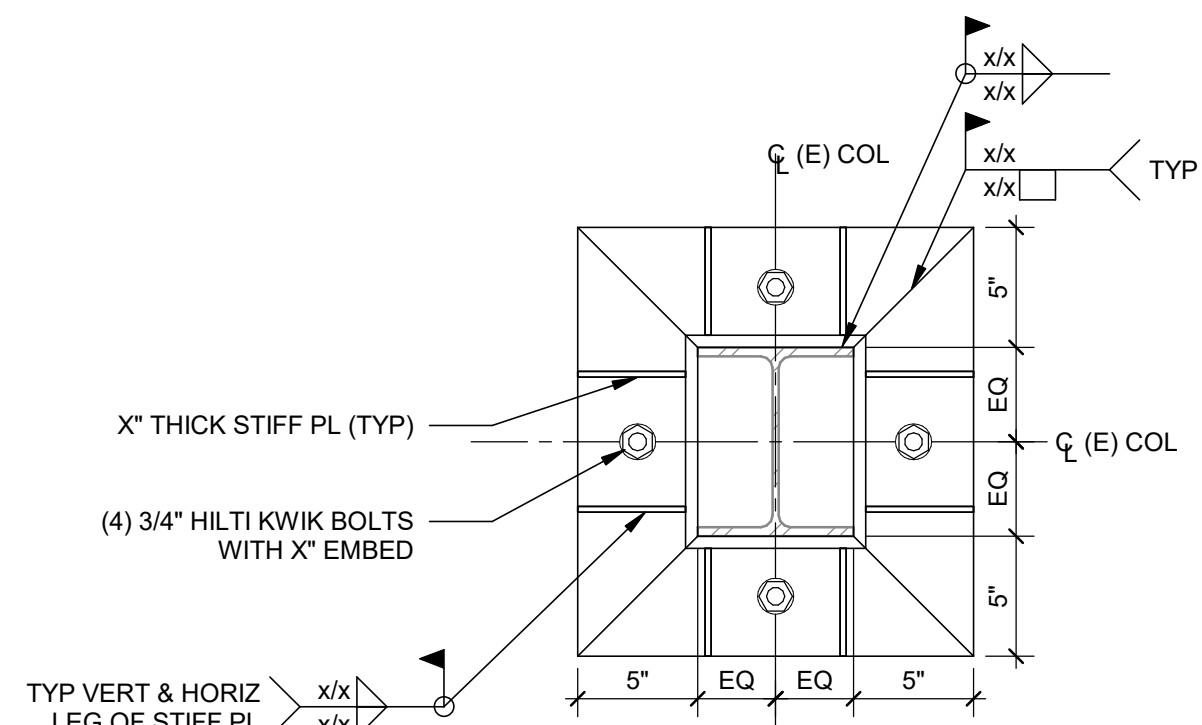
1/2" = 1'-0"



- NOTES:
1. REFER TO FOUNDATION PLAN (S-101) FOR EXISTING AND NEW FOUNDATION DIMENSIONS.
 2. REFER TO "CONCRETE REPAIR MATERIAL" SECTION FOR MORE INFORMATION.
 3. MAINTAIN 3" COVER OVER NEW BARS WHERE THE CONCRETE SURFACE IS IN CONTACT WITH SOIL.
 4. COMPACT SOIL AROUND EXISTING FOUNDATION AS SPECIFIED IN CONSTRUCTION DOCUMENTS BEFORE ADDING NEW CONCRETE SECTION.
 5. CONTRACTOR SHALL LOCATE EXISTING REINFORCEMENT PRIOR TO DRILLING INTO THE FOUNDATION TO AVOID DAMAGING EXISTING REINFORCEMENT.

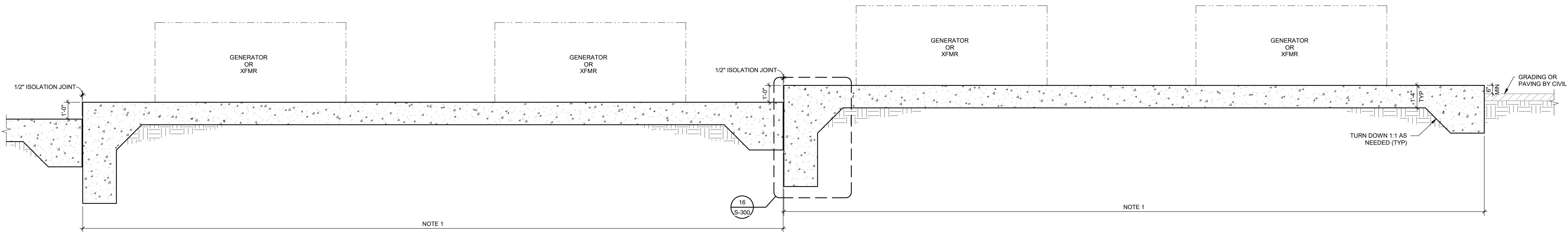
2 TYPICAL FOOTING ENLARGEMENT

1/2" = 1'-0"



6 STEEL COLLAR AT EXISTING COLUMN

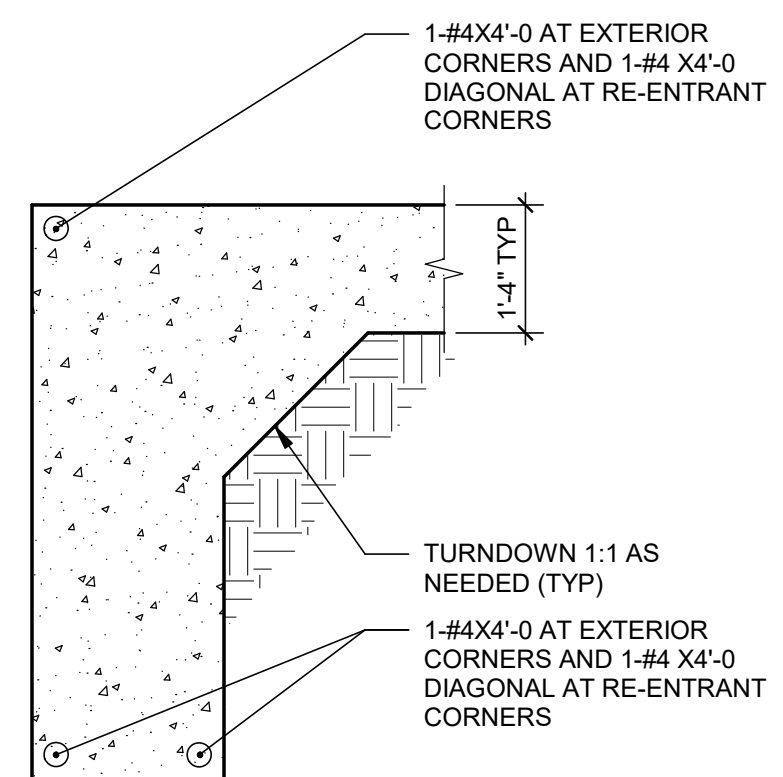
1 1/2" = 1'-0"



- NOTES:
1. REFER TO ARCHITECTURAL AND MEP DRAWINGS FOR THE EXTENT OF HOUSEKEEPING PAD.

15 TYPICAL HOUSEKEEPING PAD

3/8" = 1'-0"



16 TYPICAL HOUSEKEEPING PAD REINFORCEMENT

1/2" = 1'-0"

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S-300 - TYPICAL DETAILS

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Revisions		
02/03/26	DESIGN DEVELOPMENT /PRICING	
1 04/06/26	DD SUBMISSION	

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DD SUBMISSION
04/06/2026

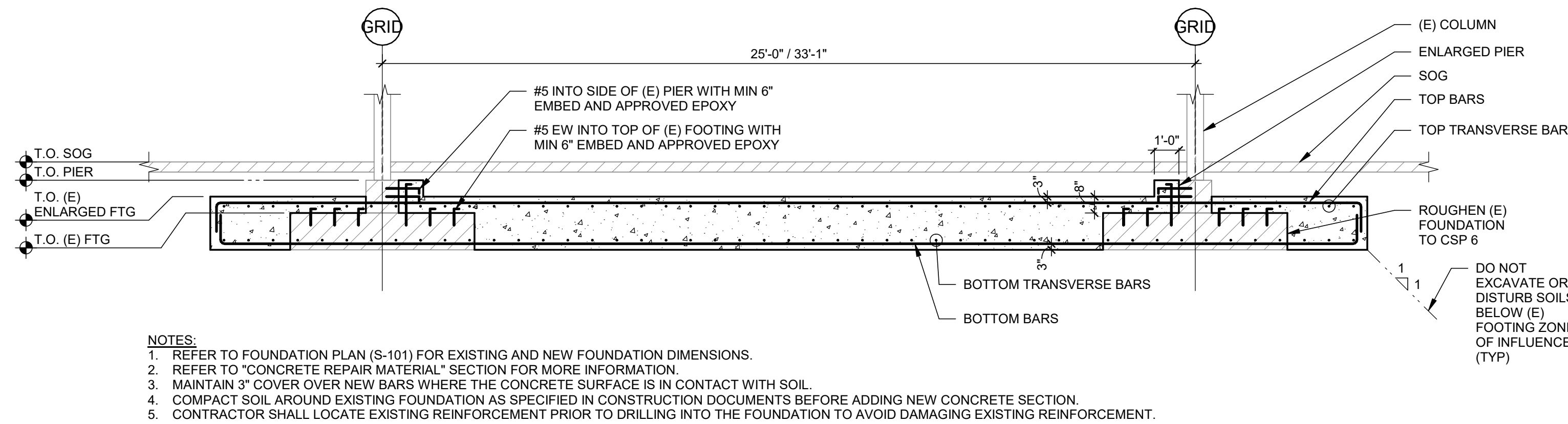
VSW JOB NO. SCALE
250202 As indicated

TYPICAL DETAILS

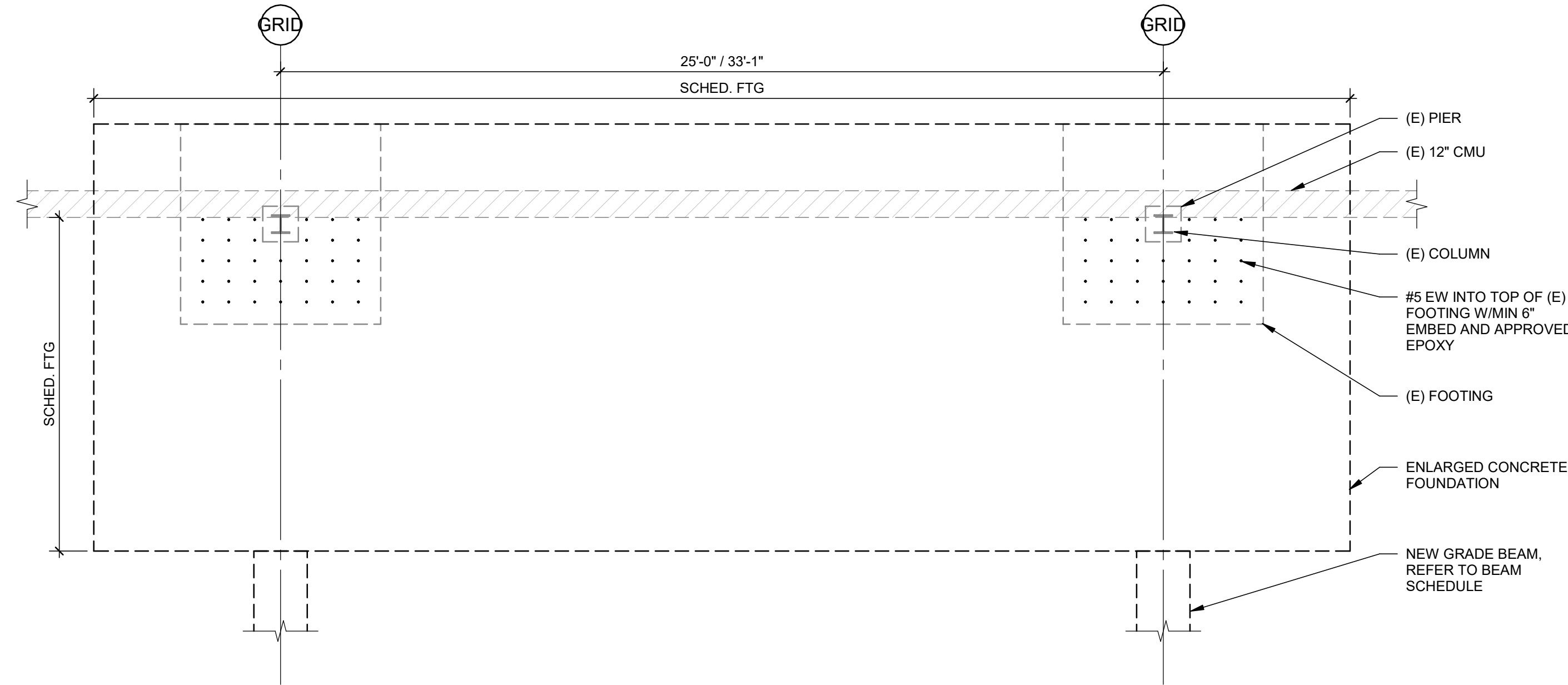
DRAWING

S-300

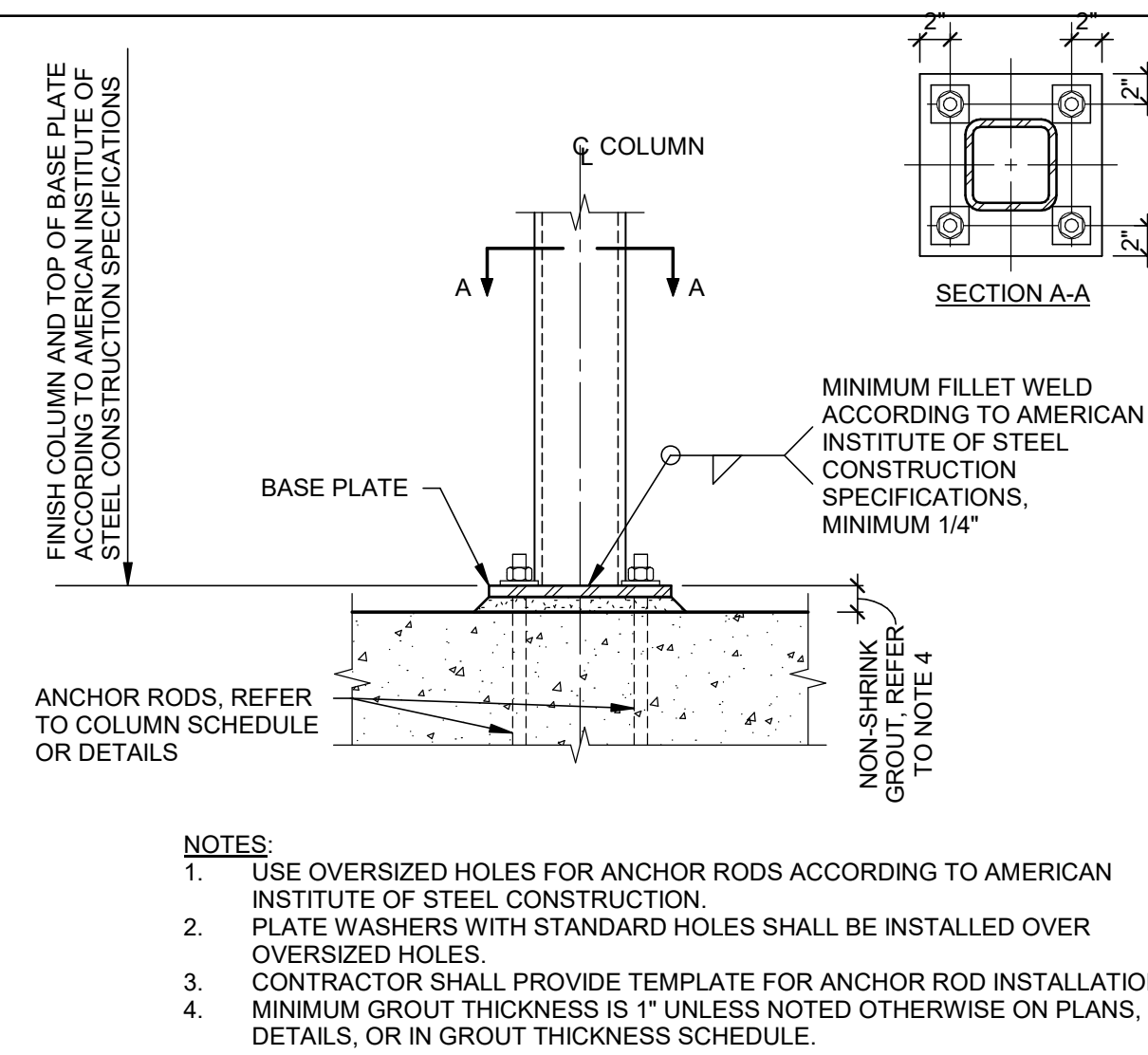
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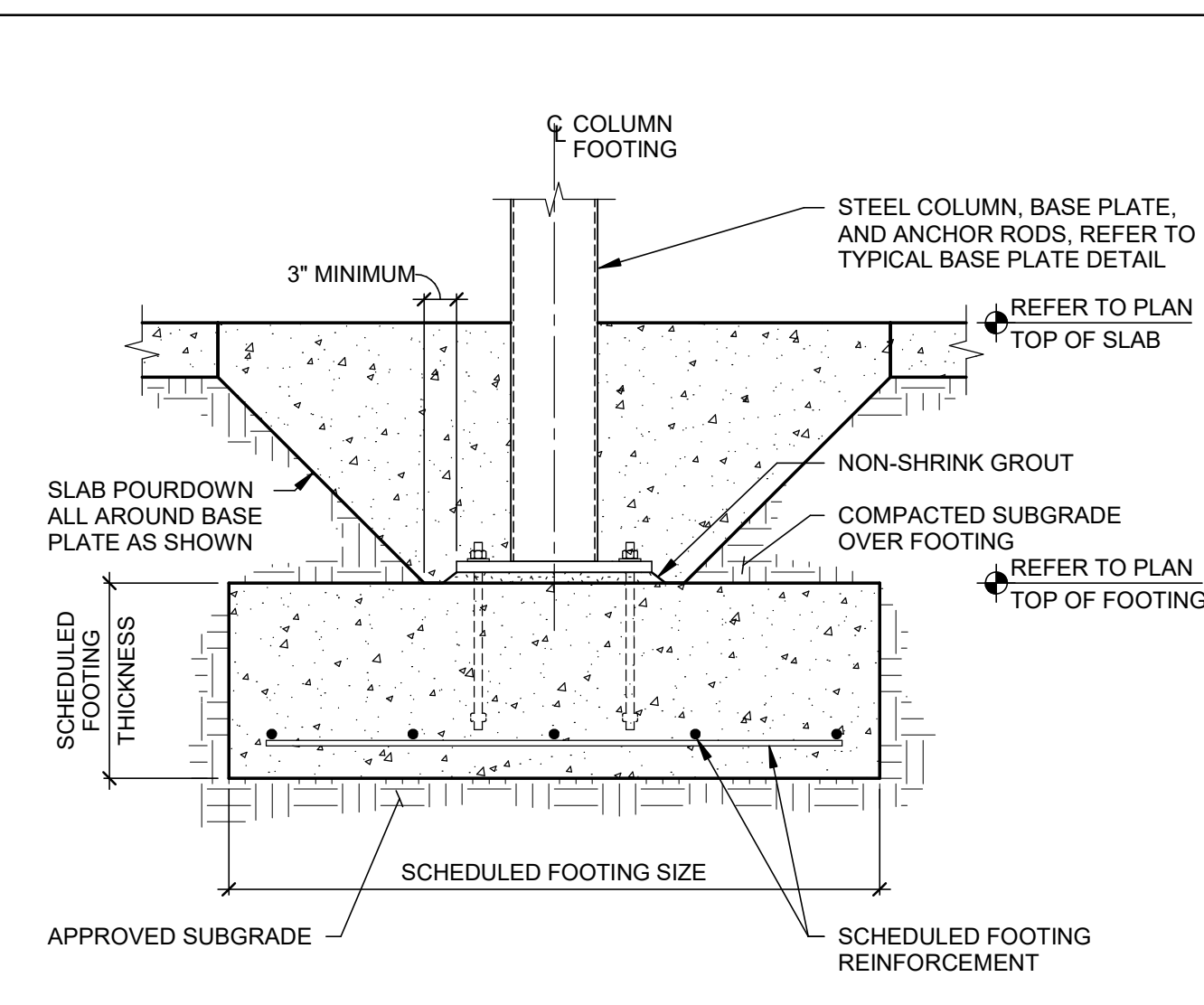
4 TYPICAL FOOTING ENLARGEMENT FOR BRACED FRAME
1/4" = 1'-0"



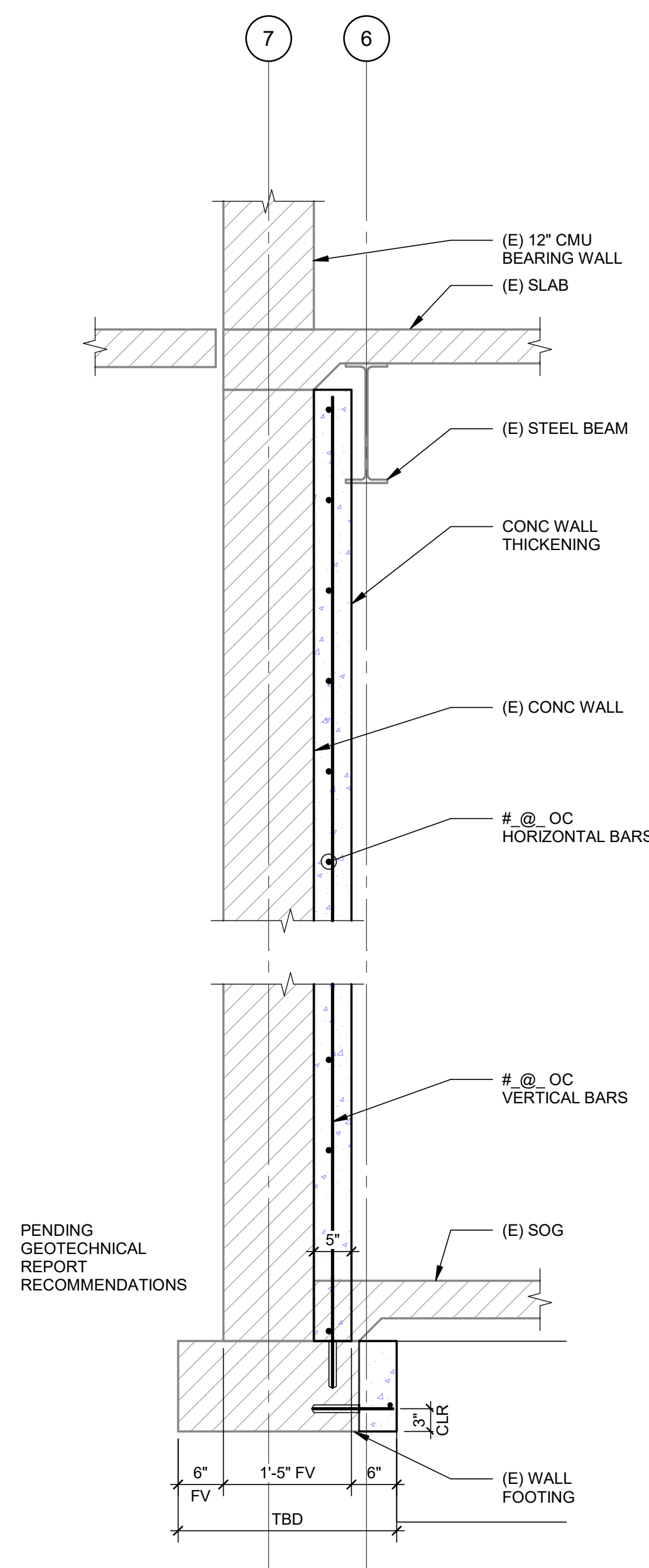
9 TYPICAL FOOTING ENLARGEMENT AT BRACED FRAME - PLAN
1/4" = 1'-0"



2 TYPICAL RECTANGULAR/SQUARE HSS COLUMN BASE CONNECTION
NO SCALE



1 TYPICAL STEEL COLUMN ON SPREAD FOOTING
NO SCALE



16 CONCRETE BASEMENT WALL STRENGTHENING DETAIL
3/4" = 1'-0"

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S-400 - FOUNDATION DETAILS AND SECTIONS

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04/06/2026

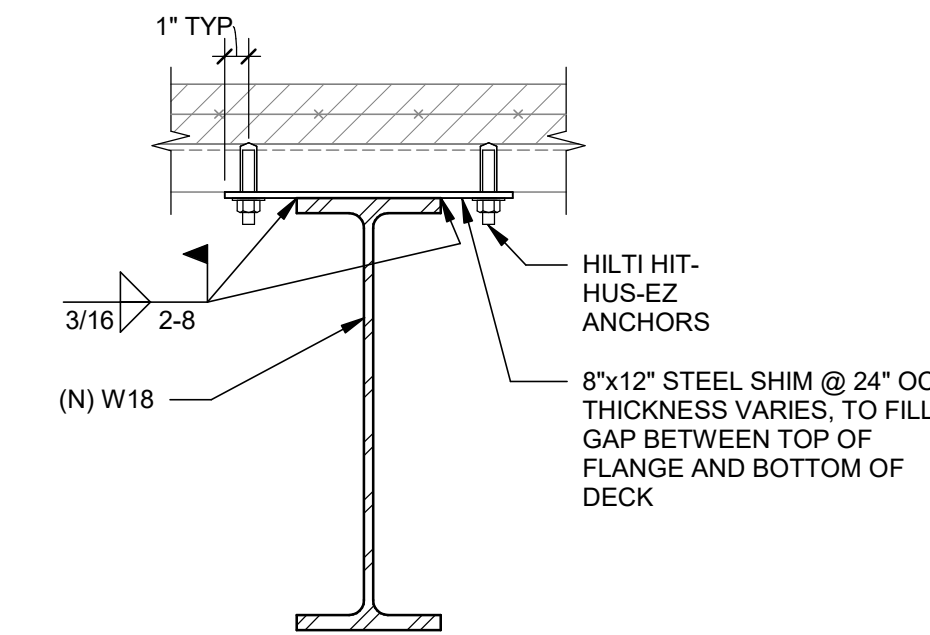
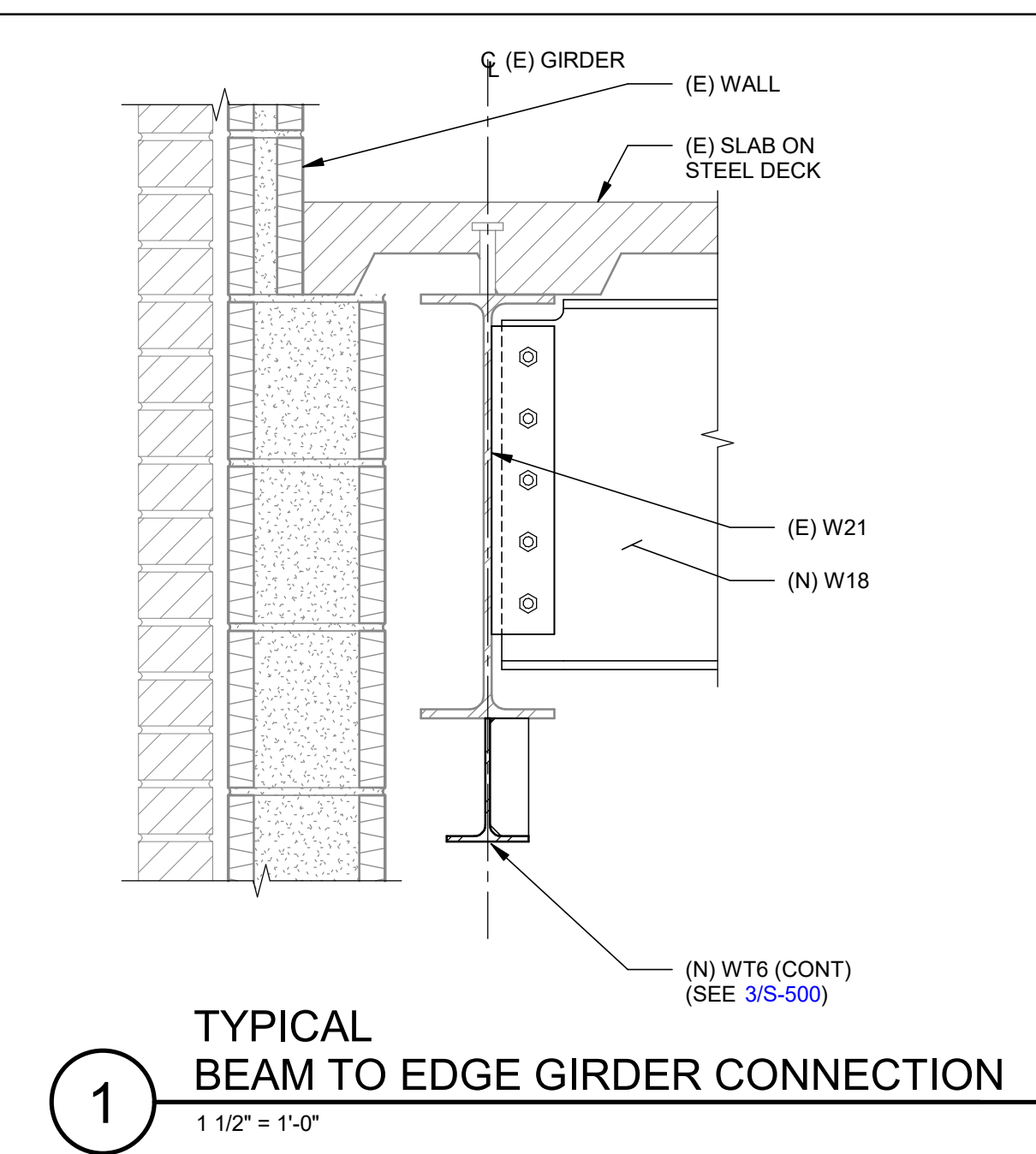
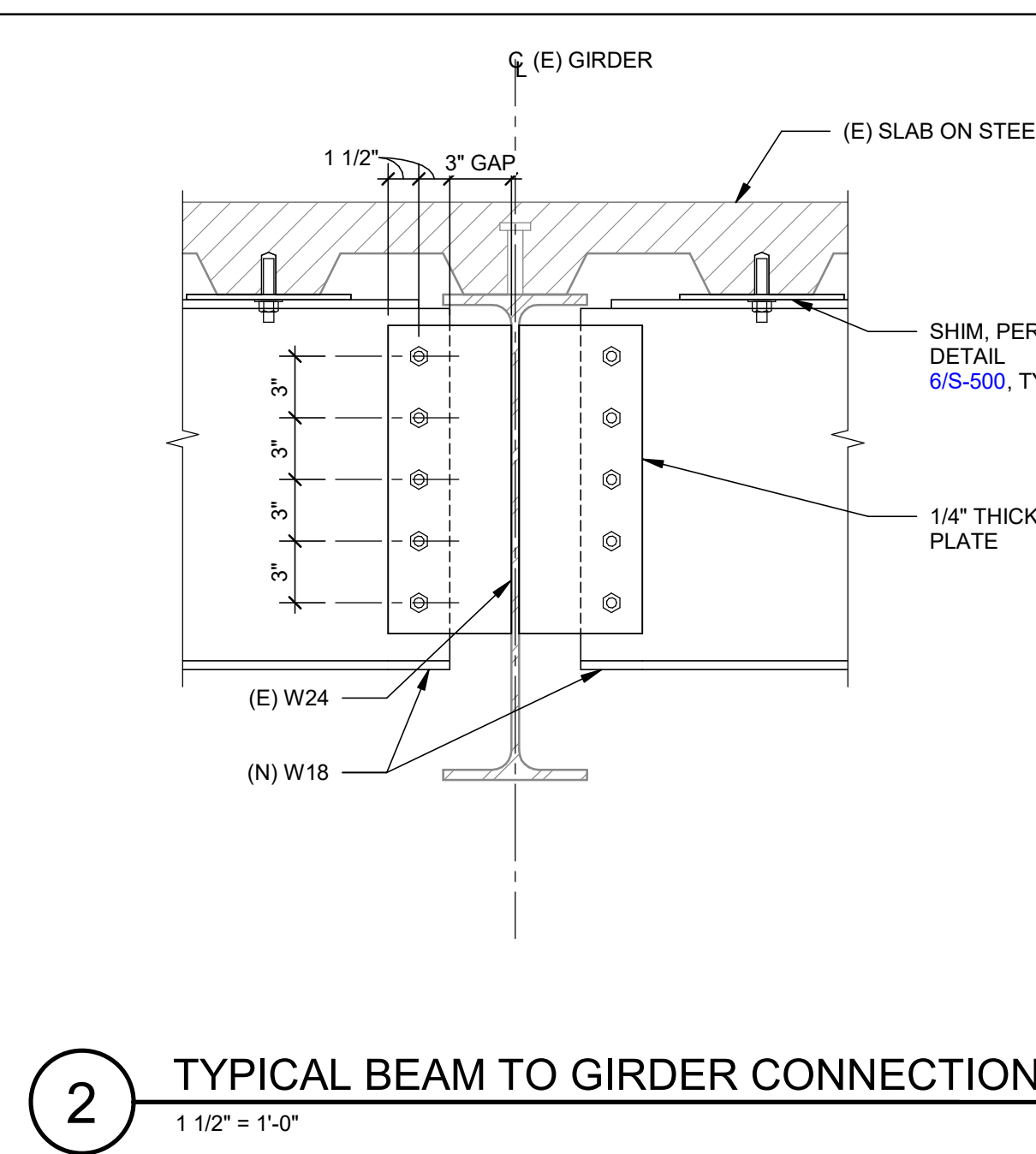
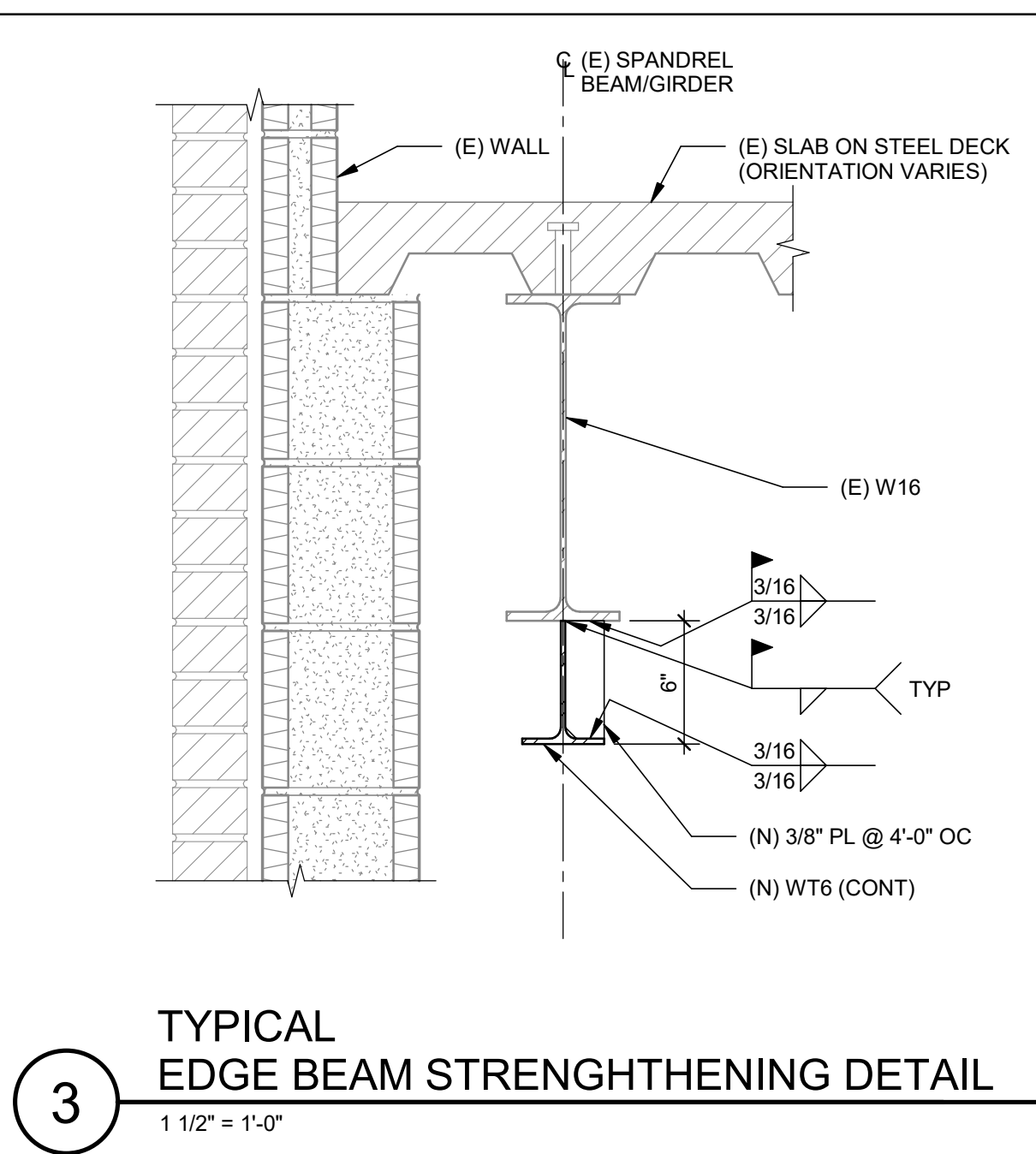
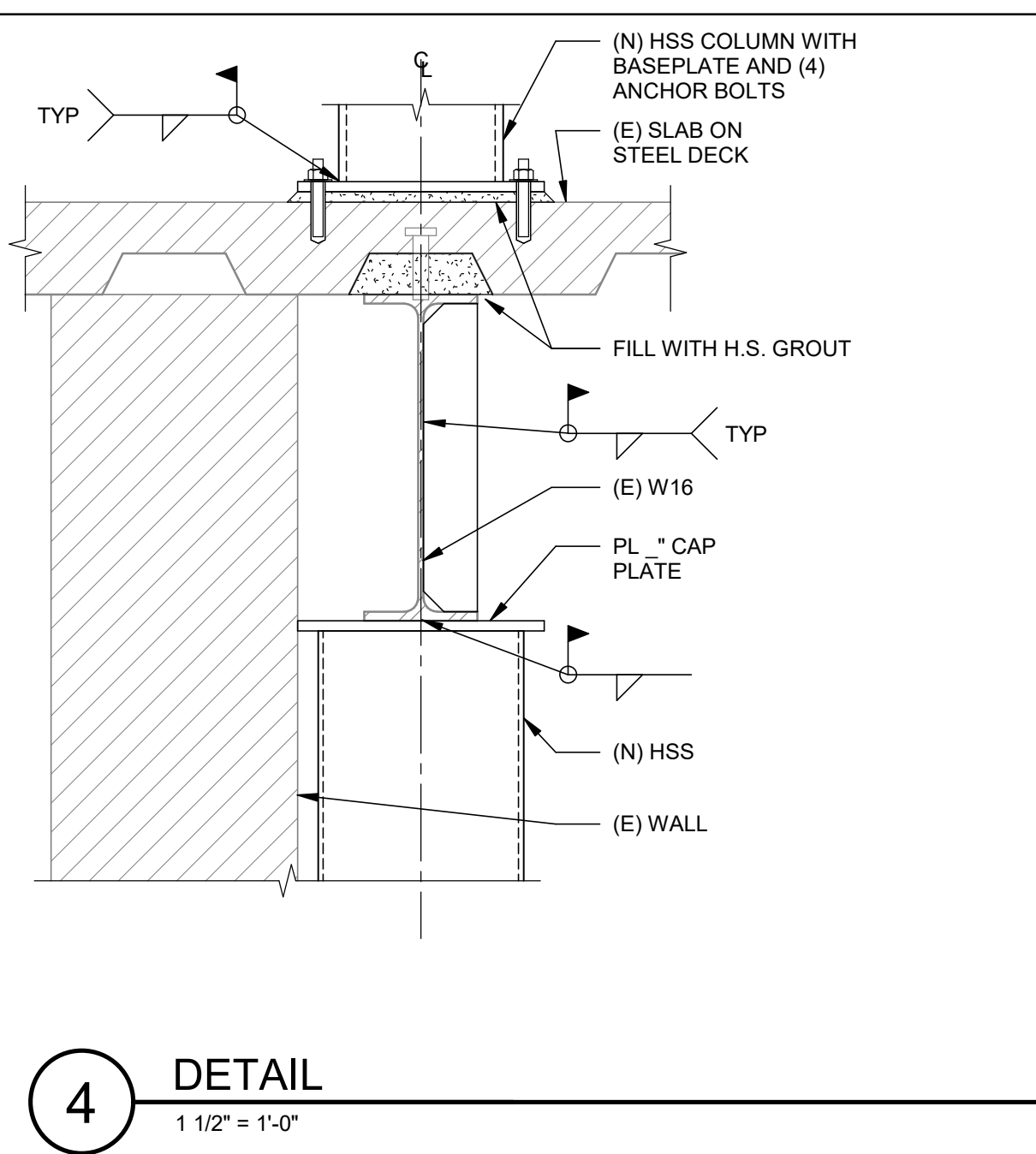
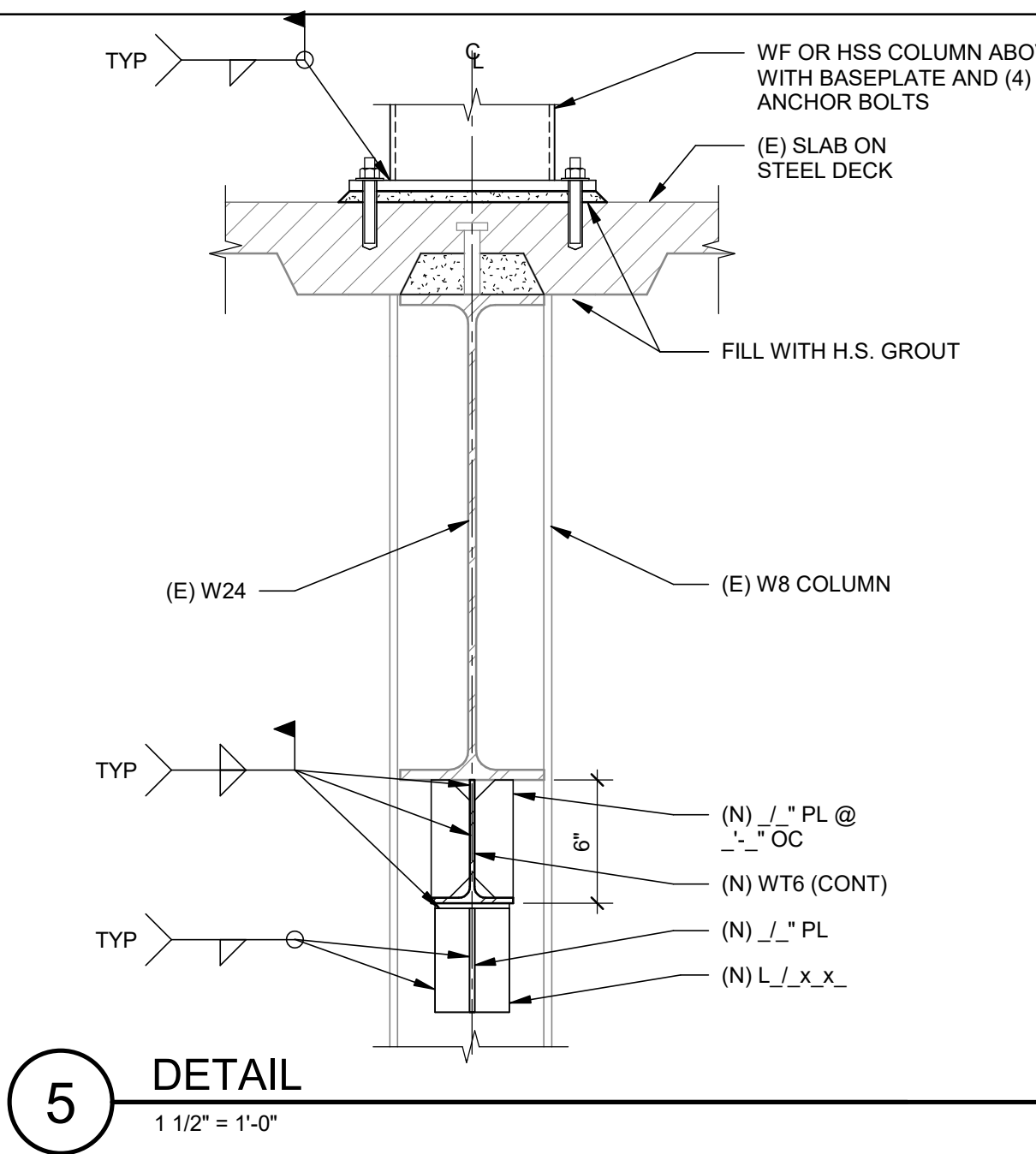
VSW JOB NO. 250202 SCALE As indicated

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S-500 - HORIZONTAL FRAMING DETAILS AND SECTIONS

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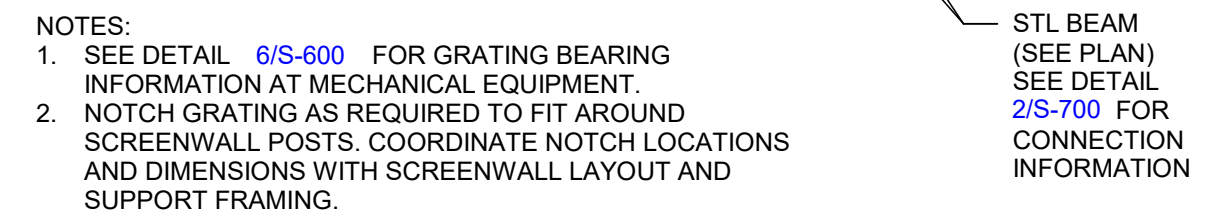
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02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

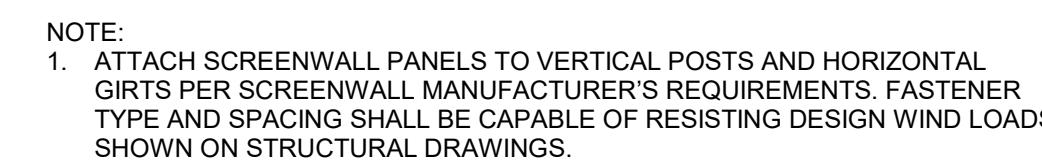
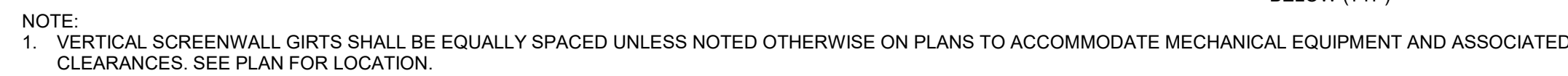
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VSW JOB NO. 250202 SCALE 1 1/2" = 1'-0"
HORIZONTAL FRAMING DETAILS AND SECTIONS
DRAWING

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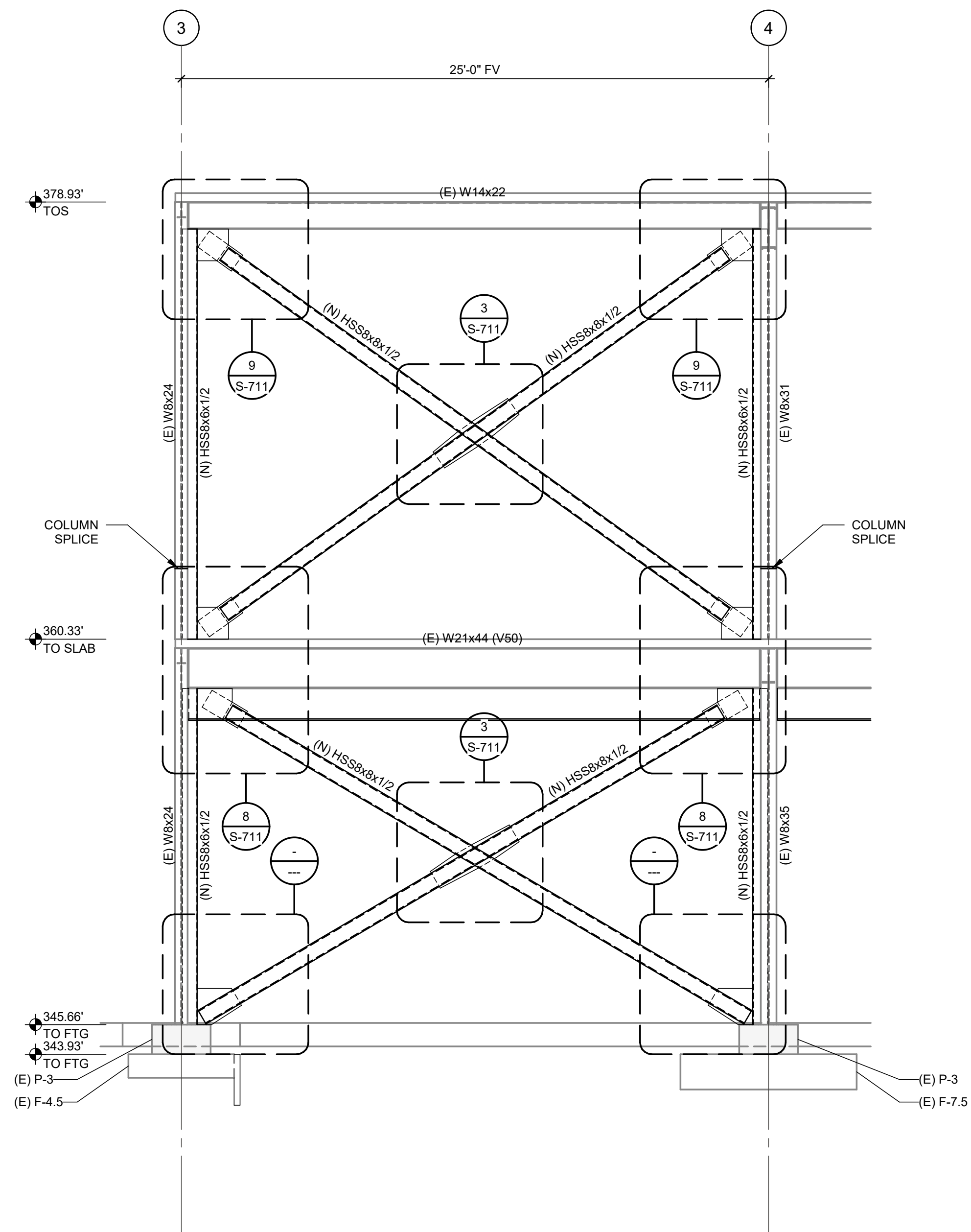
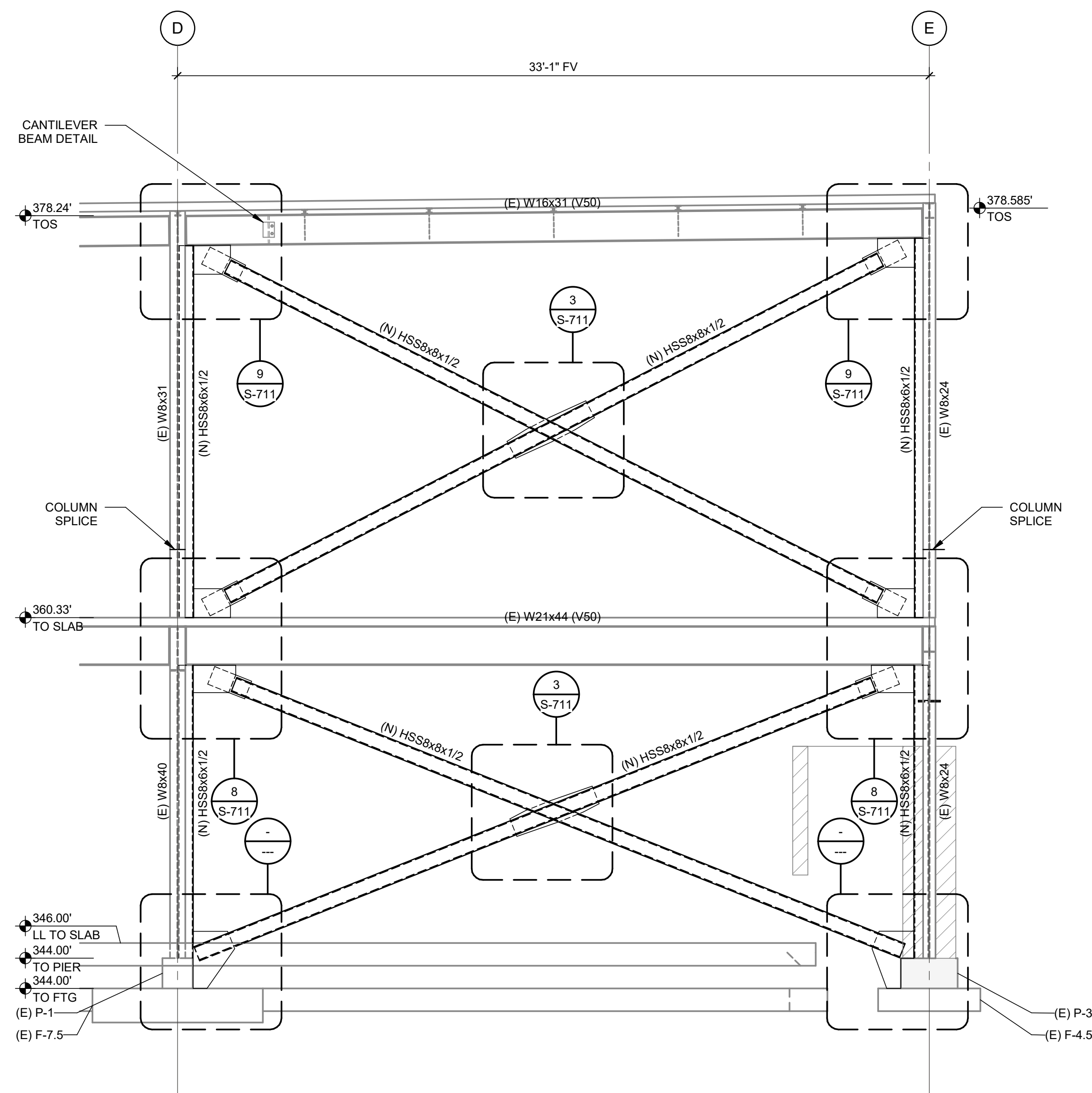
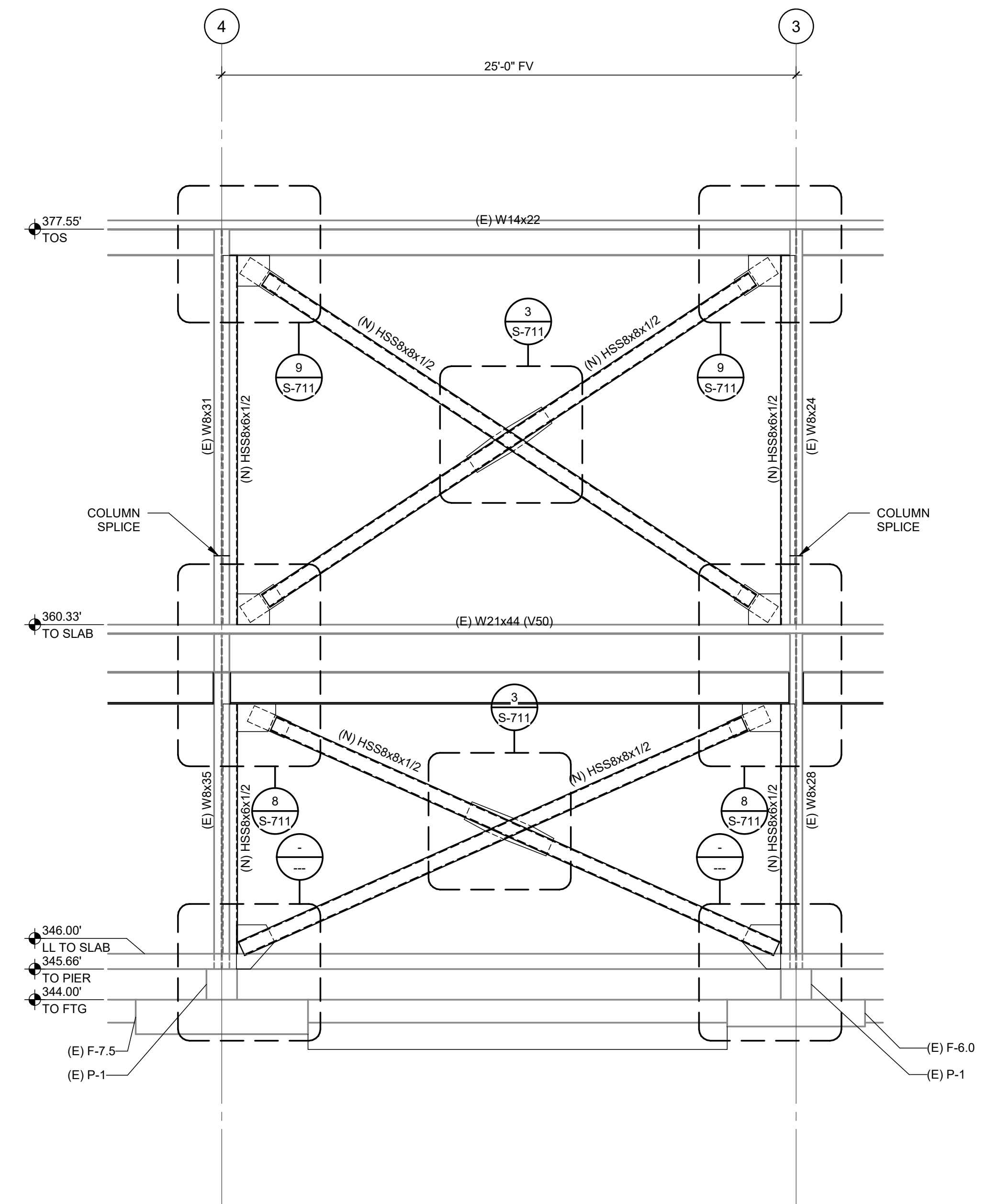
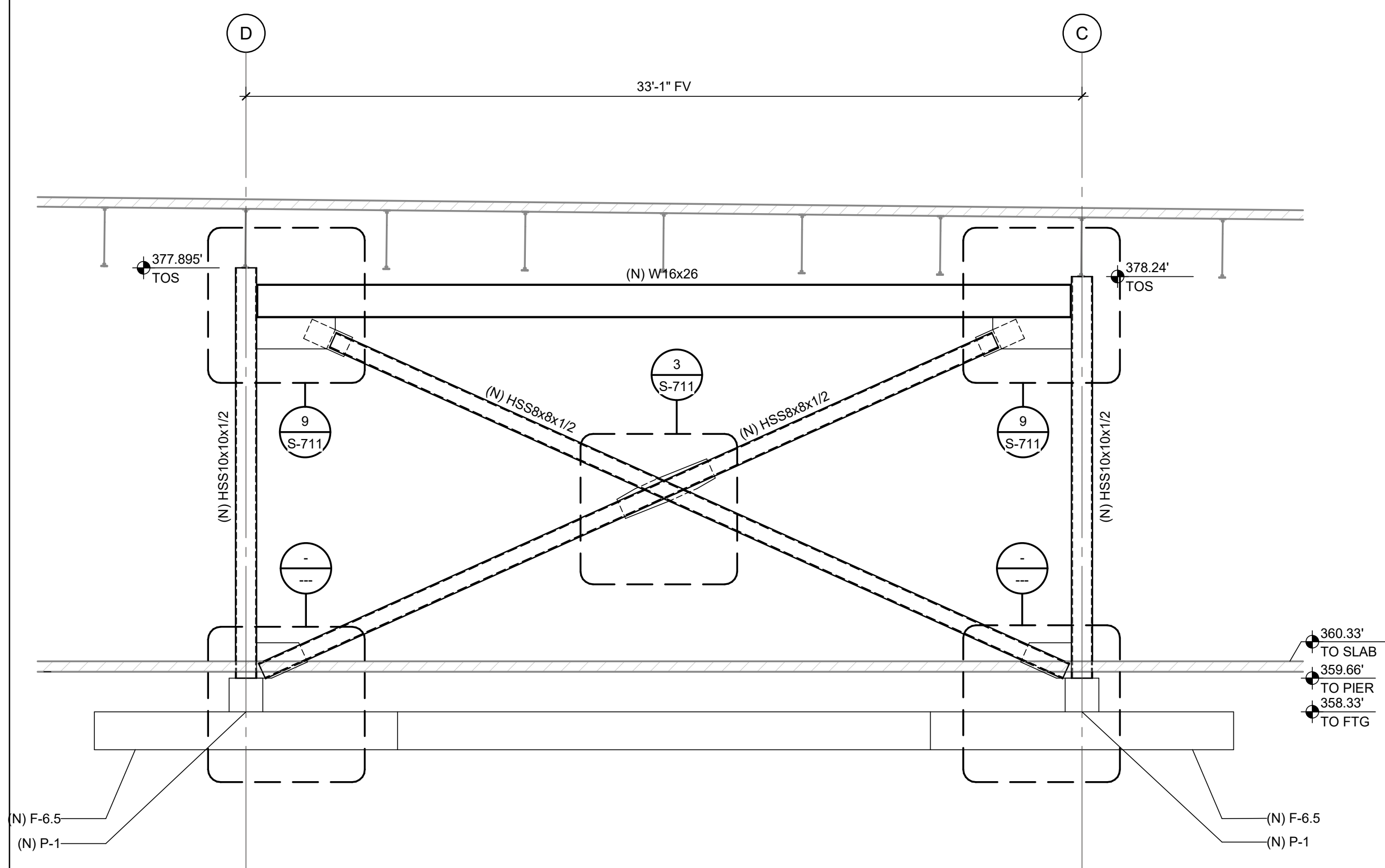
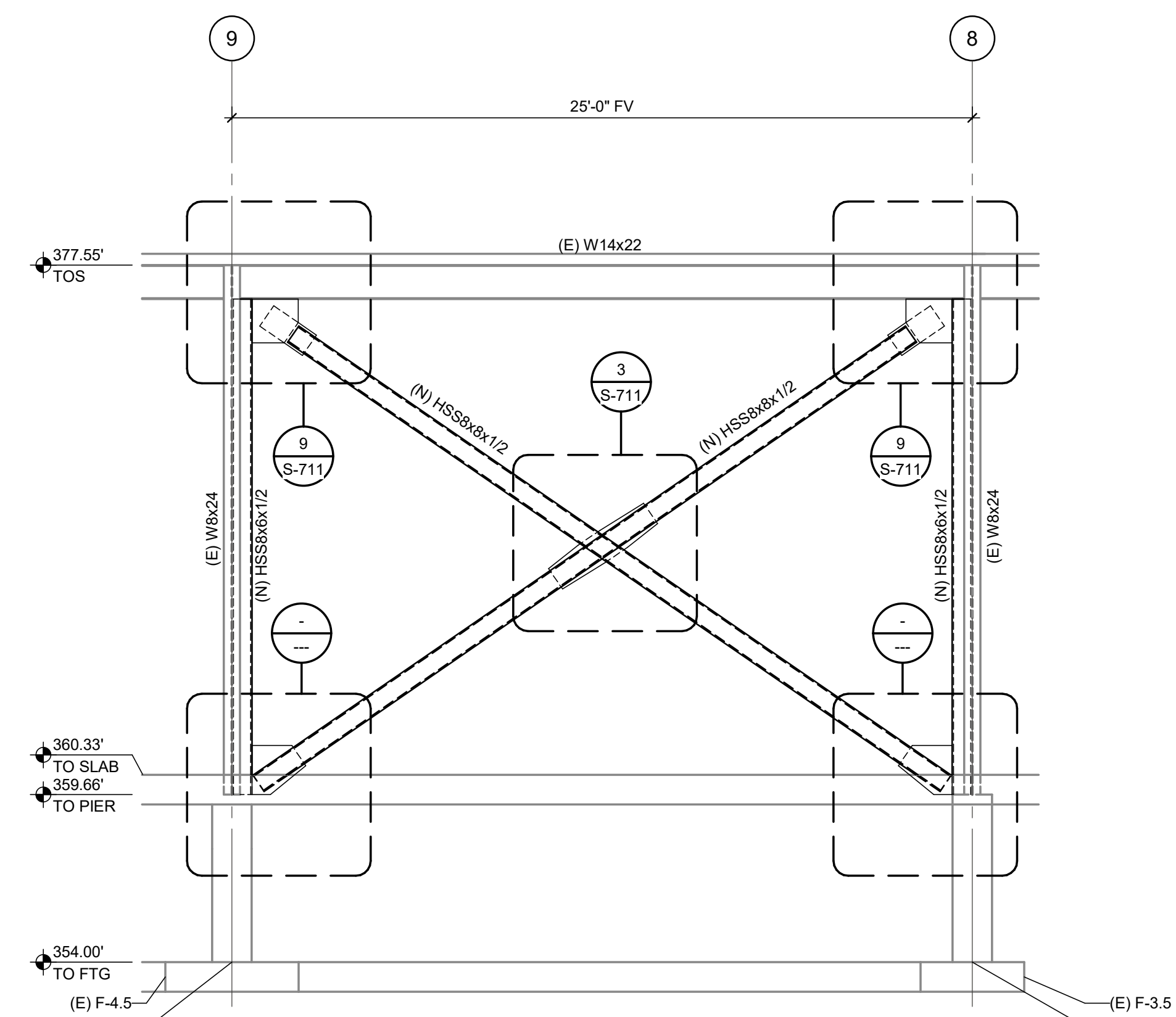
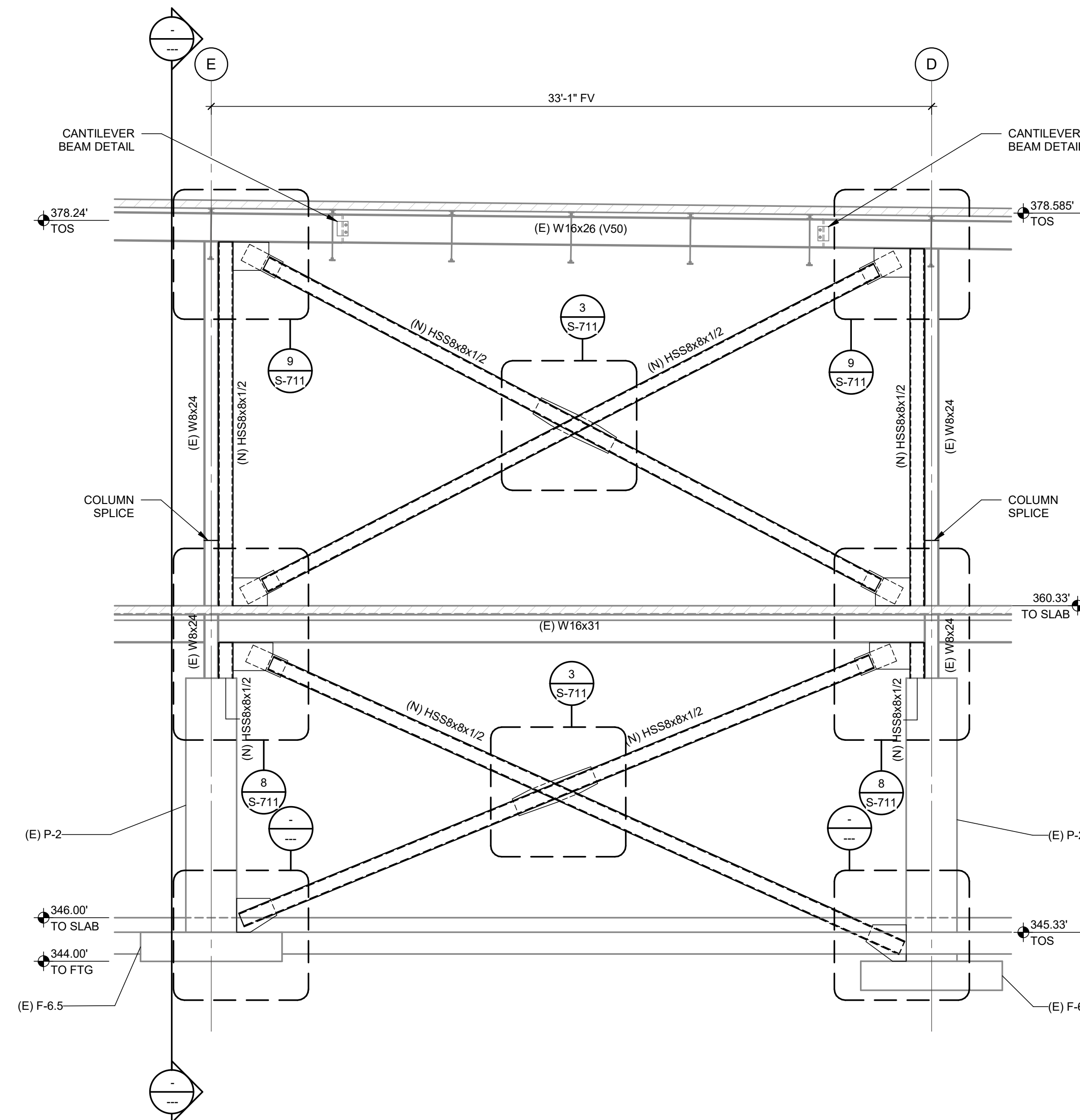
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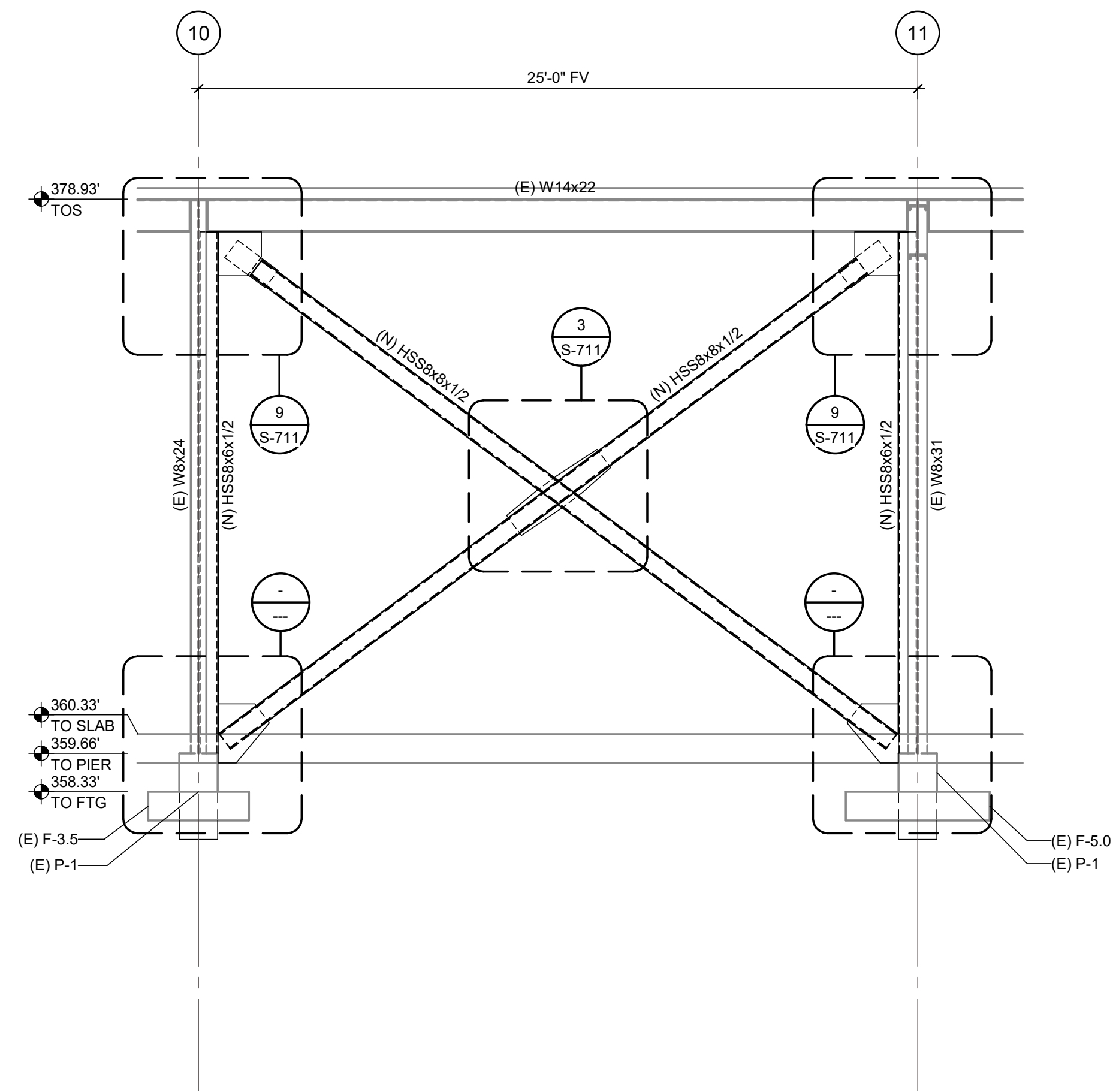
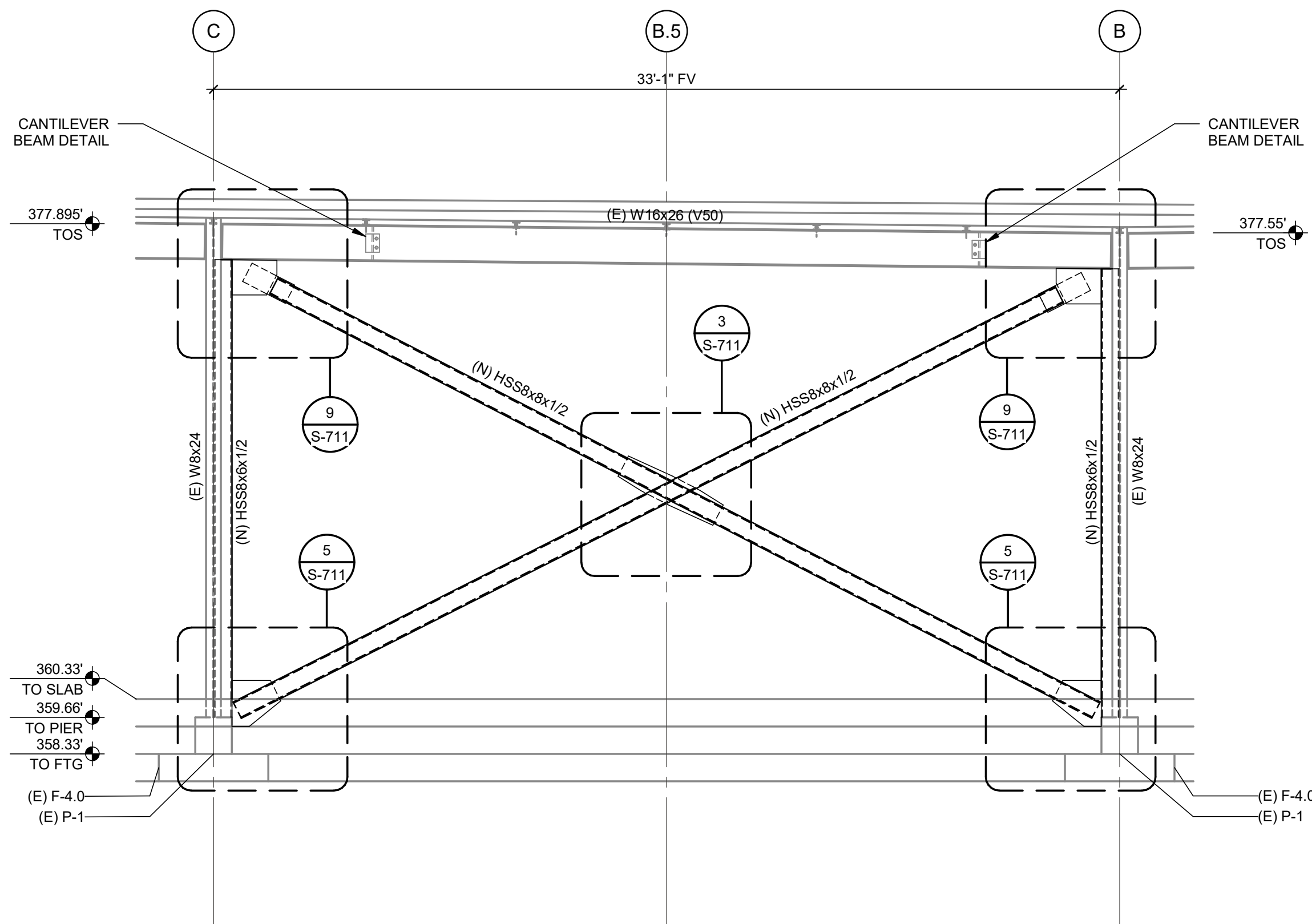
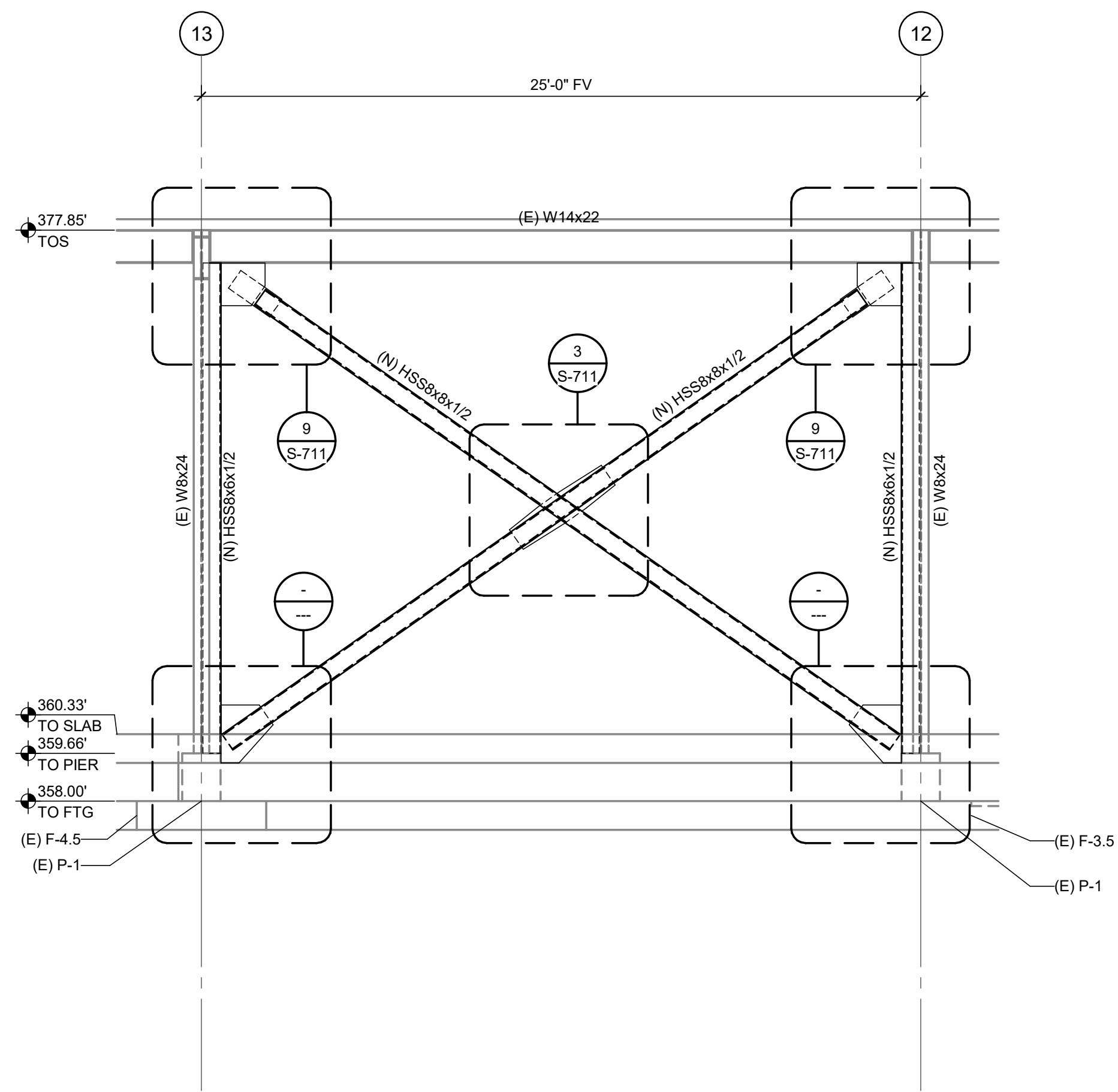

$$1'' = 1' - 0''$$

$$1^* = 1^* - 0$$

$$1'' = 1' - 0''$$


1/4" = 1'-0"



**3** SECTION BETWEEN GRIDLINES 3 AND 4 - BF-3
1/4" = 1'-0"**2** SECTION BETWEEN GRIDLINES D AND E - BF-2
1/4" = 1'-0"**1** SECTION BETWEEN GRIDLINES 3 AND 4 - BF-1
1/4" = 1'-0"**6** SECTION BETWEEN GRIDLINES C AND D - BF-6
1/4" = 1'-0"**5** SECTION BETWEEN GRIDLINES 8 AND 9 - BF-5
1/4" = 1'-0"**4** SECTION BETWEEN GRIDLINES D AND E - BF-4
1/4" = 1'-0"



3 SECTION BETWEEN GRIDLINES 12 AND 13 - BF-9
1/4" = 1'-0"

2 SECTION BETWEEN GRIDLINES B AND C - BF-8
1/4" = 1'-0"

1 SECTION BETWEEN GRIDLINES 10 AND 11 - BF-7
1/4" = 1'-0"

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S-701 - STRUCTURAL BRACING ELEVATIONS

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04/06/2026

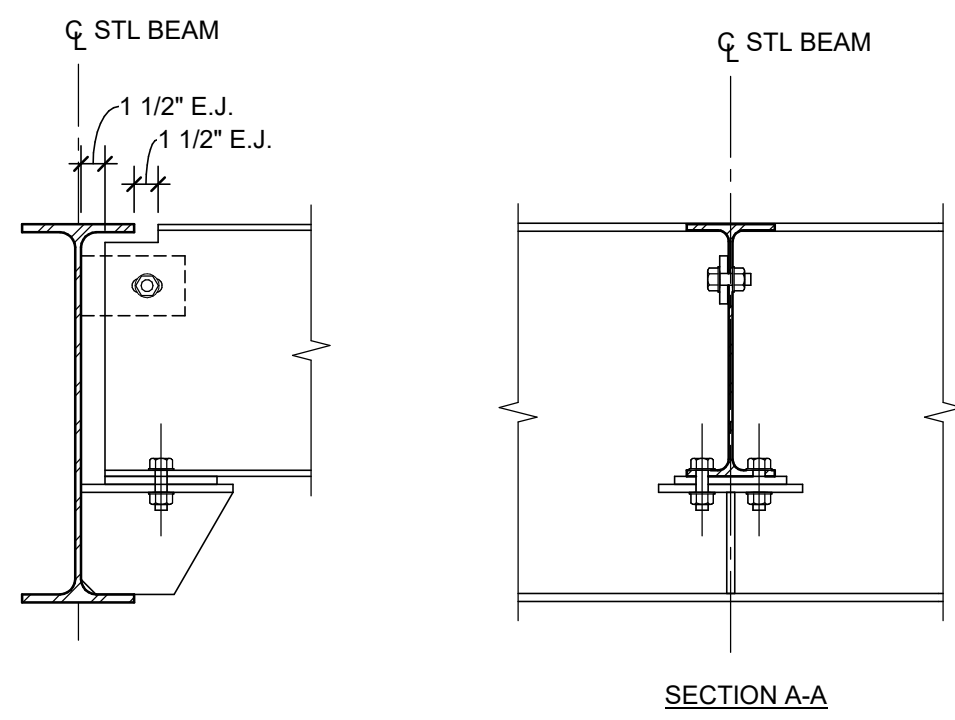
VSW JOB NO. SCALE
250202 1/4" = 1'-0"

STRUCTURAL BRACING ELEVATIONS

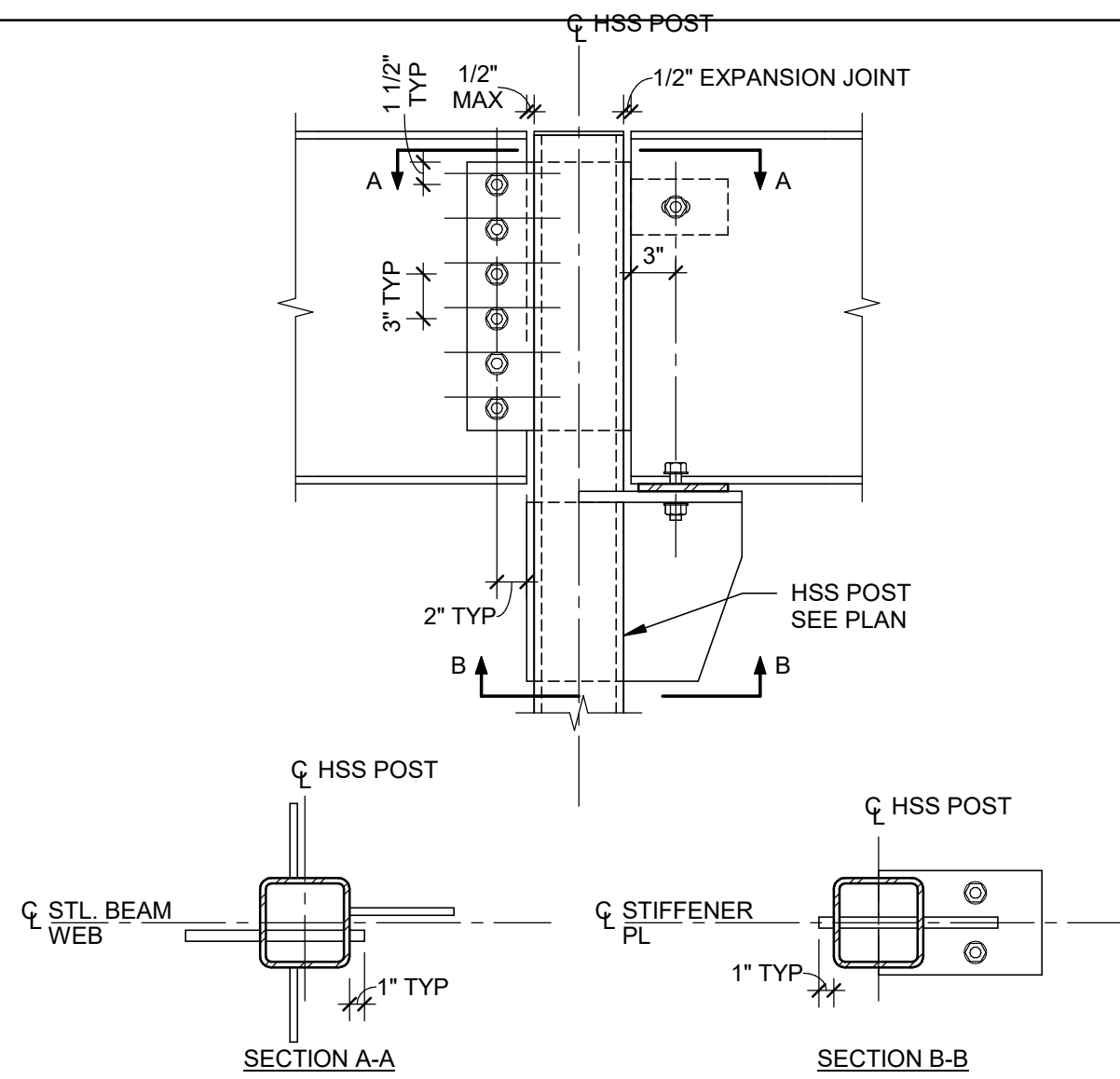
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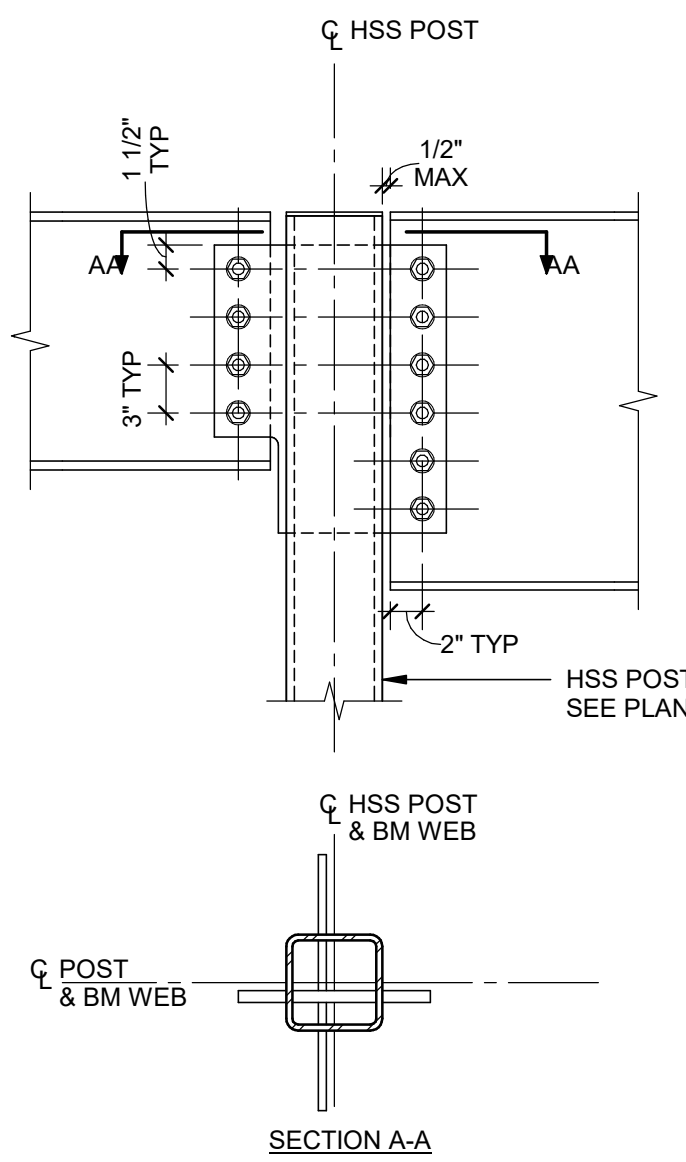
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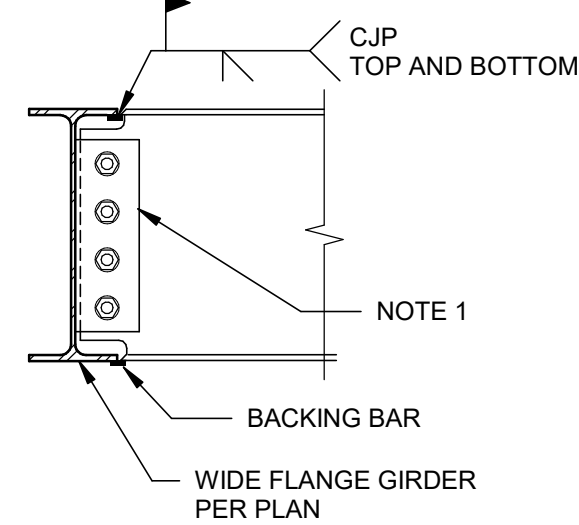
5 TYPICAL BEAM TO BEAM CONNECTION AT EXPANSION JOINT
1" = 1'-0"



4 TYPICAL BEAM TO POST CONNECTION AT EXPANSION JOINT
1" = 1'-0"

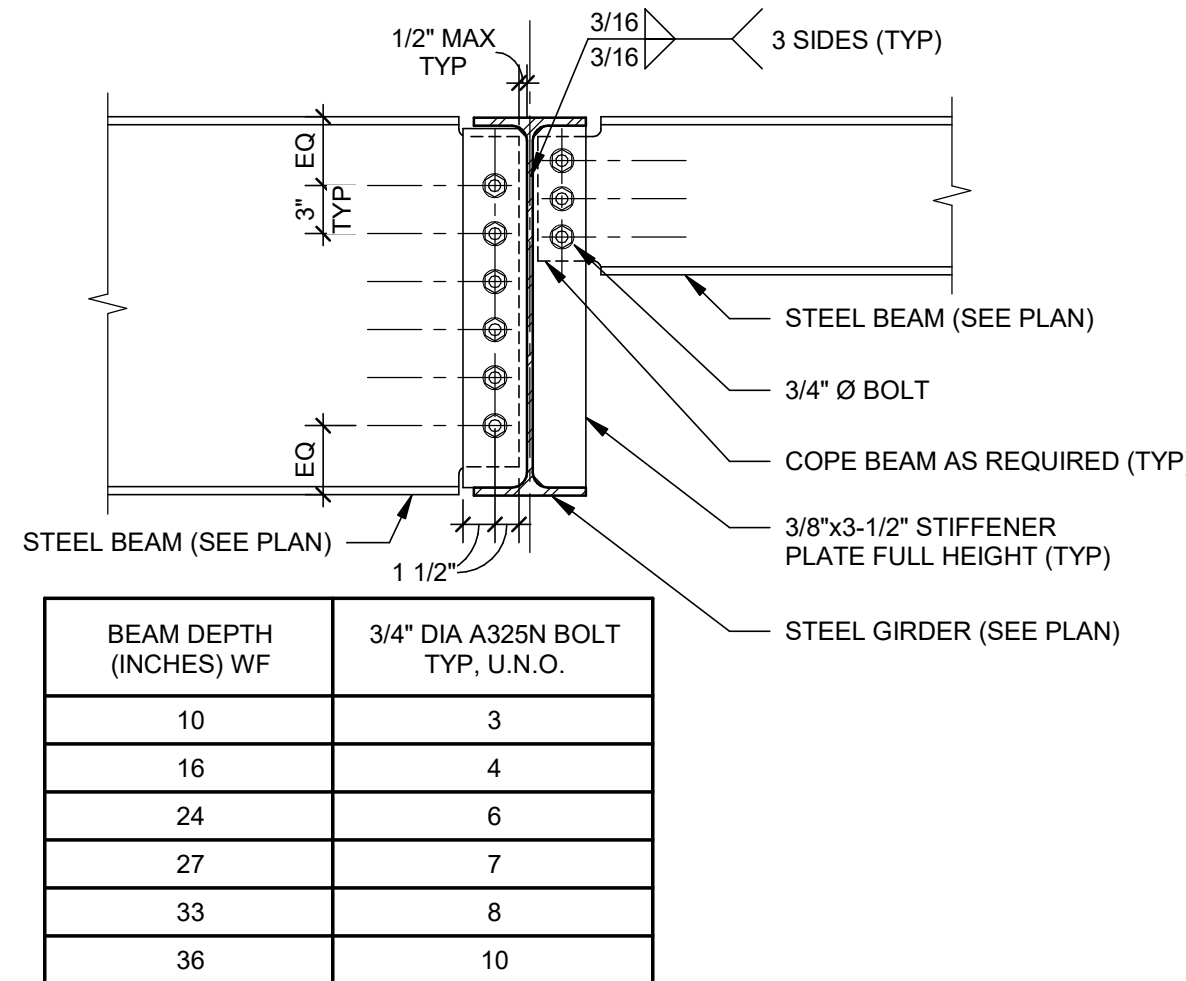


3 TYPICAL BEAM TO POST CONNECTION
1" = 1'-0"



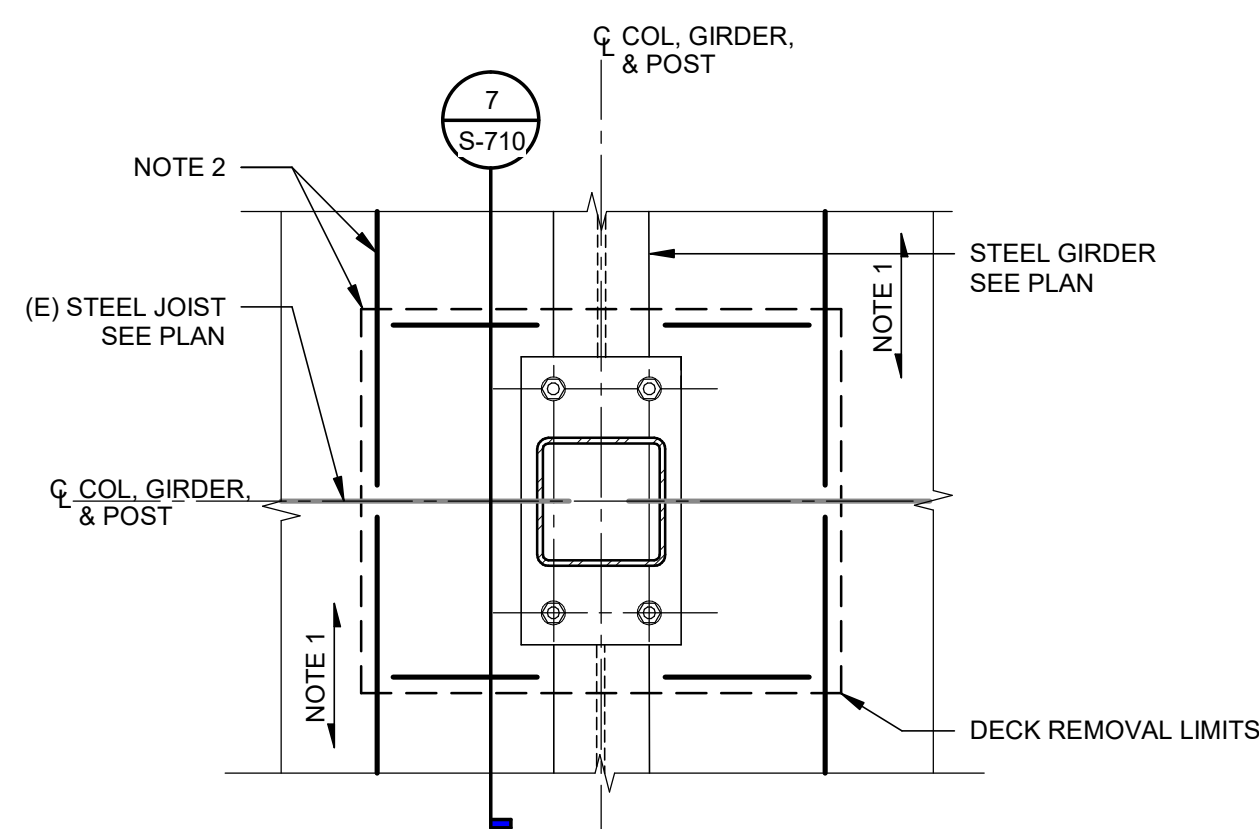
NOTE:
1. SEE DETAIL S-710 FOR SHEAR PLATE THICKNESS AND BOLT REQUIREMENTS PER BEAM SIZE. FOR MOMENT CONNECTIONS, FULL-HEIGHT SHEAR PLATES ARE NOT REQUIRED.

2 TYPICAL BEAM TO BEAM MOMENT CONNECTION
1" = 1'-0"



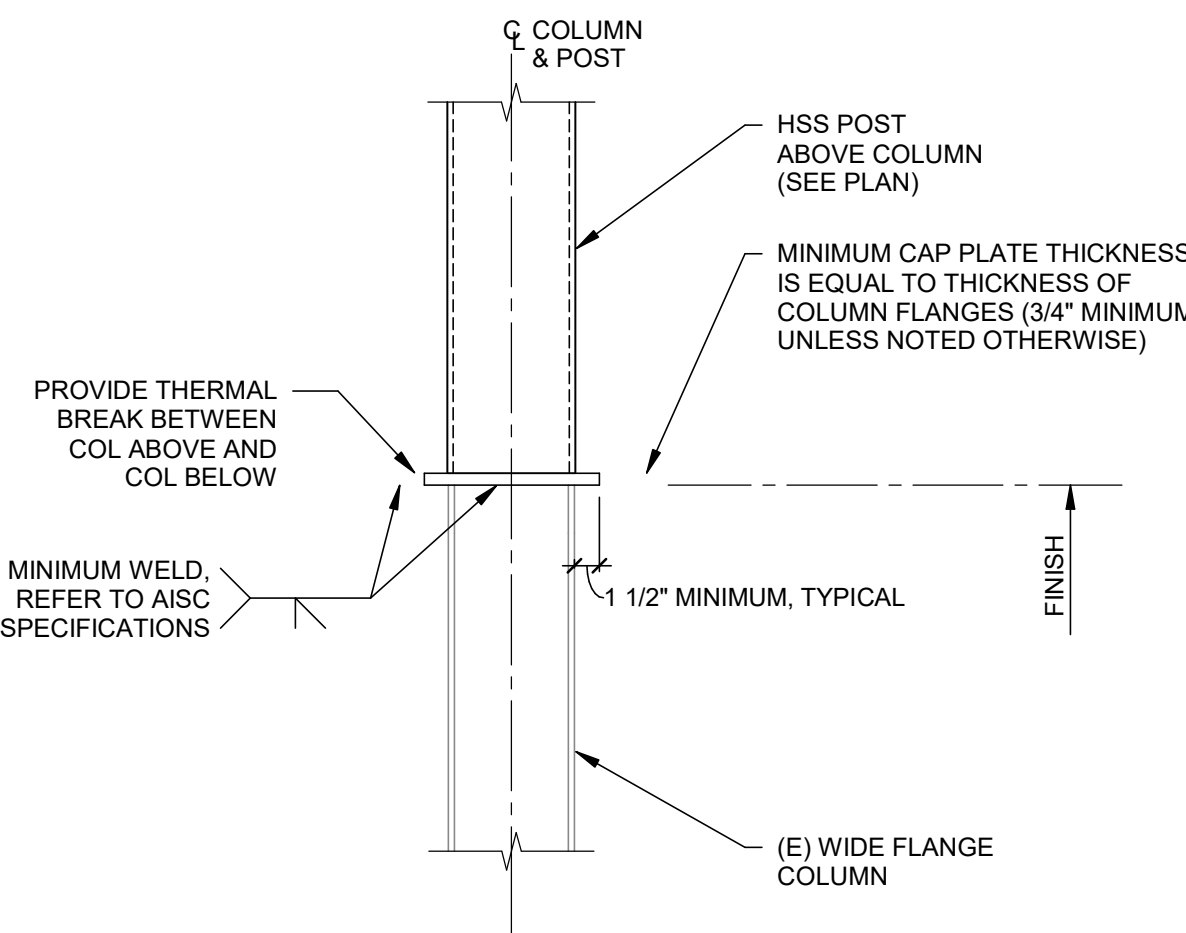
NOTE:
1. CONNECTION REFERS TO THE SHALLOWER BEAM IN A BEAM-TO-BEAM CONNECTION.

1 TYPICAL BEAM-TO-BEAM SHEAR CONNECTION
NO SCALE

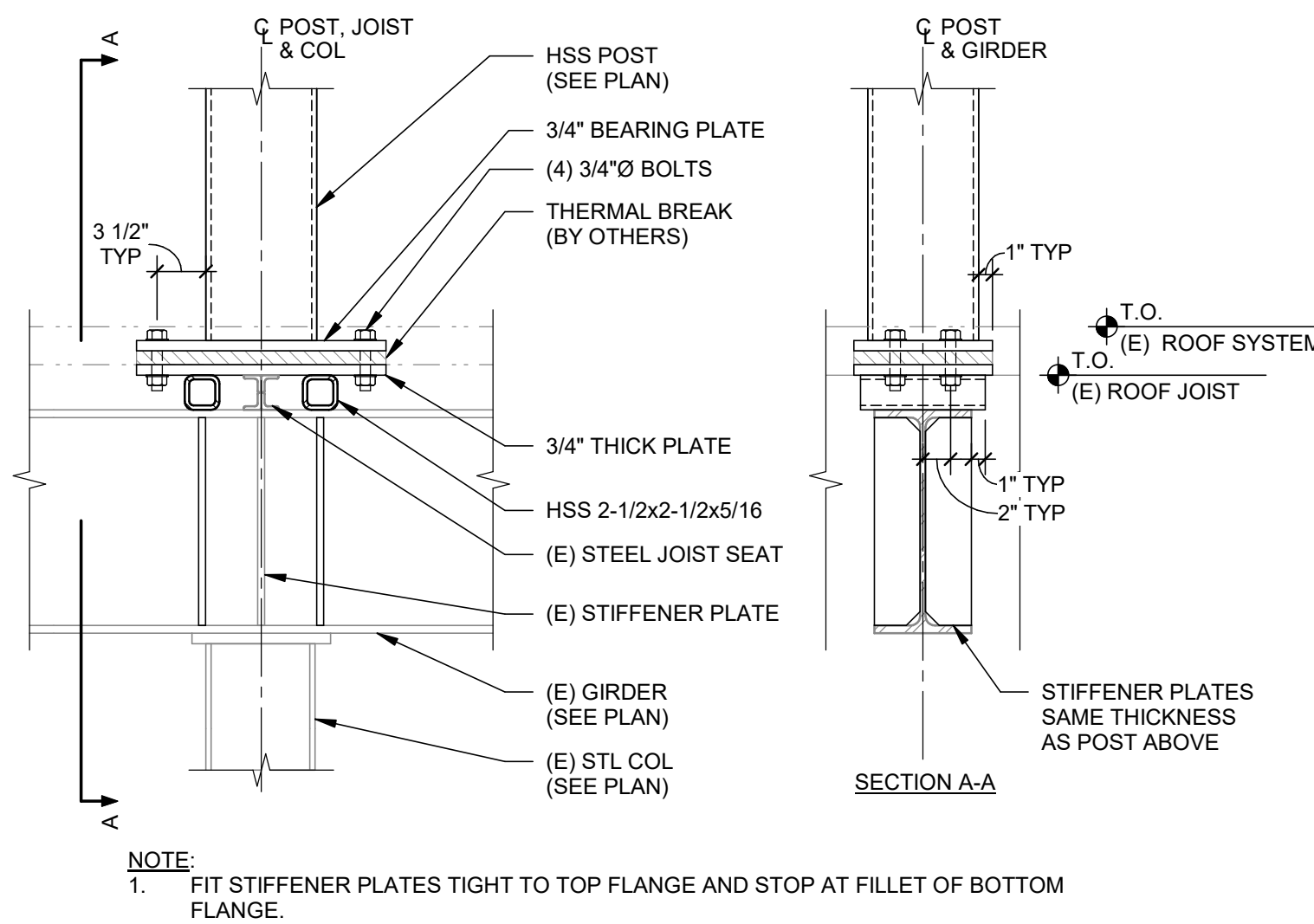


NOTES:
1. 1 1/2" STEEL DECK (FIELD VERIFY) SPANNING BETWEEN EXISTING STEEL JOISTS.
2. PROVIDE OPENING IN EXISTING ROOF DECK AS REQUIRED FOR INSTALLATION OF NEW DUNNAGE POST. FOR OPENINGS EXCEEDING 18" IN WIDTH (PERPENDICULAR TO DECK SPAN), PROVIDE SUPPLEMENTAL FRAMING AND DECK SUPPORT AROUND OPENING AS SHOWN SEE DETAIL S-501 FOR ADDITIONAL INFORMATION.

9 STEEL HSS POST TO TRANSFER GIRDER
1" = 1'-0"

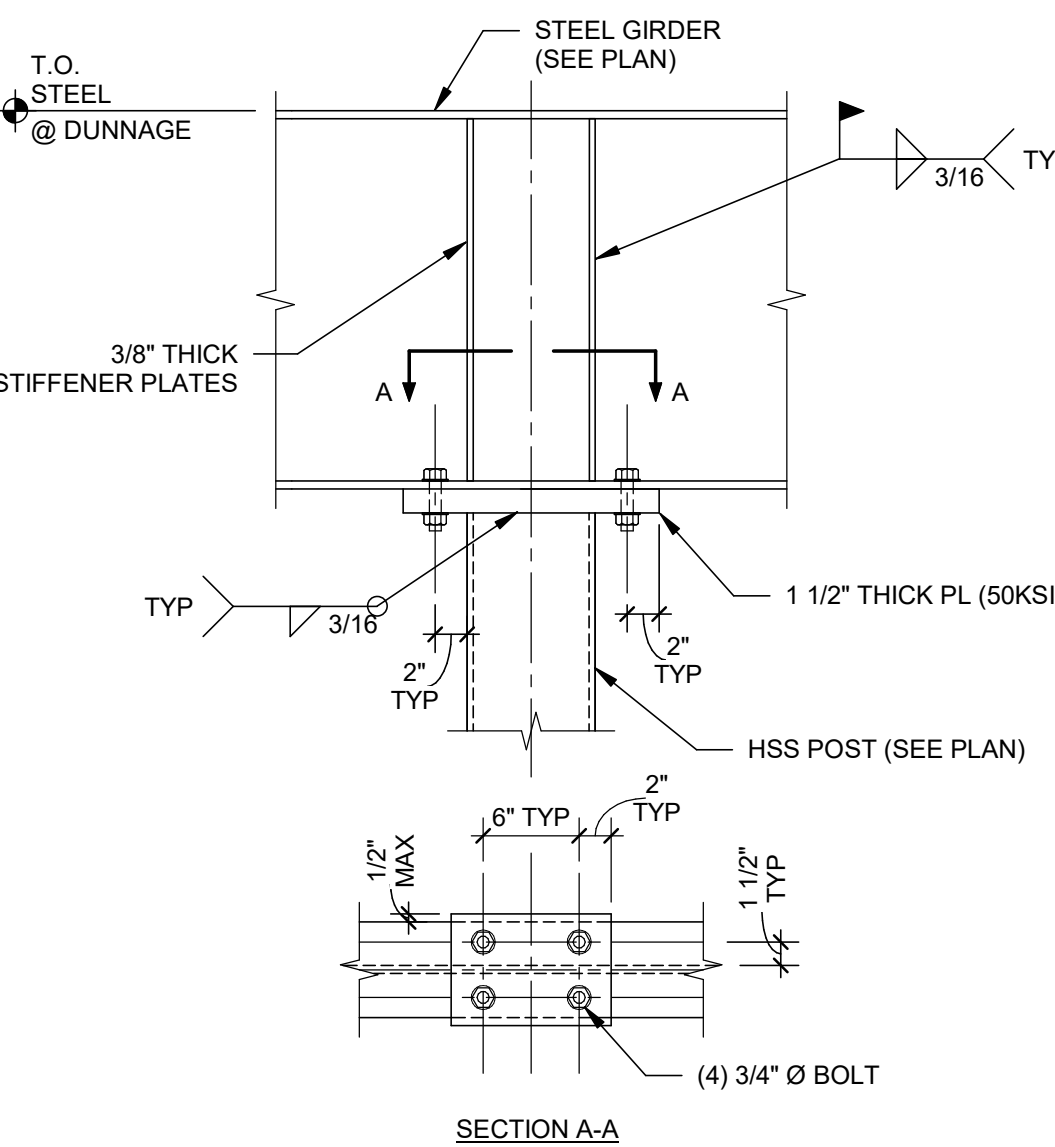


8 WIDE FLANGE COLUMN TENSION SPLICE
NO SCALE



NOTE:
1. FIT STIFFENER PLATES TIGHT TO TOP FLANGE AND STOP AT FILLET OF BOTTOM FLANGE.

7 STEEL HSS POST TO TRANSFER GIRDER
NO SCALE



6 BEAM OVER COLUMN CONNECTION
1" = 1'-0"

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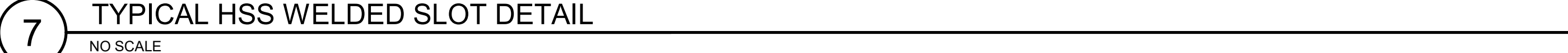
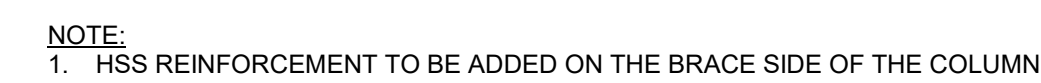
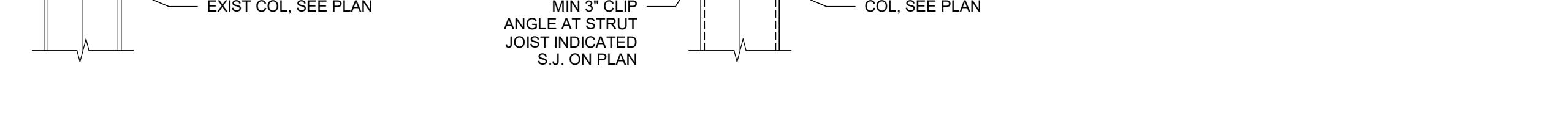
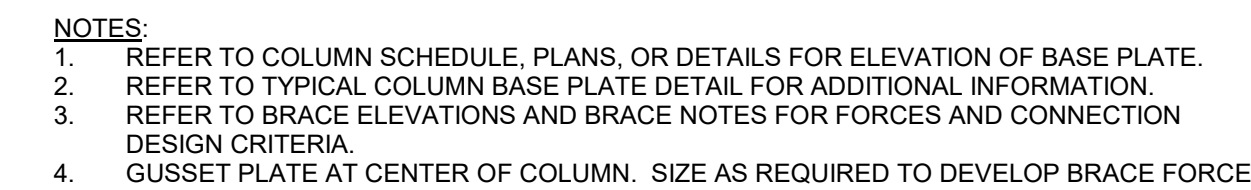
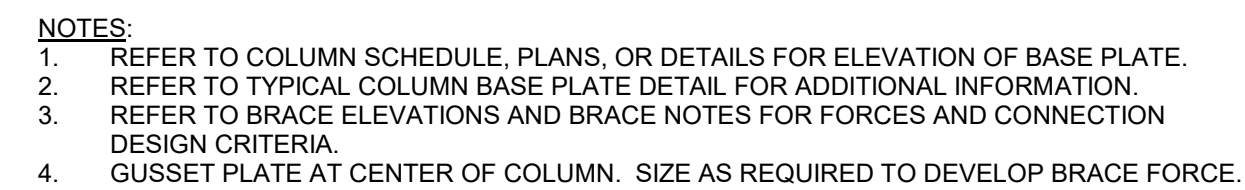
VSW JOB NO. SCALE
250202 1" = 1'-0"

STRUCTURAL DETAILS

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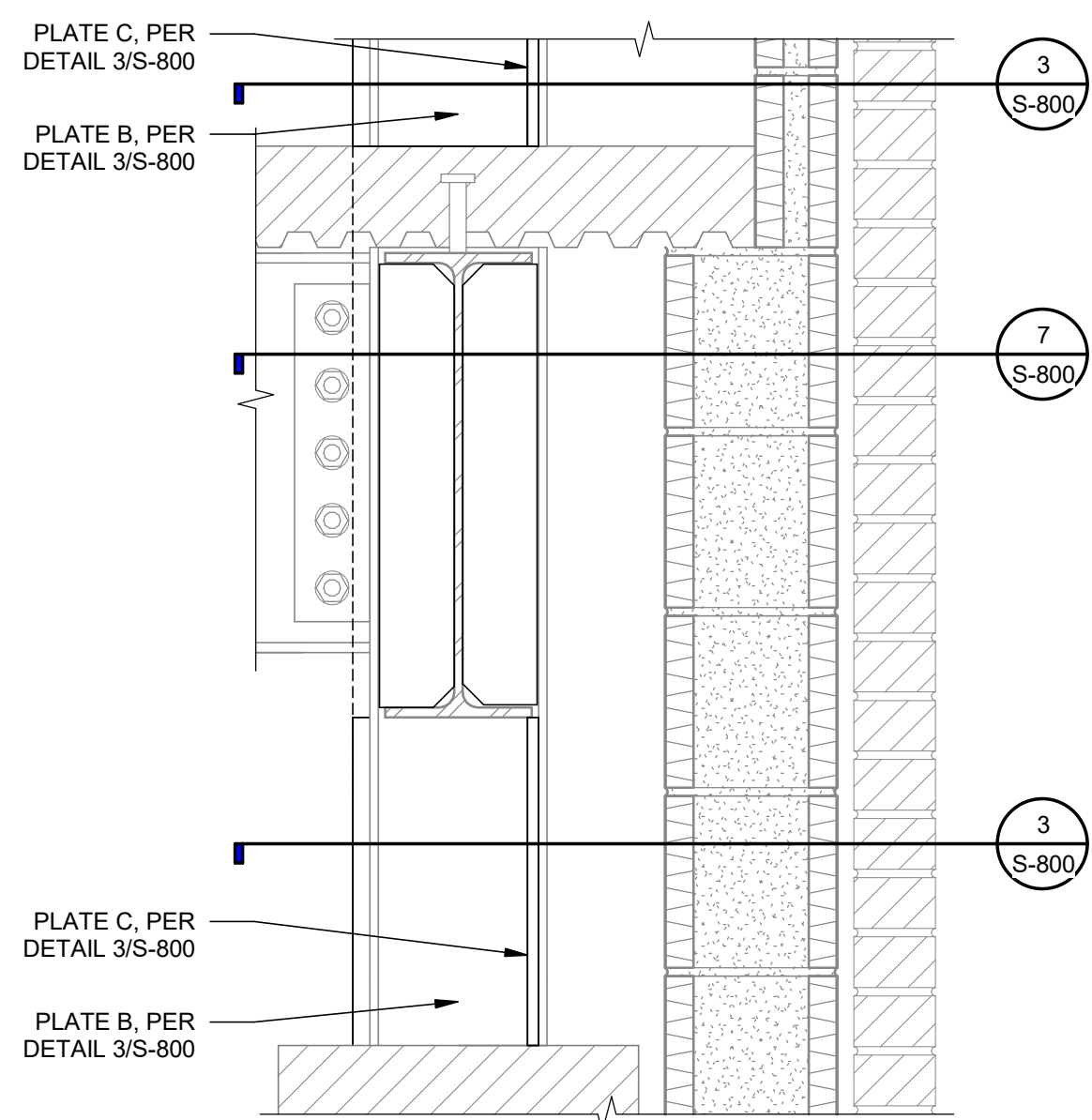
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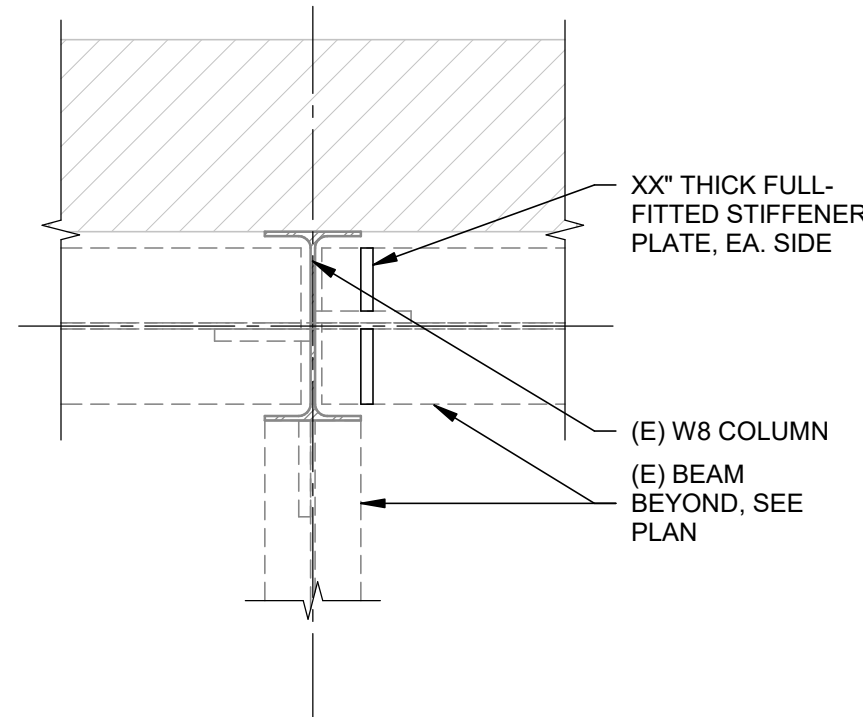
- TYPICAL BRACED FRAME NOTES
-
- NO SCALE



3 EXTERIOR COLUMN REINFORCEMENT



1 TYPICAL
INTERIOR COLUMN REINFORCEMENT



0-000